PHYSICS GRADE 10: INTRODUCTION TO SPACE PHYSICS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.2 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 4.0: Space Physics
Sub-Strand	Sub-Strand 4.2: Introduction to Space Physics
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– The Big Bang Theory and the origin of the universe – Evolution of the universe (formation of galaxies, stars, and planetary systems) – Classification of celestial bodies (stars, planets, moons, asteroids, comets, nebulae, galaxies) – The Solar System: structure, composition, and scale – Planetary motion: Kepler's Laws of orbital motion – Gravitational forces and their role in maintaining orbits – Telescopes: types, principles of operation, and their role in space exploration – History of space exploration (Sputnik, Apollo missions, Mars rovers, James Webb Space Telescope) – Kenya and Africa's role in astronomy (e.g., Square Kilometre Array, African VLBI Network) – Effects of space phenomena on Earth (solar winds, eclipses, tides, space weather) – Careers in space physics and related fields – Consolidation: connecting evidence to the origin and structure of the universe
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the Big Bang Theory as the current scientific model for the origin of the universe, b) describe the evolution of the universe including the formation of galaxies, stars, and planetary systems, c) classify celestial bodies found within and beyond the Solar System, d) explain the structure and scale of the Solar System, e) apply Kepler's Laws to describe and explain planetary motion, f) explain how gravitational forces govern the motion of celestial bodies, g) describe the types and principles of operation of telescopes used in space exploration, h) outline the history of space exploration and Kenya/Africa's emerging role in astronomy, i) investigate the effects of space phenomena (solar winds, eclipses, tides) on Earth, j) identify career opportunities in space physics and related fields, k) appreciate the significance of space science in understanding humanity's place in the universe.
Core Competencies	– Communication and Collaboration: Learners work in groups to debate competing models of the universe, present findings from simulations (e.g., PhET Gravity and Orbits), and co-construct the Driving Question Board — embracing teamwork and

	shared decision-making throughout. – Critical Thinking and Problem Solving: Learners analyse orbital data, apply Kepler's Laws to real planetary datasets, and evaluate evidence from the cosmic microwave background and redshift to assess the validity of the Big Bang Theory. – Learning to Learn: Learners practise self-directed inquiry by using NASA resources, Khan Academy, and Stellarium to investigate celestial phenomena, building metacognitive awareness of how scientific models evolve over time. – Digital Literacy: Learners interact with digital tools including Stellarium sky maps, PhET simulations, the Virtual Telescope Project livestreams, and YouTube documentary content to gather and interpret space science evidence. – Creativity and Imagination: Learners design cumulative visual models of the universe's evolution — from the Big Bang to the present Solar System — revising and improving them as new evidence is introduced across lessons.
Core Values	– Responsibility: Learners take ownership of assigned roles when researching and presenting career profiles in space physics, and responsibly curate evidence added to the shared Driving Question Board. – Respect: Learners demonstrate open-mindedness toward diverse cultural and scientific perspectives on the cosmos — including indigenous Kenyan and African astronomical traditions — while engaging with peers during discussions and presentations. – Unity: Learners recognise that space science is a global endeavour, appreciating Kenya and Africa's growing contribution to international astronomy projects such as the Square Kilometre Array (SKA), and celebrate shared human curiosity about the universe. – Social Justice: Learners confront myths and gender misconceptions about who can study or work in space physics, actively affirming that all learners — regardless of gender or background — can pursue careers in astronomy and space science.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: Learners generate and refine questions for the Driving Question Board in every lesson, progressively narrowing their inquiry from "How did the universe begin?" toward specific, testable sub-questions about planetary motion and space phenomena. – SEP 2 – Developing and Using Models: Learners build, revise, and annotate cumulative visual models of the universe — starting with the Big Bang and expanding to include the Solar System, orbital paths, and telescope fields of view — across all 12 lessons. – SEP 4 – Analysing and Interpreting Data: Learners analyse real orbital period and distance data for Solar System planets to derive and verify Kepler's Third Law, and interpret redshift spectral data as evidence for an expanding universe. – SEP 6 – Constructing Explanations and Designing Solutions: Learners construct written and oral explanations linking the Big Bang, gravitational collapse, and planetary motion to the anchoring phenomenon of the night sky visible from Nairobi. – SEP 7 – Engaging in Argument from Evidence: Learners evaluate competing historical models of the Solar System (geocentric vs. heliocentric) using observational evidence, and argue for the validity of the Big Bang Theory using cosmic microwave background radiation and Hubble's redshift data. – SEP 8 – Obtaining, Evaluating, and Communicating Information: Learners critically read NASA summaries, watch curated documentary clips, and use Stellarium to obtain credible space science information, then communicate findings through class presentations, model annotations, and the DQB.
Pertinent & Contemporary Issues (PCIs)	– Gender Disparity: Learners actively challenge myths and stereotypes about gender in science as they research and present profiles of space scientists — including female astronomers and astronauts (e.g., Mae Jemison, Margaret Hamilton, and African women in STEM) — affirming that space physics is an inclusive field open to all learners regardless of gender. – Environmental Education: Learners examine the effects of space weather (solar flares, geomagnetic storms) on Earth's environment and climate systems, connecting space phenomena to environmental sustainability concerns relevant to Kenya's agricultural and energy sectors. – Life Skills — Critical Thinking and Decision Making: Learners evaluate the credibility of online sources about space, distinguishing scientific evidence from pseudoscience and misinformation about topics such as flat Earth claims and astrology.

	– Career Guidance and Counselling: Learners explore diverse career pathways in space physics — including astrophysics, aerospace engineering, satellite communications, and meteorology — and connect these to Kenyan institutions such as the Kenya Space Agency (KSA) and universities offering related programmes.
Career Connections	– Astrophysicist / Astronomer: This sub-strand directly develops the conceptual foundations (Big Bang, stellar evolution, celestial classification) required for university-level astrophysics, with reference to African research programmes such as the SKA and the African VLBI Network. – Aerospace Engineer: Learners study the history of space exploration, rocket trajectories, and orbital mechanics, providing a foundational context for engineering careers in satellite design and launch — relevant to Kenya's growing space sector via the Kenya Space Agency. – Satellite Communications Technician: Learners explore how satellites maintain orbital paths using gravitational principles, directly linking to careers in Kenya's telecommunications and broadcasting industries that depend on geostationary satellites. – Meteorologist / Climate Scientist: Learners investigate how space phenomena such as solar wind and the Earth–Moon gravitational interaction (tides) affect Earth's weather and oceans, connecting to careers in Kenya Meteorological Department and climate research. – Secondary School Physics Teacher / Science Educator: Learners develop science communication skills throughout the unit — presenting models, explaining phenomena, and building the DQB — preparing those who may pursue education pathways. – Data Scientist / Computational Physicist: Learners analyse orbital datasets and interpret spectral data, introducing the quantitative and data-driven skills underpinning modern space science careers in big-data astronomy (e.g., SKA data pipelines).
Focus for Lessons	This unit introduces Grade 10 learners to the fundamental concepts of space physics through an inquiry-driven, phenomenon-anchored approach rooted in the Kenyan context. Beginning with the puzzle of the observable night sky above Nairobi and the question of how everything in it came to exist, learners progressively build understanding of the universe's origin, the structure and motion of the Solar System, the tools used to explore space, and Kenya's emerging role in global astronomy. The unit integrates digital simulations, NASA and ESA resources, and African space science examples to make abstract cosmic concepts tangible and locally relevant, while systematically developing NGSS Science and Engineering Practices alongside KICD core competencies.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? KICD KEY INQUIRY QUESTIONS: 1. How did the universe begin and how has it evolved over time? 2. How do the motions of celestial bodies in the Solar System affect life on Earth?
Anchoring Phenomenon	ANCHORING PHENOMENON: On a clear night in Nairobi — away from the city's light pollution, perhaps in the Rift Valley or on the shores of Lake Victoria — a student gazes at the sky and sees thousands of stars, a glowing band of the Milky Way, the Moon hanging large and bright, and (on certain nights) visible planets like Venus or Jupiter. The sky looks static and eternal, as though it has always existed exactly this way. Yet scientists claim that approximately 13.8 billion years ago, none of this existed — not the stars, not the planets, not even space or time itself. Everything observable in tonight's sky allegedly erupted from an infinitely hot, infinitely dense point in what is called the Big Bang. What is puzzling is the disconnect between the apparent permanence and stillness of the night sky and the scientific claim of a violent, dynamic origin and an ever-expanding universe. How can something as vast, structured, and seemingly unchanging as the universe we see above Nairobi have begun from nothing, and how do we know? This tension — between what we see and what science tells us — anchors every lesson in the unit and drives students to seek evidence that bridges direct observation with deep cosmic history.
Storyline Thread	Lesson 1 – The Big Bang Theory: Students are confronted with the anchoring phenomenon (the night sky above Nairobi) and

	make initial predictions about the universe's origin. Using Kurzgesagt YouTube animations and Khan Academy reading on the Big Bang, learners gather evidence for the universe's violent origin 13.8 billion years ago and add their first questions to the Driving Question Board. Lesson 2 – Evolution of the Universe: Learners trace the timeline from the Big Bang through the formation of subatomic particles, atoms, stars, and galaxies. Using NASA's interactive galaxy classification tool and a structured timeline activity, they explain how the structured universe visible tonight emerged from early chaos — updating their cumulative model for the first time. Lesson 3 – Classification of Celestial Bodies: Learners use Stellarium to identify and classify objects visible in the Kenyan night sky (stars, planets, the Moon, the Milky Way). They build a classification table distinguishing stars, planets, moons, comets, asteroids, and galaxies, and connect each class of object to the universe's evolutionary timeline from Lesson 2. Lesson 4 – The Solar System: Structure and Scale: Learners investigate the structure, composition, and extraordinary scale of the Solar System using a scaled model activity (e.g., using a toilet paper roll to represent AU distances). They explain why planets orbit the Sun rather than flying off into space, introducing the concept of gravitational force as the organising principle. Lesson 5 – Planetary Motion and Kepler's Laws: Using the PhET "Gravity and Orbits" simulation and real orbital data for Solar System planets, learners derive and apply Kepler's Three Laws. They mathematically verify Kepler's Third Law ($T^2 \propto r^3$) using data for Earth, Mars, and Jupiter, and revise their solar system models to show elliptical rather than circular orbits. Lesson 6 – Gravitational Forces in Space: Learners deepen their understanding of gravity as the universal force governing all orbital motion — from the Moon orbiting Earth to Earth orbiting the Sun. They investigate the tidal effects of the Moon on Lake Victoria and ocean coastlines in Mombasa, connecting gravitational theory to observable Earth phenomena and supporting the anchoring phenomenon thread. Lesson 7 – Telescopes: Windows to the Universe: Learners investigate how telescopes work (refraction, reflection, and radio wave detection) and why they are essential for space science — since direct travel to distant objects is impossible. Using Virtual Telescope Project livestreams and video profiles of the Hubble and James Webb Space Telescopes, learners explain how humans gather evidence about the universe without leaving Earth, directly addressing the driving question. Lesson 8 – History of Space Exploration: Learners trace the timeline of human space exploration from Sputnik (1957) through the Apollo Moon landings, Mars rovers, and the James Webb Space Telescope, examining what each mission revealed. They also explore Kenya's connection through the Kenya Space Agency (KSA) and Africa's role in the Square Kilometre Array (SKA) project in South Africa. Lesson 9 – Space Phenomena and Effects on Earth: Learners investigate how space phenomena — solar flares, geomagnetic storms, eclipses, and tidal forces — affect life on Earth, using the supporting phenomenon of the 2024 geomagnetic storm's effect on Kenyan radio communications. They construct explanations linking solar activity to real impacts on Kenyan infrastructure, weather, and agriculture. Lesson 10 – Careers in Space Physics: Learners research and present profiles of space scientists and engineers — with emphasis on African and Kenyan professionals — connecting sub-strand content to career pathways through the Kenya Space Agency, universities, meteorological services, and telecommunications. Gender equity in space careers is explicitly addressed as a PCI

	focus. Lesson 11 – Consolidation and Evidence Review: Using a teacher-provided three-column template (Lesson Title | What Was Learned | How It Explains the Phenomenon), learners collaboratively synthesise all 11 lessons into a coherent evidence-based narrative answering the driving question. The Driving Question Board is reviewed, with all remaining questions addressed or flagged for future study. Lesson 12 – Model Building and Final Presentation: Learners complete and present their cumulative visual models of the universe — from the Big Bang to the present night sky above Nairobi — incorporating all key concepts: the Big Bang, stellar evolution, Solar System structure, orbital mechanics, gravitational forces, telescopic evidence, and space exploration history. Peers provide structured feedback and learners reflect on how their understanding has changed from Lesson 1.
Supporting Phenomena	– A time-lapse video of the night sky filmed from Amboseli National Park shows stars tracing arc paths across the sky over several hours — raising the question of whether the stars are moving or the Earth is rotating, and connecting to planetary motion and reference frames. – Photographs of the cosmic microwave background (CMB) radiation taken by the Planck satellite show faint temperature ripples spread uniformly across the entire sky — puzzling because this "afterglow" of the Big Bang looks the same in every direction, as if the entire universe shares a single origin point. – A Kenya Meteorological Department report on a geomagnetic storm that disrupted radio communications across East Africa prompts the question: how can an event occurring 150 million kilometres away on the Sun affect everyday technology in Nairobi? – During a total solar eclipse visible from parts of East Africa, the Moon perfectly covers the Sun's disc — a striking coincidence of apparent size that invites inquiry into the relative distances and sizes of celestial bodies and the geometry of the Earth–Moon–Sun system.

LESSON 1 (80 minutes): The Big Bang Theory: Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To confront learners with the anchoring phenomenon — the night sky above Nairobi — and launch the unit-long investigation into the origin, structure, and evidence of the universe.
Knowledge	Students will know that the Big Bang Theory proposes that the universe originated approximately 13.8 billion years ago from an infinitely hot, infinitely dense singularity, and that the universe has been expanding and evolving ever since.
Skills	Students will be able to: (a) make evidence-based predictions about the origin of the universe; (b) gather and interpret information from multimedia sources (Kurzgesagt animations and Khan Academy readings) about the Big Bang; (c) construct and post initial questions on the Driving Question Board (DQB); (d) distinguish between direct observation and scientific inference as ways of knowing.
Attitudes	Students will appreciate that scientific knowledge about the universe — including the Big Bang Theory — is built on evidence, reasoning, and the collaborative work of scientists across generations and cultures, and that wonder and curiosity are valid starting points for scientific inquiry.
Key Inquiry Question	How did the universe begin and how has it evolved over time?
Purpose in Storyline	This is the opening lesson of the unit. It confronts learners with the anchoring phenomenon (the Nairobi night sky appearing eternal and static) and creates cognitive dissonance by introducing the scientific claim that everything observable began 13.8 billion years ago from a singularity. Learners make initial predictions, gather first evidence via the Big Bang multimedia resources, and populate the Driving Question Board with questions they will investigate across all 7 lessons. All subsequent lessons will refer back to the questions and predictions generated here.
Safety Notes	No physical hazards in this lesson. Ensure digital devices (tablets/laptops) are handled responsibly. If using a darkened room to simulate night-sky viewing, ensure all learners can move safely and that any extension cords are taped down. Be sensitive to learners whose cultural or religious backgrounds may frame the origin of the universe differently — establish a classroom norm that science explains natural phenomena through evidence, and that multiple ways of knowing can coexist respectfully.

B. LESSON OVERVIEW

In this opening 80-minute lesson, learners are introduced to the anchoring phenomenon: a vivid description and imagery of the clear night sky as seen from outside Nairobi — perhaps from the Rift Valley escarpment or the shores of Lake Victoria — where thousands of stars, the Milky Way band, the Moon, and visible planets like Venus and Jupiter are observable. The sky appears permanent and unchanging. Yet science claims that 13.8 billion years ago, none of it existed. Learners first predict what they think the origin of the universe looks like, then gather evidence by watching a Kurzgesagt animation on the Big Bang and reading a Khan Academy article on the expansion of the universe. Through structured discussion and a sensemaking routine (Claim–Evidence–Reasoning), learners construct an initial explanation for the universe's origin. The lesson closes with learners posting their burning questions onto the class Driving Question Board (DQB), which will anchor the remaining 6 lessons in the unit. NGSS Science and Engineering Practices embedded: Asking Questions (Practice 1), Obtaining and Evaluating Information (Practice 8), and Constructing Explanations (Practice 6).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Wave-Particle Theory (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a high-resolution image of the night sky as seen from a dark site in the Rift Valley — a sweeping panorama showing the Milky Way band, a bright Moon, and visible planets. The teacher reads aloud a short vivid narrative: 'Imagine you are standing on the escarpment above Nakuru on a moonless night. You switch off your torch. Above you, thousands of stars blaze, the Milky Way stretches like a river of light from horizon to horizon, and Jupiter hangs low in the east. The sky feels ancient and unchanging — as if it has always been exactly this way.' Learners then respond individually and in silence (3 minutes) to two written prompts in their exercise books: (1) 'Where do you think all of this — the stars, the Moon, the planets, even space itself — came from?' and (2) 'How confident are you in your answer, and what would change your mind?' Pairs then share predictions with each other (Think–Pair) before the teacher cold-calls 4 pairs to share with the class. All predictions are recorded visibly on the board without judgment, labelled 'Our Starting Ideas.'	1. Display the Rift Valley night-sky image on the projector/screen in silence for 30 seconds before speaking — allow the image to land. 2. Read the narrative slowly and expressively. Then say: 'In your exercise books, write your honest prediction — not the textbook answer, YOUR idea. You have 3 minutes. I am not looking for right answers yet.' 3. Circulate silently during individual writing; do NOT correct or affirm — only nod to signal effort. 4. After Think–Pair (2 min), cold-call exactly 4 pairs using a random name stick. For each, ask: 'What did your pair predict, and what is one reason you believe that?' WAIT 10–15 seconds after each response before commenting. 5. Record every prediction on the board verbatim under 'Our Starting Ideas.' Circle any prediction that mentions 'explosion,' 'beginning,' or 'nothing' as these will resurface. 6. Bridge to the phenomenon tension: 'Some of you said the universe has always existed. Some said it was created. Scientists have a very specific — and very strange — answer. Let us find out what evidence they use.'	Think–Pair–Share with Written Prediction Anchoring. Learners first commit ideas to writing (preventing groupthink), then refine through peer talk, then share publicly. Recording all predictions on the board creates a classroom 'prediction wall' that can be revisited and revised — modelling how scientists revise ideas when confronted with new evidence. This directly mirrors NGSS Practice 1 (Asking Questions and Defining Problems).	Observable indicator: Every learner has at least two sentences written in their exercise book before the pair-share begins. Teacher scans books during circulation. Listen for and note: (a) learners who invoke a creator/deity — these need careful, respectful handling; (b) learners who mention 'Big Bang' already — probe: 'What evidence do you have for that?'; (c) learners who say 'I don't know' — prompt: 'What does the sky make you wonder about?' Record 2–3 key misconceptions on a sticky note for targeted follow-up in Phase 3.
Observe Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Wave-Particle]	Learners gather evidence from two multimedia sources. First (15 min), the class watches the Kurzgesagt YouTube animation 'The Big Bang — How Did Everything Begin?' (approx. 6–8 min). While watching, learners complete a structured Evidence Collection Table in their exercise books with three columns: 'What I Observed/Heard,' 'What It Suggests About the Universe's Origin,' and 'A Question	1. Before playing the video, say: 'As you watch, you are scientists gathering evidence. Your job is NOT to memorise — it is to notice: What does this video claim? What evidence does it offer? Write in your table as you watch.' 2. Play the Kurzgesagt video in full without pausing. If internet is unavailable, use the downloaded offline version on the ARES platform. 3. After the video, give exactly 2 minutes of	Structured Evidence Collection Table + Annotated Reading. The two-source evidence gathering (visual animation + academic text) models how scientists triangulate evidence from multiple sources — a core scientific practice. The Evidence Collection Table structures observation so learners do not passively consume the video but actively extract claims and questions. Annotated reading	Observable indicator: Each learner's Evidence Collection Table has at least 2 rows completed (one from the video, one from the reading). During circulation while learners read, teacher checks for ★ and ? annotations — a blank page signals disengagement. Listen during cold-call sharing: target misconception is 'the Big Bang was an explosion in space' (correct

Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	It Raises.' After the video, learners have 2 minutes to complete their tables silently. Second (12 min), learners individually read the Khan Academy article 'The Big Bang' (printed handout or on device), annotating with two symbols: a star (★) next to any statement they find surprising, and a question mark (?) next to anything they do not understand. Learners then share their stars and question marks with a partner. The teacher cold-calls 3 learners to share one surprising statement each, and these are written on the board under 'Evidence We Found.'	silent writing time. Say: 'Silence please — finish your table. What did you observe? What does it suggest? What question does it raise?' 4. Distribute the Khan Academy article handout (or direct learners to ARES reading). Say: 'Read carefully. Mark a star for surprising, a question mark for confusing. You have 10 minutes.' 5. After reading, cold-call 3 learners (use name sticks): 'What one thing surprised you most? Tell us the sentence from the article and why it surprised you.' WAIT 10–15 seconds after each. 6. Write their surprising statements on the board. Then say: 'Look at our Starting Ideas from Phase 1. Which of those predictions does this evidence support? Which does it challenge?' Pause 15 seconds. Cold-call 2 learners.	with ★ and ? symbols (a modified Cornell reading strategy) develops scientific reading literacy and surfaces genuine confusion for the teacher to address. This enacts NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information).	to: it was an expansion OF space itself). If this misconception surfaces, note it on the board with a ?: 'Was it an explosion IN space, or an expansion OF space? We will return to this.'
Explain Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Wave-Particle Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Using their predictions (Phase 1) and evidence (Phase 2), learners now construct an initial scientific explanation using the Claim–Evidence–Reasoning (CER) framework, written individually in their exercise books. The prompt is: 'Make a claim about how the universe began. Use at least two pieces of evidence from the video and reading to support it. Explain the reasoning that connects your evidence to your claim.' Learners write for 8 minutes. Pairs then compare CER statements: 'Where do our explanations agree? Where do they differ? What is still unclear?' The teacher facilitates a whole-class discussion by projecting 2–3 anonymised CER examples (written on the board or typed quickly) and asking the class to evaluate: 'Is this evidence strong? Does the reasoning	1. Write the CER prompt on the board. Say: 'You have 8 minutes. Write your own explanation — not your partner's. Use the words: CLAIM, EVIDENCE, REASONING as headings.' 2. Circulate and read over shoulders. Identify 2 strong and 1 developing CER to share anonymously. Note: do NOT put a weak learner's work on the board without their permission. 3. After pair comparison (3 min), bring class together. Say: 'I found some interesting explanations. Let us evaluate them together as scientists.' Write the 2–3 examples on the board. 4. Ask: 'Does this evidence actually support this claim?' WAIT 15 seconds. Cold-call 3 different learners (not those who spoke in Phase 1 — track participation). 5. After CER discussion, shift register: dim the lights if possible, show the Rift	Claim–Evidence–Reasoning (CER) Framework. CER is a structured scientific argumentation strategy that separates what a learner believes (claim) from what they can point to as support (evidence) and how they connect the two (reasoning). Using CER in Lesson 1 establishes the epistemic norms for the entire unit — science is not about reciting facts but about constructing arguments from evidence. Evaluating anonymised peer CER statements develops critical scientific literacy and models peer review. This enacts NGSS Practice 6 (Constructing Explanations) and Practice 7 (Engaging in Argument from Evidence).	Observable indicator: Each learner's exercise book contains a CER with all three labelled components. Quality check: Does the evidence section contain specific data (e.g., '13.8 billion years ago,' 'temperature of universe dropped as it expanded') rather than vague restatements ('the video said the universe started')? Collect 5–6 exercise books at random at end of lesson to review CER quality and identify learners needing targeted support in Lesson 2. Key misconception to flag: learners who write 'the Big Bang proves God does not exist' — address sensitively: 'Science describes HOW, not necessarily WHY or WHO. Both questions can coexist.'

	connect to the claim?' The lesson then zooms out: the teacher connects the CER back to the phenomenon by asking, 'If the Big Bang happened 13.8 billion years ago, why does the night sky above Nairobi look so calm and permanent tonight?' This question is left deliberately open — it will not be resolved today. It is the engine of the unit.	Valley night-sky image again. Say quietly: 'Here is what puzzles me — and what I want you to carry with you for the rest of this unit: If everything began 13.8 billion years ago in a violent event, why does this sky look so still? So permanent? So eternal? How do we KNOW what we claim to know, if we cannot travel back in time or touch a star?' 6. Pause for 15 seconds of silence. Do NOT answer. Say: 'Hold that question. That is exactly what this unit is about.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Wave-Particle Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives 2 sticky notes (or small paper squares if sticky notes unavailable). They write ONE genuine question per sticky note — questions that genuinely puzzle them after today's lesson. Sentence starters are provided on the board: 'I wonder why…', 'How do scientists know…', 'What would happen if…', 'Is it possible that…'. Learners then post their sticky notes onto the class Driving Question Board (a large sheet of manila paper or a section of the classroom wall labelled 'OUR DRIVING QUESTION BOARD'). The teacher reads aloud 6–8 of the posted questions and, with learner input, clusters them into emerging themes (e.g., 'Questions about the beginning,' 'Questions about evidence,' 'Questions about stars and planets,' 'Questions about us/Earth'). The teacher then reveals the unit's overarching Driving Question — 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' — and	1. Distribute 2 sticky notes per learner. Say: 'You now know more than you did an hour ago — but real scientists never stop asking questions. Write two questions that you GENUINELY want answered. Not questions you think I want. Real ones. You have 3 minutes.' 2. Play soft background music (optional) during question writing. 3. Say: 'Bring your questions and post them on the board. Read your neighbour's questions as you post.' Allow 2–3 minutes of movement and reading. 4. Stand at the DQB and read 6–8 questions aloud expressively. Say: 'Listen to these. These are YOUR questions. Scientists ask questions exactly like these.' 5. Cluster questions visually on the board with learner input: 'Where does this question belong? Is it about the beginning? About evidence? About stars?' Use a marker to draw circles around clusters and label them. 6. Reveal the unit Driving Question by posting it at the top of the DQB. Read it together as a class. Say: 'Every lesson in this unit is one step toward answering THIS question. Your questions on this	Driving Question Board (DQB) — Wonder Wall Protocol. The DQB is a collaborative sensemaking artefact that externalises the class's collective curiosity and tracks how understanding grows across a unit. By clustering learner questions into themes, the teacher models how scientists organise inquiry around big ideas rather than isolated facts. Returning to the DQB each lesson and annotating answered questions creates a visible narrative of intellectual progress — a powerful motivator and a metacognitive tool. This enacts NGSS Practice 1 (Asking Questions) at the unit level.	Observable indicator: Every learner posts at least one sticky note. Quality check: Are questions specific and scientifically tractable ('How do scientists measure the age of the universe?') or vague and unfocused ('What is space?')? For vague questions, prompt the learner quietly: 'That is a great start — can you make it more specific? What exactly about space puzzles you?' Photograph the DQB and note the distribution across clusters — if most questions cluster around 'the beginning' and few address 'evidence,' plan to emphasise the evidence-gathering theme explicitly in Lesson 2.

	posts it at the top of the DQB. Learners record the Driving Question in their exercise books.	board will be answered — one by one — over the next six lessons. We will return to this board every single day.' 7. Take a photo of the DQB for the ARES platform record.		
Model Building Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Wave-Particle Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final phase, learners create their first individual scientific model — a labelled timeline diagram — in their exercise books. The prompt is: 'Draw a timeline of the universe from the Big Bang to tonight's sky above Nairobi. Label at least 4 key events you learned about today. Use arrows, drawings, and labels — this is a thinking tool, not an art competition.' Learners work individually for 6 minutes. Pairs then compare timelines: 'What did your partner include that you missed? What is different about how you drew it?' Learners add at least one new element to their timeline based on peer feedback, using a different colour pen. The teacher tells learners: 'This is Version 1 of your Universe Model. You will revise it in every lesson as you learn more. By Lesson 7, your model will be much richer. Keep it — do not tear it out.' The lesson closes with the teacher restating the anchoring phenomenon tension and issuing a brief homework prompt.	1. Say: 'Scientists do not just explain with words — they build models. Models are thinking tools that help us see patterns and gaps. Your first model is a timeline. Open a fresh page in your exercise book. Title it: MY UNIVERSE MODEL — VERSION 1. Date it.' 2. Write the prompt on the board. Clarify: 'Four events minimum. Arrows showing direction of time. Labels with approximate ages if you know them. Go.' 3. Circulate during the 6 minutes. Prompt struggling learners: 'What is the very first event you know about? Put that at the left. What happened next?' 4. After pair comparison (2 min), say: 'Add at least ONE new thing your partner had that you did not — use a different colour. This shows your model growing.' 5. Hold up 2–3 exercise books (with permission) to show diverse approaches: 'Notice how [learner's name] drew the expansion of space with widening lines. Notice how [another learner] included the formation of the Milky Way. Both are valid models.' 6. Close: 'Every lesson, we will add to this model. By the time we reach Lesson 7, you will be able to explain everything in the night sky above Nairobi. Tonight, if you can, look at the sky — or look at a photo of the Rift Valley night sky on your phone. Ask yourself: What am I seeing? When did it form? How do I know?' 7. Homework prompt: 'Find out ONE fact about the	Scientific Model Building — Iterative Timeline (Version 1 of 7). Using a versioned model throughout the unit is a deliberate metacognitive strategy: learners can see their own understanding grow visually, lesson by lesson. The timeline format is cognitively appropriate for Lesson 1 (chronological, linear) and will be expanded into more complex diagrams in later lessons (orbital models, wave diagrams, stellar life-cycle models). Peer comparison with a different-colour annotation models the peer-review process in science. This enacts NGSS Practice 2 (Developing and Using Models).	Observable indicator: Every learner has a labelled timeline with at least 4 events on a titled, dated page. Quality check: Does the timeline include approximate time scales (e.g., '13.8 billion years ago — Big Bang,' '4.6 billion years ago — Solar System forms')? Are the events in correct chronological order? Collect 4–5 timelines at random to review before Lesson 2 — identify learners who have correct chronology but missing time scales (needs reinforcement in Lesson 2) versus learners with chronological errors (needs targeted re-teaching of the Big Bang sequence at the start of Lesson 2).

		Cosmic Microwave Background Radiation — what it is and why scientists say it is evidence for the Big Bang. Bring it to Lesson 2.'		

D. TEACHER REFLECTION
After the lesson, consider the following reflective questions: (1) Did the anchoring phenomenon — the Rift Valley night sky image and narrative — genuinely create cognitive dissonance, or did learners accept the Big Bang claim without tension? If learners accepted it too quickly, the 'Why does the sky look permanent?' closing question may need more structured unpacking in Lesson 2. (2) Review the DQB photograph: How many questions are scientifically tractable? Are there clusters of questions that reveal deep misconceptions about the Big Bang (e.g., 'Did God make the Big Bang?', 'Was there a universe before the Big Bang?')? These are valuable — plan brief acknowledgements at the start of Lesson 2. (3) Review the 5–6 collected exercise books: What is the quality of CER reasoning — are learners connecting evidence to claims, or simply restating the video? Adjust the scaffolding of the CER framework in Lesson 2 accordingly. (4) Were all learners reached in cold-calling? Track participation with a class list — note learners who did not speak publicly today and intentionally target them in Lesson 2. (5) Was the multimedia (Kurzgesagt video) accessible to all learners on the ARES platform offline? Note any technical issues for the platform administrator.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A high-resolution image of the night sky above the Rift Valley showing the Milky Way, Moon, and visible planets; a Kurzgesagt animation showing the Big Bang and early universe expansion; a Khan Academy article describing the Big Bang, the expansion of the universe, and the timeline from singularity to the present day.
What did I learn?	The universe began approximately 13.8 billion years ago from an infinitely hot, infinitely dense singularity in an event called the Big Bang. The universe has been expanding and cooling ever since. Everything observable in tonight's sky — stars, planets, galaxies, even space and time itself — originated from this event. Scientists use evidence such as the expansion of galaxies and the Cosmic Microwave Background Radiation (CMB) to support this theory, even though no human was present and we cannot travel back in time to observe it directly.
How does this explain the phenomenon?	The night sky above Nairobi appears permanent and unchanging, but this is because the time scales of cosmic change are far longer than a human lifetime or even recorded human history. The Big Bang Theory explains the origin of everything observable in the universe — not as an explosion within existing space, but as the expansion of space itself from a singularity. The tension between the static appearance of the night sky and the scientific evidence of a dynamic, expanding universe is the central puzzle of this unit, and every subsequent lesson will add evidence to help explain and model this phenomenon.

LESSON 2 (80 minutes): Evolution of the Universe: From the Big Bang to the Structured Night Sky	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to trace the evolutionary timeline of the universe from the Big Bang through the formation of subatomic particles, atoms, stars, and galaxies, and to connect this cosmic history to the structured night sky observed above Nairobi.
Knowledge	Learners will know that: (a) the universe began approximately 13.8 billion years ago in an event called the Big Bang; (b) the universe evolved in distinct stages — from a quark-gluon plasma, to the formation of protons and neutrons, to the first atoms (hydrogen and helium), to the first stars and galaxies; (c) galaxies are classified by shape (elliptical, spiral, irregular) and the Milky Way is a barred spiral galaxy; (d) the Cosmic Microwave Background (CMB) radiation is key evidence for the Big Bang; (e) the universe continues to expand today.
Skills	Learners will be able to: (a) construct and annotate a scaled timeline of the universe's evolution; (b) classify galaxy images using NASA's galaxy classification tool; (c) explain how early chaos gave rise to the structured universe visible tonight; (d) use evidence (CMB, galaxy formation, expansion) to support the Big Bang model; (e) update their cumulative explanatory model to incorporate evolutionary stages.
Attitudes	Learners will appreciate that the seemingly static and eternal night sky above Nairobi is the product of 13.8 billion years of dynamic cosmic evolution, and will develop curiosity and humility in the face of scientific evidence that challenges everyday appearances.
Key Inquiry Question	How did the universe begin and how has it evolved over time?
Purpose in Storyline	In Lesson 1, learners established what they can directly observe in the night sky above Nairobi — stars, the Moon, the Milky Way band, and planets — and raised the puzzle of how something so vast and structured could have a beginning. Lesson 2 now provides the first layer of scientific explanation: learners trace the cosmic timeline from the Big Bang to the formation of the galaxies they see tonight, explaining HOW the structured universe emerged from early chaos. This is the first major evidence update to their cumulative model and directly advances the driving question by showing that the night sky has a deep evolutionary history.
Safety Notes	This lesson involves digital devices and screen-based activities. Ensure all devices are fully charged or connected to power. If using a shared computer lab, assign device pairs in advance to avoid crowding. No physical hazards are present. Remind learners to respect shared equipment and handle tablets/laptops with care.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 observational sketches of the Nairobi night sky and their initial predictions about the universe's origin. Through a structured Big Bang timeline activity, a NASA galaxy classification task, and a short video visualisation, they trace the 13.8-billion-year evolutionary story from the Big Bang to the stars and galaxies visible tonight. Sensemaking is anchored in the contrast between the apparent stillness of the night sky and the violent, dynamic processes that produced it. By the end of the lesson, learners update their cumulative explanatory model for the first time, adding the evolutionary stages of the universe as the foundation layer beneath everything they can observe.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Rogue Particles and Waves (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners individually examine their Lesson 1 night-sky sketch (or a shared class sketch of the Nairobi sky) and respond in writing to two prompt questions in their science notebooks: (1) 'If the universe began 13.8 billion years ago from a single point, what do you think existed in the first few minutes after that event? Write or sketch your best guess.' (2) 'The Milky Way band you can see from the Rift Valley contains over 200 billion stars. Where do you think all of those stars came from?' Learners share predictions with a partner using a Think-Pair-Share before three pairs are cold-called to share with the class. Predictions are recorded on a class 'Prediction Wall' (sticky notes or a section of the whiteboard) and left visible for the remainder of the lesson so they can be revisited and revised.	Display the anchoring photograph of the Nairobi/Rift Valley night sky on the projector. Point to the Milky Way band and say: 'Last lesson we observed this sky carefully. Today I want you to think not about what you SEE — but about what MADE all of this.' Distribute science notebooks and read both prompts aloud. Say: 'Take 3 minutes — write or sketch silently. There are no wrong answers here; I want your honest thinking.' [WAIT 3 minutes — circulate and read responses without commenting.] After 3 minutes: 'Turn to your partner. Share what you wrote. You have 90 seconds each.' [WAIT 3 minutes for pair discussion.] Cold-call exactly 3 pairs. For each, ask: 'What did you and your partner agree on? What did you disagree on?' Record key prediction phrases verbatim on the Prediction Wall. Do NOT correct errors — say: 'Hold that thought. We will come back to these predictions at the end of the lesson and see how our thinking has changed.'	Think-Pair-Share with a Prediction Wall. This strategy activates prior knowledge and surfaces misconceptions (e.g., 'the universe has always existed,' 'stars were created by God/were always there') in a low-stakes environment. Recording predictions publicly creates cognitive accountability — learners are motivated to reconcile their initial ideas with evidence gathered during the lesson.	Teacher circulates during individual writing and scans notebooks. Observable indicators of readiness: (a) learner can name at least one thing that might have existed early in the universe (even if incorrect); (b) learner attempts a sketch or written description. Flag any learner who writes nothing — seat them next to a more confident partner. Use cold-call responses to gauge the range of prior conceptions present in the class before instruction begins.
Observe Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Rogue Particles and Waves (PLIX) Source: CK-12 🔍 Search ARES for similar	Learners engage in TWO evidence-gathering activities run in sequence: ACTIVITY A — Big Bang Timeline Construction (25 min): Working in groups of 4, learners receive a printed 'Cosmic Events Card Set' (12 cards, each describing a key event: Big Bang, quark-gluon plasma, protons/neutrons form, first atoms form [380,000 yrs], first stars ignite [~200 million yrs], first galaxies form [~1 billion yrs], Milky Way forms [~13.6 billion yrs ago], Solar System forms [~4.6 billion	Distribute card sets and ropes. Say: 'Your job is to become the universe's historians. These 12 cards describe events across 13.8 billion years. Your challenge: put them in order AND mark on the rope roughly WHERE each event falls — not just the order, but how long after the Big Bang it happened. Pay attention to the numbers.' [WAIT 8 minutes for initial ordering — circulate and observe.] After 8 minutes, prompt groups who are struggling with scale: 'If the Big Bang is at one end and tonight is at the other end — where on this 2-metre rope	Concrete Timeline Manipulation + Image-Based Classification. The physical rope timeline makes the abstract scale of cosmic time tangible — learners viscerally experience that human history is an almost invisible sliver of cosmic time, which directly challenges the perception that the night sky is 'eternal.' The galaxy classification activity connects the abstract evolutionary story to real observable objects, reinforcing that galaxies — including our own Milky Way visible from the Rift Valley — are the physical products of that evolutionary process.	Teacher photographs each group's timeline arrangement during circulation. Observable success indicators: (a) cards are in correct chronological order; (b) first atoms are placed significantly far from the Big Bang on the rope (380,000 years vs. 13.8 billion — a very small distance); (c) Solar System and Earth are placed in the final third of the rope; (d) learners correctly identify the Milky Way as a barred spiral. Misconception flag: if groups place 'first stars' very close to the Big Bang (confusing minutes with millions of years), pause the class and ask: 'What

readings	yrs ago], Earth forms, first life on Earth, modern humans, tonight above Nairobi). Groups arrange the cards in chronological order along a 2-metre rope stretched across their desk (representing 13.8 billion years on a log scale). They annotate each card with: what existed at that stage, and what did NOT yet exist. Groups photograph their completed timeline with a tablet. ACTIVITY B — NASA Galaxy Classification (15 min): Pairs of learners access NASA's Galaxy Zoo / galaxy classification interface (or a teacher-prepared offline image gallery of 10 galaxy images). They classify each galaxy as elliptical, spiral, barred spiral, or irregular, and record their classifications in a shared table. Learners identify which type our Milky Way belongs to (barred spiral) and locate an example galaxy image that resembles what they see as the Milky Way band from the Rift Valley.	would the entire history of human civilisation fit? Think about that.' [WAIT 15 seconds.] After timeline completion, direct learners to Activity B. Say: 'You have just seen how galaxies took about 1 billion years to form. Now let us look at what galaxies actually look like. Open the galaxy image gallery. Your task is to classify each image and find our home galaxy.' Circulate during classification. Prompt deeper thinking by asking pairs: 'Why do you think some galaxies are smooth and round while others have spiral arms? What might cause those differences?' [WAIT 10 seconds before taking responses.] Cold-call 4 pairs to share their Milky Way identification and reasoning.		does 200 million years look like on a 13.8-billion-year timeline? Work that out as a fraction.'
Explain Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Rogue Particles and Waves (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners watch the short video 'Evolution of the Universe in 5 Minutes' (YouTube, teacher pre-downloaded for offline use). While watching, they complete a structured note-catcher with three columns: STAGE | WHAT FORMED | EVIDENCE WE HAVE TODAY. After the video, groups compare their timeline rope arrangement with the video's sequence and make any corrections. Each group then produces a 'Universe Origin Explanation Paragraph' responding to the prompt: 'A farmer in Kericho looks up at the	Before the video: 'As you watch, fill in your note-catcher. Focus on three things: what formed at each stage, and what evidence scientists use TODAY to know this happened.' Play video (pre-downloaded). After video: 'Compare your timeline rope to what the video showed. Do you need to move any cards? You have 2 minutes.' [WAIT 2 minutes.] Introduce the Kericho farmer prompt. Say: 'This is the real test of understanding — can you explain it to someone who was not in this classroom? Write your paragraph now. 5 minutes,	Structured Note-Catcher + Explanation-to-a-Novice Writing. The note-catcher organises the video content into the STAGE → FORMATION → EVIDENCE framework that underpins scientific reasoning in this unit. The 'explain to a Kericho farmer' task forces learners to reconstruct the explanation in their own words for a non-specialist audience, which is the highest indicator of genuine understanding. The 'sky time-travel' questions connect the evolutionary timeline directly back to the anchoring phenomenon — the night sky above Nairobi — and	Collect or photograph the Kericho farmer explanation paragraphs. Success criteria: (a) paragraph mentions the Big Bang as a starting point; (b) paragraph includes at least two intermediate stages (e.g., formation of atoms, formation of stars); (c) paragraph connects the timeline to the visible Milky Way; (d) explanation avoids purely supernatural framing (learner uses causal, evidence-based language). Misconception to watch for: learners claiming 'the Big Bang created the planets directly' — probe with 'What had to exist before planets could form?'

	Milky Way on a clear night and asks you: Where did all those stars come from? Using what you have learned today, write a clear explanation — in plain language a farmer would understand — of how the universe evolved from the Big Bang to the galaxy they are looking at tonight.' Groups share paragraphs with the class. Whole-class discussion follows, anchored by the question: 'What would the sky above Nairobi have looked like 13 billion years ago? 1 billion years ago? 4.6 billion years ago?' Learners record a refined explanation in their own notebooks.	individually first.' [WAIT 5 minutes.] 'Now share with your group and agree on the best version.' [WAIT 3 minutes.] Cold-call 3 groups to read their paragraphs aloud. After each, ask the class: 'What did they explain well? What is missing or could be clearer?' Then pose the sky time-travel questions one at a time. For each, say: 'Think silently for 10 seconds, then share with your partner.' [WAIT 10 seconds between each question.] Cold-call 2 pairs per question. Conclude by saying: 'The sky above Nairobi looks permanent and eternal — but every single star you see is the product of billions of years of cosmic evolution. That is what the evidence shows us.'	make the abstract timeline personally relevant.	
Driving Question Board (DQB) Creation 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Rogue Particles and Waves (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board established in Lesson 1. Individually, each learner writes on a sticky note: (a) ONE thing they now know (GREEN note) that helps answer the driving question 'How did everything in the night sky above Nairobi come to exist?', and (b) ONE new question that today's lesson has raised for them (ORANGE note). Learners post their notes on the DQB under the relevant driving question column. The teacher leads a 5-minute whole-class 'tour' of the new notes, grouping emerging themes aloud (e.g., 'Several of you are now asking about WHERE the energy for the Big Bang came from — that is a question even professional scientists are still investigating'). The class collectively identifies which parts of the driving question today's lesson has partially answered and which parts remain open.	Hand out sticky notes (two colours). Say: 'Take a green note and write one thing you now KNOW — something from today — that helps us answer our driving question. Take an orange note and write the new question today's lesson has opened up for you. You have 3 minutes — write independently.' [WAIT 3 minutes.] Direct learners to post notes on the DQB. Lead a gallery-walk-style verbal tour: read several notes aloud, group related ideas, and say: 'Look at how many new questions we have generated. In science, good evidence does not just answer questions — it creates new ones. That is exactly what is happening here.' Point to the driving question and ask: 'Show of hands — who feels we have partially answered the first key inquiry question today? Who feels we need more evidence?' Acknowledge both responses as valid. Say: 'We will keep building.	Driving Question Board (DQB) — a Research-Practice Framework tool. The DQB functions as a public, cumulative record of the class's growing understanding. Posting both knowledge gains AND new questions models authentic scientific practice, where evidence simultaneously resolves and opens questions. The physical act of adding to the board each lesson makes the incremental nature of evidence-building visible and motivating.	Teacher reads all posted sticky notes after class. Observable indicators: (a) green notes reference specific content from the lesson (Big Bang, galaxy formation, CMB, timeline stages); (b) orange notes show genuine intellectual curiosity and are not simply restatements of lesson content; (c) absence of green notes from a learner signals incomplete understanding and requires follow-up at the start of Lesson 3. Tally the ratio of 'answered' vs. 'new questions' to calibrate the depth of coverage needed in subsequent lessons.

		Every lesson adds a new layer.'		
Model Building Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Rogue Particles and Waves (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 cumulative explanatory model (a sketch/diagram of what they think the universe looks like and how it works, drawn at the end of Lesson 1). Using a different coloured pen, they make their FIRST MODEL UPDATE by adding: (a) a timeline arrow showing the Big Bang at one end and 'tonight above Nairobi' at the other, with at least 4 labelled stages; (b) arrows showing that galaxies formed from earlier, simpler matter; (c) a label identifying the Milky Way as a barred spiral galaxy containing our Solar System; (d) a note indicating the universe is approximately 13.8 billion years old. Learners write a 'Model Update Note' at the bottom of their model page: 'I changed/added _______ because the evidence from today shows _______.' Models are stored in learners' science portfolios for use in all subsequent lessons.	Distribute learners' Lesson 1 models. Say: 'This is your living model of the universe. It will grow every lesson as we gather more evidence. Today, use a different colour pen so we can always see what you knew in Lesson 1 versus what you know now.' Display a sample model update on the projector (teacher-prepared example) showing the four required additions. Say: 'You have 8 minutes. Your model must show the timeline, the formation stages, the Milky Way label, and the 13.8-billion-year age. Write your Model Update Note at the bottom — this is your evidence log.' [WAIT 8 minutes — circulate and provide targeted feedback.] Prompt learners who only draw: 'Your model needs words too — label what each stage produced.' Before collecting, ask 2–3 volunteers to hold up their models and describe ONE change they made and WHY. Close the lesson by connecting back to the phenomenon: 'When you next look at the Milky Way from the Rift Valley or the shores of Lake Victoria, you will know that the light reaching your eyes left those stars millions or even billions of years ago — and that everything you see is the product of 13.8 billion years of cosmic evolution. That is what today's evidence tells us.'	Cumulative Model Revision with Evidence Annotation. The use of a contrasting pen colour to mark updates makes the evolution of learner understanding literally visible over time — both to the learner and the teacher. The mandatory 'Model Update Note' (I changed ___ because the evidence shows ___) enforces the link between evidence and explanation, which is the core NGSS Science and Engineering Practice of 'Constructing Explanations.' The model revision also creates continuity across the 12-lesson sequence, so every lesson builds visibly on the last.	Collect models for teacher review after class. Success indicators: (a) timeline arrow is present with at least 4 correctly labelled stages; (b) Milky Way is identified as barred spiral; (c) 13.8 billion years is noted; (d) Model Update Note uses evidence-based language ('the evidence shows…' rather than 'I think…'). Models with no updates, or updates that contradict lesson content (e.g., 'Big Bang happened 6,000 years ago'), should be flagged for individual follow-up at the start of Lesson 3.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your teaching journal:\n\n1. TIMELINE ACTIVITY: Were learners able to correctly place events on the rope at approximately the right proportional distances, or did they simply order them without grasping the scale? If scale was difficult, consider opening Lesson 3 with a brief 'cosmic calendar' analogy (compressing 13.8 billion years into one calendar year — humans appear on 31 December at 11:59 PM).\n\n2. PREDICTION WALL:

How many of the initial predictions on the Prediction Wall were revised during the lesson? Were there predictions that remained unchanged? This signals which misconceptions were most resistant — address these explicitly in Lesson 3.\n\n3. KERICHO FARMER PARAGRAPHS: Did learners use causal, evidence-based language, or did explanations remain vague ('everything just appeared')? If vague, the Explain phase may need more scaffolding in future lessons — consider providing a sentence-starter frame.\n\n4. DQB QUALITY: Were the orange 'new question' sticky notes genuinely curious and lesson-generated (e.g., 'What caused the Big Bang?', 'How do scientists know the age of the universe?'), or were they shallow? Shallow questions may indicate learners are not yet deeply engaged with the phenomenon — revisit the anchoring photograph at the start of Lesson 3.\n\n5. MODEL UPDATES: Were model updates substantive and evidence-referenced, or cosmetic (learners added a label without changing their underlying model)? Learners who made only cosmetic changes may not have integrated today's evidence — pair them with stronger model-builders in Lesson 3's group work.\n\n6. KENYA CONTEXT: Did the use of the Kericho farmer scenario and the Rift Valley/Lake Victoria sky references make the content feel personally relevant to learners? Note which contextual references generated the most engagement and use those entry points in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed: (1) a physical rope timeline showing 12 key cosmic events across 13.8 billion years, making the immense scale of cosmic time tangible; (2) a gallery of real galaxy images (elliptical, spiral, barred spiral, irregular) classified using NASA's galaxy classification framework; (3) a short video ('Evolution of the Universe in 5 Minutes') visualising the sequential stages from the Big Bang to the formation of galaxies, stars, and planetary systems.
What did I learn?	Learners learned that: (1) the universe began approximately 13.8 billion years ago in the Big Bang — before which neither matter, energy, space, nor time existed as we know them; (2) the universe evolved through distinct stages: quark-gluon plasma → protons and neutrons → first atoms (hydrogen and helium, ~380,000 years after Big Bang) → first stars (~200 million years) → first galaxies (~1 billion years) → our Milky Way (~13.6 billion years ago) → our Solar System (~4.6 billion years ago); (3) the Milky Way — the glowing band visible from the Rift Valley and Lake Victoria — is a barred spiral galaxy containing over 200 billion stars, including our Sun; (4) the Cosmic Microwave Background radiation is one of the key pieces of evidence scientists use to confirm the Big Bang model; (5) the universe continues to expand today.
How does this explain the phenomenon?	Learners explained that: (1) the apparently static and eternal night sky above Nairobi is in fact the product of 13.8 billion years of dynamic cosmic evolution — from a hot, dense, structureless state to the complex, galaxy-filled universe visible tonight; (2) the structured universe (stars, galaxies, planets) emerged progressively as matter cooled and gravity caused it to clump together over vast timescales; (3) scientists gather evidence for this evolutionary history through CMB radiation, galaxy observation and classification, and the measured expansion of the universe — without needing to physically travel to or touch the objects they study; (4) each star visible from Nairobi is itself a product of this evolutionary process, and the light reaching our eyes left those stars millions or billions of years ago.

LESSON 3 (80 minutes): Classification of Celestial Bodies: Reading the Kenyan Night Sky	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners classify the objects visible in the Kenyan night sky — stars, planets, moons, comets, asteroids, and galaxies — and connect each class of object to the universe's evolutionary timeline established in Lesson 2, advancing the claim that the structured sky above Nairobi is the product of 13.8 billion years of cosmic evolution.
Knowledge	Learners will know: (a) the defining physical characteristics that distinguish stars, planets, moons, comets, asteroids, and galaxies; (b) which of these objects are visible from Nairobi with the naked eye or basic tools; (c) how each class of object fits into the chronological sequence of cosmic evolution (Big Bang → first stars → galaxies → solar systems → rocky bodies).
Skills	Learners will be able to: (a) use Stellarium (digital telescope platform) to identify and label at least five classes of celestial objects in a simulated Kenyan night sky; (b) construct a classification table that organises objects by type, key characteristics, distance scale, and position in the cosmic timeline; (c) ask investigable questions about observed objects and record them on the Driving Question Board.
Attitudes	Learners will appreciate that the apparently chaotic and overwhelming night sky above the Rift Valley is deeply ordered — each object belongs to a category with predictable properties — and that this order is itself evidence of the universe's structured evolutionary history.
Key Inquiry Question	How did the universe begin and how has it evolved over time?
Purpose in Storyline	In Lesson 1, learners anchored on the puzzling disconnect between a static-looking night sky and a violent cosmic origin. In Lesson 2, they traced the Big Bang timeline and the formation of matter, stars, and galaxies. Lesson 3 now asks: what exactly ARE those objects in tonight's sky, and when in cosmic history did each type appear? By classifying objects using Stellarium set to Nairobi's coordinates, learners transform the night sky from an undifferentiated field of lights into a structured catalogue of evidence — evidence they will continue interrogating through Lessons 4–12 to answer the driving question.
Safety Notes	No physical hazards in this lesson. If devices are shared, remind learners to handle tablets/laptops with care. If the session extends to an actual outdoor observation, caution learners about uneven ground in low-light conditions. Ensure screen brightness is kept comfortable to avoid eye strain during extended Stellarium use.

B. LESSON OVERVIEW

Learners open Stellarium set to Nairobi (1.2921° S, 36.8219° E) and the current date/time (or a clear pre-set night), then explore the simulated sky to identify objects they recognise from the anchoring phenomenon. Using guided observation prompts, they gather evidence about each object's appearance, movement, and brightness. They then read a short structured text and reference infographic that introduces the six major classes of celestial bodies and their cosmic-timeline origins. Working in pairs, learners build a classification table connecting each object class to its defining characteristics and its place in the evolutionary sequence from Lesson 2. The lesson closes with a Driving Question Board update, where new questions generated by the classification activity are posted, and a brief whole-class model-revision discussion asks: "Does classifying these objects help us explain how everything in the night sky came to exist?"

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Without opening Stellarium yet, learners individually examine a printed or on-screen wide-angle photograph of the night sky taken from the Rift Valley (outside Nairobi's light pollution). They study the image for 90 seconds in silence. Each learner then writes answers to three prompt questions in their science notebook: (1) 'Name every different TYPE of object you think you can see in this image — not individual objects, but categories.' (2) 'How would you decide whether two objects belong to the same category or different categories?' (3) 'Pick one object in this image that puzzles you most. What do you already know about it, and what do you NOT know?' Learners then share their type-lists with a neighbour (Think-Pair) for 2 minutes before the teacher takes a cold-call sample.	Display the Rift Valley night-sky photograph on the projector. Say: 'Before we use any tools today, I want you to be scientists working only with your own eyes and prior knowledge — just like the student in our phenomenon, standing outside Nairobi looking up. Study this image in silence for 90 seconds. Then answer the three prompts in your notebook.' [Start 90-second timer — visible on screen.] Circulate silently; do not answer questions. After writing time, say: 'Now share your type-list with the person next to you. You have 2 minutes. Argue if you disagree.' After pair-share, cold-call FIVE different students: 'Amina — what is one category you listed?' Tally all unique categories on the board without commentary. Then ask the class: 'Which categories appear on almost everyone's list? Which are surprising or controversial?' [WAIT 10 seconds before calling on anyone.] Do NOT confirm or correct — leave all predictions visible as a class record to revisit.	Think-Pair-Share with public tally: By forcing individual commitment before peer discussion, this strategy surfaces prior knowledge and misconceptions (e.g., learners often list 'satellites' or 'shooting stars' but not 'asteroids' or 'galaxies'). The public tally on the board creates a shared reference point that learners will compare against their classification table at the end of the lesson, making learning gains visible.	Observable indicator: Every learner's notebook contains at least three written category names and a written 'puzzle' question before discussion begins. Teacher scans notebooks during the pair-share. Red flag: If most learners list fewer than three categories or cannot articulate any distinguishing criterion, the teacher should prompt: 'Think about objects that MAKE their own light versus those that only REFLECT light — does that help you split your list?'
Observe Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar	Learners open Stellarium on paired devices, set location to Nairobi (or use the teacher's pre-saved 'Nairobi Night' session) and set time to a pre-selected clear night (e.g., a recent date with Venus and Jupiter visible). They complete a structured Stellarium Observation Sheet with four columns: Object Name | Type (my best guess) | How I can tell (evidence from Stellarium data panel) | Questions this raises. Learners must log at least eight distinct objects, including: the Moon, Venus, Jupiter, a named star (e.g., Sirius or Canopus,	Before releasing to Stellarium, demonstrate for 2 minutes on the projector: click on the Moon → show the data panel (type, distance, magnitude) → say: 'Stellarium is not just a pretty picture — it is a data source. Every object has a data panel. Your job is to BE A SCIENTIST: use that data to justify your type classification.' Release learners to paired devices. Circulate every 3–4 minutes. Use these specific prompts when visiting pairs: 'What is the data panel telling you about whether this object makes its own light?' / 'You classified that as a	Data-Panel Evidence Logging with Video Timestamp Matching: Stellarium's data panel gives learners real observational data (distance in light-years, object type, magnitude) rather than labels, pushing them to infer classification from evidence. Pausing the timeline video at cosmic milestones and asking learners to match their logged objects to those moments explicitly connects the classification task to the Lesson 2 evolutionary sequence — a core thread of the storyline.	Observable indicator: Each learner's Observation Sheet contains at least eight objects with a TYPE entry AND at least one piece of Stellarium data cited as evidence in the 'How I can tell' column. Teacher collects or photographs two or three sheets mid-activity to check for common error: students writing 'it looks bright' as evidence rather than citing data-panel values (magnitude, distance, spectral type). If this pattern is widespread, pause the class for a 2-minute mini-lesson: 'Brightness to your eye is NOT the same as what type

readings	visible from Nairobi), the Milky Way band, and at least one object from Stellarium's deep-sky catalogue (e.g., the Andromeda Galaxy or the Orion Nebula). After individual exploration (15 minutes), pairs compare their sheets and reconcile any disagreements about type-classification. The class then watches a 3-minute clip from 'Evolution of the Universe in 5 Minutes' (YouTube) paused at the moment each new class of object first appears in cosmic history, with the teacher asking learners to place a checkmark next to any object on their sheet that they spot in the video.	star — Venus is also bright, why is it NOT a star? What does the data say?' / 'Zoom out to maximum — what structure do you see that you didn't notice before?' [WAIT 12 seconds after each question before accepting an answer.] At the 15-minute mark, say: 'Freeze your screens. Do not close Stellarium.' Play the YouTube video. Pause at 0:42 (first stars form), 1:15 (galaxies), 2:05 (solar system formation). At each pause, ask the class: 'Raise your hand if you have an object on your sheet that was JUST born in the universe at this moment.' Cold-call THREE learners at each pause to justify their claim.		of object it is — show me the data panel value that tells you the difference.'
Explain Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive (or access on device) a one-page structured reference infographic: 'The Six Classes of Celestial Bodies and Their Place in Cosmic History.' The infographic presents stars, planets, moons, comets, asteroids, and galaxies in a visual timeline alongside their key defining characteristics (energy source, composition, size range, orbital behaviour). Working in pairs, learners construct a Classification Table in their notebooks with the following columns: Class | Defining Characteristics (2–3) | How to Identify in Stellarium | Approximate Cosmic Age (when this class first appeared) | Example visible from Nairobi. They must complete all six rows. Pairs then join into groups of four to compare tables, resolve disagreements using the infographic as arbiter, and generate one 'Big Explanation Claim' statement per group: a sentence that begins 'The variety of objects we see in the Nairobi	Distribute the infographic and say: 'Your classification table is your EVIDENCE ORGANISER — it turns what you observed in Stellarium into a structured argument you can use to answer our driving question.' Give pairs 12 minutes for table construction. Circulate and use Socratic prompts: 'You wrote that a planet defines its own light — look again at column two of the infographic. What is the actual energy source for a planet?' / 'Your cosmic-age column says planets formed at the Big Bang — does the infographic support that? What year in cosmic history does it say?' [WAIT 10 seconds after each prompt.] At the group-of-four stage (8 minutes), say: 'Your Big Claim must mention BOTH what the objects are AND when they appeared. A claim that only describes without explaining the timeline is incomplete.' Cold-call FOUR groups to read their claims. After each, ask the class: 'Thumbs up if this claim would help	Structured Classification Table with Claim Writing: The table provides a cognitive scaffold that forces learners to think across four dimensions simultaneously (characteristics, observability, timeline, local example). The transition from table to claim-writing is a disciplinary literacy move — it requires learners to synthesise descriptive information into an explanatory statement, mirroring how scientists move from data to argument. Scribing group claims on the board creates a shared 'claim wall' that will be refined in later lessons.	Observable indicator: Each pair's table has all six rows completed with at least two characteristics per class AND a cosmic-age entry that is consistent with the Lesson 2 timeline (e.g., stars: ~200 million years after Big Bang; planets: ~9 billion years after Big Bang for our solar system). Teacher checks specifically that learners are NOT conflating 'age of the universe' with 'age of Earth/solar system.' Red flag: If groups' Big Claims are purely descriptive ('There are many types of objects'), prompt: 'Your claim describes WHAT is there. Our driving question asks HOW it came to exist — what word is missing from your sentence?'

	night sky exists because…' Groups share their claims aloud; teacher scribes key phrases on the board.	us answer our driving question. Thumbs sideways if it is missing something. What is missing?' [WAIT 12 seconds.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or types into the class DQB digital board). On the FIRST sticky note, they write one piece of NEW EVIDENCE from today's lesson that helps answer the driving question: 'How did everything we see in the night sky above Nairobi come to exist…?' — starting their note with 'We now know that…' On the SECOND sticky note, they write one NEW QUESTION that today's classification activity has raised for them — a question they did NOT have before this lesson. Learners post both notes on the class DQB (physical or digital), grouping their new-question notes into emerging theme clusters: 'Questions about Stars,' 'Questions about Planets,' 'Questions about Galaxies,' 'Questions about How We Know.' The class spends 5 minutes doing a gallery walk of the new questions, placing a dot sticker next to any question they also want answered. The top three dot-voted questions are circled by the teacher and labelled 'Lessons 4–6 targets.'	Say: 'Every lesson in this unit, we update our Driving Question Board. The DQB is our collective brain — it tracks what we know and what we still need to find out. Today you have classified the objects in our phenomenon sky. That is new knowledge. And new knowledge always raises new questions.' Hand out sticky notes. Give 4 minutes of individual silent writing. Circulate and prompt learners who are stuck: 'Look at your classification table — is there a class of object whose cosmic-age surprised you? That surprise is a new question.' After posting, facilitate the gallery walk with the instruction: 'Read silently. Dot-vote for questions that YOU also want answered and that you think will help us crack our driving question.' After voting, announce the top three questions: 'These three questions will guide our next three lessons. Write them at the top of your notebook page for Lesson 4.' Explicitly connect to the storyline: 'We started with a student standing on the shores of Lake Victoria seeing a sky full of lights. Today we gave those lights their names and their cosmic birthdays. But knowing WHAT they are is not the same as knowing HOW they got there or HOW WE KNOW. That is where we go next.'	Public Question Tracking with Dot Voting: The DQB is a living sensemaking artefact — it externalises and democratises the class's epistemic state, making the gap between current knowledge and the driving question visible and motivating. Dot voting adds metacognitive value: learners must evaluate which questions are most scientifically productive, not just most interesting, developing the NGSS practice of Asking Questions and Defining Problems.	Observable indicator: Every learner has posted one 'We now know…' evidence note and one new question note. Teacher reads 'We now know…' notes to check that stated knowledge is accurate and lesson-derived (not recycled prior knowledge). Red flag: If evidence notes are vague ('We now know there are stars and planets'), the teacher pulls two or three examples and reads them aloud asking: 'Is this specific enough to help us answer the driving question? What detail could we add to make this more powerful as evidence?' New-question notes are also screened for testability — questions like 'Why does God make stars?' are acknowledged respectfully and redirected: 'That is a philosophical question — can anyone help us turn it into a scientific question we could investigate?'
Model Building Phase 📹 **VIDEO:** [Big bang]	Learners return to the cumulative model they began building in Lesson 1 (a visual representation of the night sky above Nairobi and how it came to be). In Lesson 1,	Say: 'Pull out your cumulative model from Lesson 2. Models in science are not finished — they grow as evidence grows. Today you have new evidence: a full	Cumulative Annotated Model Revision with Update Statement: Requiring learners to physically revise — not redraw — their model makes learning progression visible	Observable indicator: Every learner's revised model contains colour-coded labels for at least four of the six celestial object classes, at least two cosmic-era

introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	they drew what they could see. In Lesson 2, they added a Big Bang timeline arrow. Today, they perform a REVISION: they add labels to every object in their drawing, colour-coding by class (e.g., yellow = stars, blue = planets, white = moons, orange = comets/asteroids, purple = galaxies), and annotate each class with its approximate cosmic birth era. They then add a small legend box titled 'What each object type tells us about cosmic history.' Finally, each learner writes a one-sentence model update statement at the bottom of their diagram: 'My model has changed since Lesson 2 because I now understand that…' Pairs briefly share their update statements (1 minute each). Two or three volunteers share with the whole class.	classification system and a cosmic timeline for each class of object. Your job is to revise your model to reflect what you now know.' Give 10 minutes for individual model revision. Circulate with specific feedback prompts: 'I can see you have coloured stars yellow — great. Now show me in your annotation WHEN stars first appeared in cosmic history.' / 'Your legend says planets reflect light. Can you add a note about where the light they are reflecting COMES from?' / 'Look at your original Lesson 1 drawing — what did you draw that you cannot now classify? Is it possible it belongs to a class you didn't know about before today?' [WAIT 10 seconds between prompt and expected response.] For whole-class share: cold-call TWO or THREE learners to read their model-update statements. After each, ask: 'How does this change help us get closer to answering our driving question?' Conclude by saying: 'Your model is now more powerful than it was 80 minutes ago. Every lesson, we will revise it again. By the end of this unit, your model will be your answer to the driving question.'	and tangible. The colour-coding system provides a shared visual language across the class. The written model-update statement is a metacognitive move (NGSS Practice: Developing and Using Models) that forces learners to articulate the delta between prior and current understanding, reinforcing the idea that science is a process of successive approximation rather than sudden revelation.	annotations, and a completed legend. The model-update statement explicitly names a conceptual change ('I used to think… now I know…' structure is acceptable). Teacher checks that cosmic-era annotations are consistent with the Lesson 2 timeline and the infographic. Red flag: If learners' update statements are superficial ('I added more labels'), prompt: 'Labels are data. What does your model now EXPLAIN that it didn't explain before? What does your model still NOT explain about our driving question?' Collect model booklets at end of lesson to provide written feedback before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before preparing Lesson 4: (1) Did learners successfully distinguish between objects that generate their own light (stars, galaxies) and objects that reflect light (planets, moons, comets, asteroids) — and is this distinction now stable enough to build on in Lesson 4's exploration of stellar properties? (2) Which of the top three dot-voted DQB questions most surprised you, and what does this reveal about where your learners' genuine conceptual gaps lie? (3) Were learners able to connect classification to the cosmic timeline, or did the two tasks (classify vs. date) feel disconnected to them? If disconnected, consider opening Lesson 4 with a brief 5-minute re-visit of the infographic timeline to re-establish the link. (4) Did the Stellarium data-panel evidence logging work as intended — were learners citing actual data, or reverting to visual impressions? If the latter, consider designing a more scaffolded Stellarium worksheet for Lesson 4 that explicitly points learners to the data fields they need. (5) Were any learners' DQB questions about HOW WE KNOW rather than WHAT IS OUT THERE? These are the most epistemically sophisticated questions and should be celebrated aloud at the start of Lesson 4 — they go to the heart of the driving question's second clause: 'how do scientists gather evidence about a universe they cannot directly touch or travel to?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Using Stellarium set to Nairobi's coordinates, learners identified and logged at least eight distinct celestial objects (including the Moon, Venus, Jupiter, Sirius, and the Milky Way band) by consulting each object's Stellarium data panel for type, distance, and magnitude evidence.
What did I learn?	Learners constructed a six-class Classification Table (stars, planets, moons, comets, asteroids, galaxies) linking each class to its defining characteristics, its method of identification in Stellarium, and its approximate first-appearance era in the 13.8-billion-year cosmic timeline established in Lesson 2.
How does this explain the phenomenon?	Learners revised their cumulative night-sky models with colour-coded class labels and cosmic-era annotations, and wrote a group Big Claim statement explaining that the variety of objects visible in the Nairobi night sky exists because different classes of objects formed at different stages of the universe's evolutionary history — advancing the class's collective answer to the driving question.

LESSON 4 (40 minutes): The Solar System: Structure and Scale

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate the structure, composition, and extraordinary scale of the Solar System and explain why gravitational force keeps planets in orbit around the Sun rather than flying off into space.
Knowledge	Learners will know: (1) the names, order, and key characteristics of the eight planets; (2) that the astronomical unit (AU) is the standard measure of distance within the Solar System; (3) that the Solar System is overwhelmingly empty space and that scale models reveal this dramatically; (4) that gravitational force between the Sun and each planet is the organising principle that maintains orbital motion.
Skills	Learners will be able to: construct a scaled model of Solar System distances using toilet paper squares as AU proxies; calculate and compare planet distances in AU; interpret a data table of planetary properties; explain the role of gravitational force in orbital mechanics using evidence from the model.
Attitudes	Learners will appreciate the humbling scale of the universe and develop curiosity about humanity's place within it, recognising that even our entire Solar System is a tiny speck in the Milky Way galaxy visible above Nairobi.
Key Inquiry Question	How do the motions of celestial bodies in the Solar System affect life on Earth? (This lesson establishes the structural and gravitational foundation needed to answer this question in depth.)
Purpose in Storyline	Lessons 1 to 3 established that the universe began 13.8 billion years ago and that the night sky contains stars, planets, moons, and galaxies. Lesson 4 zooms in: the planets visible from Nairobi such as Venus and Jupiter are part of our own Solar System. Before learners can study planetary motion (Lesson 5) or gravitational effects on Lake Victoria (Lesson 6), they must first grasp what the Solar System actually is, how vast it is, and what force holds it together. This lesson provides that structural and conceptual foundation.
Safety Notes	No hazardous materials are used. If the scaled model activity is conducted in a corridor or outdoors, learners must remain in supervised pairs and not run. Remind learners to be cautious of tripping on the toilet paper roll. If conducted indoors, push desks to one side to create a clear walkway.

B. LESSON OVERVIEW

B. LESSON OVERVIEW
Learners begin by predicting Solar System structure from memory and connecting it to objects they identified in Stellarium in Lesson 3. They then conduct a toilet-paper scaled model activity in which each square represents approximately 0.5 AU, physically pacing out the distances between planets along a corridor or outside the classroom. The dramatic emptiness of space revealed by the model drives a class explanation of why planets do not fly off: gravitational force from the Sun. Learners update their cumulative model and add new evidence to the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Galaxies and](#)	Learners are shown a projected image of the night sky above the Rift Valley (the anchoring	Display the Rift Valley night sky image and say: Last lesson we identified Venus in Stellarium and	Predict-then-test anchoring: learner sketches serve as a baseline mental model that the	Teacher circulates and notes which learners include the asteroid belt, dwarf planets, or the Kuiper

gravity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	phenomenon) with Venus clearly visible near the horizon. The teacher reminds learners that in Lesson 3 they classified Venus as a planet in Stellarium. The teacher poses the prompt: On your whiteboard or paper, sketch what you think the Solar System looks like from above. Include the Sun, all planets you can name, and anything else you think belongs there. Guess the order and rough spacing. Learners work individually for 2 minutes then compare sketches with a partner. Almost all learners will draw planets in a neat row with roughly equal spacing. This prediction is deliberately preserved for later contradiction by the scale model.	placed it in our classification table as a planet. Venus is part of our Solar System, the same system Earth belongs to. Before we explore how big and structured that system really is, I want to see what picture you already have in your head. Sketch the Solar System from above. You have 2 minutes, no looking at phones. After sketches are done, select three volunteers to hold up their sketches and ask the class: What do you notice about how everyone has drawn the spacing between planets? Wait 10 seconds for responses. Listen for the assumption that planets are evenly or closely spaced. Say: Keep that sketch. We are going to test whether your mental picture matches reality.	scaled model activity will productively disrupt, creating cognitive dissonance that motivates deeper engagement with scale data.	Belt unprompted. These learners can be called upon later as expert voices. Note which learners draw equal spacing: this misconception is the primary target for the Observe phase.
Observe Phase 📹 **VIDEO:** Galaxies and gravity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher distributes one toilet paper roll per group of four learners plus a printed Solar System Distance Data Card showing each planet's distance from the Sun in AU, where 1 AU equals the Earth-to-Sun distance of approximately 150 million kilometres. The data card shows: Mercury 0.39 AU, Venus 0.72 AU, Earth 1.00 AU, Mars 1.52 AU, Asteroid Belt approximately 2.2 to 3.2 AU, Jupiter 5.20 AU, Saturn 9.58 AU, Uranus 19.22 AU, Neptune 30.05 AU. Each toilet paper square represents 0.5 AU. Learner groups calculate how many squares each planet is from the Sun (e.g. Earth is 2 squares, Jupiter is approximately 10 squares, Neptune is approximately 60 squares). One learner holds the roll at the Sun position. The group then walks along a corridor or outside, unrolling the paper and	Before going outside say: We are going to build the Solar System in our corridor. Each toilet paper square is 0.5 AU. First, calculate how many squares each planet is from the Sun using the data card. You have 3 minutes. Circulate and check calculations. For Neptune, confirm learners get approximately 60 squares. Then say: Let us walk this out. Hold the roll at the Sun. Do not tear the paper. Just walk and unroll. After the activity, gather learners at the Neptune position and say: Look back toward the Sun. All four inner planets, Mercury, Venus, Earth, and Mars, are within the first 3 squares back there. What does that tell you about the structure of the Solar System? Wait 15 seconds before taking responses. If learners are quiet, prompt: What fraction of this whole corridor is just empty space? After responses, connect	Physical embodied scale modelling: kinaesthetic experience of walking AU distances makes the abstract data emotionally and spatially real in a way that a diagram on a page cannot achieve. The corridor becomes a data display.	Teacher checks calculation accuracy on data cards during the 3-minute computation window. After returning to class, each group submits their annotated toilet paper strip (or a photograph if paper is kept) with planet labels. Teacher checks that Neptune is approximately 60 squares and that the inner four planets cluster within the first 3 squares.

	placing a sticky note label at each planet position. Learners observe and record: (1) which section of the corridor contains all four inner planets; (2) how far they must walk to reach Neptune; (3) how much of the toilet paper is just empty space. The teacher stands at the Neptune end and calls back: Can you still see Mercury and Venus from here? Learners record three observations in their notebooks using the sentence stem: I noticed that ...	to the phenomenon: Venus, which you saw shining above the Rift Valley in Lesson 3, is squeezed into that tiny inner zone right next to the Sun compared to the full scale of the system.		
Explain Phase 📹 **VIDEO:** Galaxies and gravity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Back in the classroom, the teacher poses the central explanatory question: If the Solar System is mostly empty space and the planets are so far from the Sun, what stops them from simply flying off in a straight line into that emptiness? Learners discuss in pairs for 2 minutes, drawing on prior knowledge of forces from Grade 9. Most will suggest gravity. The teacher then leads a structured class explanation using a projected diagram of the Sun and Earth with labelled force arrows. Learners are guided to articulate: (1) the Sun has enormous mass; (2) mass produces gravitational force; (3) gravitational force pulls each planet toward the Sun; (4) the planet is also moving sideways at high speed; (5) the combination of inward gravitational pull and sideways velocity produces a curved orbital path rather than a straight line or a crash into the Sun. The teacher introduces the analogy of a panga tied to a rope being swung in a circle: the rope tension acts like gravitational force, keeping the panga in a circular path. If the rope breaks,	After returning from the corridor, say: You have just walked the scale of our Solar System. Now I have a problem for you. The planets are tiny compared to all that empty space. Nothing physically connects them to the Sun. So why do they stay in orbit? Give pairs 2 minutes to discuss. Then call on three pairs, asking each: What force are you thinking of, and where does it come from? Wait 10 seconds between responses. After gravity is established, say: Exactly. Now let me show you how gravity and motion combine. Display the force diagram. Then say: Think about swinging a panga on a rope. The rope is like gravity. The panga is like a planet. What happens if you cut the rope? Wait 8 seconds. After the response, say: Write your answers to the two questions on the board. You have 4 minutes. After sharing responses, connect to the phenomenon: This same gravitational force that keeps Jupiter in its orbit is also what keeps the Moon orbiting Earth, and what causes tides on Lake Victoria and in Mombasa. We will explore that in Lesson 6.	Analogical reasoning using a locally familiar tool (the panga) to make orbital mechanics concrete. Structured written questions consolidate the explanation before model updating.	Teacher collects the two written responses. Acceptable answer for (a): Jupiter orbits because the Sun has massive gravitational force that continuously pulls Jupiter inward, and Jupiter has enough sideways velocity that it keeps falling around the Sun rather than into it or away from it. Acceptable answer for (b): Jupiter formed farther from the Sun where the solar nebula was cooler and richer in ice and gas, giving it more material to accumulate; gravitational force still acts across that greater distance but is weaker, meaning Jupiter needs less sideways velocity to maintain its orbit.

	the panga flies off in a straight line. Learners then answer two written questions: (a) Why does Jupiter orbit the Sun rather than flying away into space? (b) Why is Jupiter much farther from the Sun than Venus, yet both orbit the same Sun? Learners share responses with a partner before a whole-class debrief.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [DNA as the "transforming principle"] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Solar Power (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board posted at the front wall. They receive two sticky notes each. On the first note (green) learners write one piece of new evidence from this lesson that helps answer the driving question: How did everything in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot touch? On the second note (yellow) learners write one new question this lesson has raised for them. Suggested learner responses for green notes might include: The Solar System formed from the same cloud of gas and dust left after the Big Bang and stellar evolution, so the planets above Nairobi are direct products of cosmic history. Suggested yellow-note questions might include: What created the original force that started the Solar System spinning? Why are there exactly eight planets and not more? How do we know the distances to other planets without going there? The teacher reads three green notes and three yellow notes aloud, placing them on the board and explicitly linking the green notes to earlier DQB entries from Lessons 1 to 3.	Say: We have just added a new layer to our understanding of the night sky above Nairobi. The planets you can see from the Rift Valley are part of an organised system held together by gravity, stretching 30 AU across space. Green note: what new evidence does this give us toward answering our driving question? Yellow note: what new question has today raised for you? You have 2 minutes to write. After posting, read three of each aloud. For green notes, say: This connects to what we established in Lesson 2 about the solar nebula. For yellow notes, say: That question about measuring distances without going there is going to be answered in Lesson 7 when we study telescopes. Hold that thought. After posting, point to the board and say: Look at how our evidence wall is growing. Each lesson adds a new layer.	Progressive evidence accumulation on the DQB makes the storyline visible to learners as a building argument rather than disconnected lessons. The colour coding separates evidence from questions, training scientific literacy habits.	Teacher scans green notes to check whether learners can connect Solar System structure to the broader cosmic timeline. Misconceptions flagged here (e.g. learners who write that the Solar System has always existed) are addressed at the start of Lesson 5.
Model	Learners open their cumulative	Say: Update your cumulative	Cumulative model building creates	Teacher checks models for five

Building Phase 📹 **VIDEO:** [Galaxies and gravity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Solar Power (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	visual model begun in Lesson 2 and updated in Lesson 3. They add a new labelled panel titled Solar System Structure showing: (1) the Sun at the centre; (2) the eight planets in correct order with relative (not exact) scaled spacing, distinguishing inner rocky planets from outer gas and ice giants; (3) the asteroid belt between Mars and Jupiter; (4) labelled arrows showing gravitational force from the Sun acting on each planet; (5) a note in the bottom corner reading: Scale note: all inner planets fit within 1.52 AU. Neptune is 30 AU from the Sun. The Solar System is mostly empty space. Learners also add a connection arrow from their Lesson 3 classification table entry for planets to this new Solar System panel, annotating: These are the planets we identified in Stellarium. They orbit the Sun because of gravitational force. Learners who finish early add a small inset showing where Earth sits in the Milky Way galaxy (approximately two thirds of the way out from the centre of a spiral arm), previewing the scale introduced in Lesson 2.	model. Add a Solar System panel. You need five elements: correct planet order, inner versus outer distinction, asteroid belt, gravity arrows from the Sun, and your scale note from today. You have 5 minutes. Circulate and prompt with: Where is the asteroid belt? Are your gravity arrows pointing in the right direction, toward the Sun or away from it? For learners who rush: Is your scale note accurate? What does the toilet paper activity tell us about spacing? After 5 minutes, ask two learners to share their models briefly under the document camera. Prompt the class: What does this model help us explain about the planets visible above Nairobi on a clear night? Wait 10 seconds. Then say: Next lesson we are going to look at exactly how these planets move. Are the orbits circular or something else? That will change your model again.	a visible epistemic artefact that grows across all 12 lessons. Each update requires learners to integrate new evidence with prior knowledge, building scientific modelling skills explicitly required by the CBE framework.	required elements. Common errors to flag: gravity arrows pointing away from the Sun (misconception that the Sun pushes planets), equal spacing between planets (scale misconception not fully resolved), and missing asteroid belt. These are corrected in-person during the circulation window or noted for the Lesson 5 opening recap.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did the toilet-paper scale activity genuinely disrupt learners beliefs about equal planet spacing, or did some learners still draw equal spacing in their updated models? If so, what additional representation might help? (2) Were learners able to articulate the combination of gravitational inward pull and sideways velocity as the mechanism for orbit, or did they rely only on the word gravity without mechanistic understanding? (3) Did any learners make the connection between the gravitational force discussed today and the Moon-Earth system or tides, previewing Lesson 6? If so, honour those contributions publicly at the start of Lesson 5. (4) Were there gender equity patterns in who answered aloud and who stayed silent during the panga analogy discussion? The Kenya Space Agency profile in Lesson 10 will specifically highlight women scientists, but micro-interventions in cold-calling and pair structures should begin here. (5) Did the 40-minute window allow time for all five phases? If the corridor activity ran long, the DQB and model phases are the most compressible, but the scale insight must not be rushed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	I walked the scaled Solar System in the corridor and observed that all four inner planets fit within the first 3 squares of toilet paper while Neptune required approximately 60 squares, revealing that the Solar System is overwhelmingly empty space and that the outer planets are separated by enormous distances.
What did I learn?	I learned that the Solar System has eight planets arranged in two distinct zones separated by the asteroid belt, that distances are measured in astronomical units, and that gravitational force from the Sun is the organising principle that keeps all planets in orbital motion rather than flying off into space.
How does this explain the phenomenon?	This explains the phenomenon because the planets visible above Nairobi such as Venus and Jupiter are part of a structured, gravity-governed system that formed from the same solar nebula described in Lesson 2, connecting what we see in the night sky to the cosmic history established by the Big Bang and stellar evolution.

LESSON 5 (2 lessons (80 minutes)): Planetary Motion and Kepler's Laws	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive and apply Kepler's Three Laws of Planetary Motion using simulation-based evidence and real orbital data, and to mathematically verify Kepler's Third Law ($T^2 \propto r^3$) using data for Earth, Mars, and Jupiter.
Knowledge	Learners will know: (1) Kepler's First Law — planets orbit the Sun in ellipses with the Sun at one focus; (2) Kepler's Second Law — a planet sweeps equal areas in equal times; (3) Kepler's Third Law — the square of the orbital period is proportional to the cube of the semi-major axis ($T^2 \propto r^3$); (4) the distinction between circular and elliptical orbits; (5) how orbital data constitutes indirect evidence about the structure and history of the Solar System.
Skills	Learners will be able to: (1) manipulate the PhET 'Gravity and Orbits' simulation to observe elliptical orbits and varying orbital speeds; (2) collect, tabulate, and analyse real orbital data for Solar System planets; (3) calculate T^2/r^3 ratios and interpret their constancy as verification of Kepler's Third Law; (4) construct and revise scaled orbit diagrams replacing circular paths with ellipses; (5) use evidence from simulation and data to build scientific arguments.
Attitudes	Learners will appreciate: (1) that mathematical patterns in nature — visible even from Nairobi's night sky — reveal deep truths about how the Solar System is structured; (2) that indirect evidence (orbital timing, simulation modelling) is a powerful and legitimate scientific tool; (3) the collaborative nature of scientific knowledge-building, connecting Kepler's 17th-century observations to modern space exploration.
Key Inquiry Question	How do the motions of celestial bodies in the Solar System affect life on Earth?
Purpose in Storyline	In Lessons 1–4, learners established that the universe began with the Big Bang, that it is expanding, and that stars and galaxies form the large-scale structure of the cosmos. Now, in Lesson 5, the storyline zooms inward — from the universe and galaxies down to our own Solar System. The student gazing at Nairobi's night sky notices that planets like Venus and Jupiter appear to move against the fixed stars night after night. Why do they move the way they do? Kepler's Laws provide the mathematical 'rules of the road' for all orbiting bodies. By deriving these laws from simulation data and verifying them with real planetary data, learners gain their first piece of precise, quantitative evidence about Solar System structure — evidence that will later connect to gravitational theory (Lesson 6) and to how astronomers study distant exoplanet systems without ever visiting them.
Safety Notes	This lesson is simulation and data-analysis based. No physical hazards apply. Ensure all learners use school-approved devices and internet access points only. If using shared devices, remind learners not to alter system settings. Supervise screen time and ensure posture and eye-distance guidelines are followed during extended simulation use.

B. LESSON OVERVIEW

Learners begin by predicting the shape of planetary orbits and whether all planets take the same time to complete one orbit. They then use the PhET 'Gravity and Orbits' simulation to observe orbital shapes, compare orbital speeds at different points in an orbit, and qualitatively discover all three of Kepler's Laws. In the Explain phase, learners use a provided data table of real orbital periods (T) and semi-major axes (r) for Mercury, Venus, Earth, Mars, Jupiter, and Saturn to calculate T^2/r^3 for each planet, verifying the near-constant ratio that is Kepler's Third Law. They then revise the Solar System models they built in earlier lessons, replacing circular orbits with ellipses. The lesson concludes with learners updating their Driving Question Board, connecting planetary motion patterns to the anchoring phenomenon: the night sky above Nairobi looks static, but planets are in constant, mathematically precise motion — motion that has been occurring since the Solar System formed from the same nebula that produced the Sun approximately 4.6 billion years ago.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [3D divergence theorem intuition] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Kepler's Laws of Planetary Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a time-lapse photograph of the night sky taken from the shores of Lake Victoria, in which star trails appear as arcs and Venus traces a slow path relative to the stars over several weeks. Each learner independently writes answers to three prompt questions in their science journals: (1) 'What shape do you think the path of a planet like Venus makes as it orbits the Sun — a perfect circle, an oval, or something else? Sketch it.' (2) 'Do you think all planets take the same amount of time to complete one orbit? Why or why not?' (3) 'Does a planet move at the same speed throughout its entire orbit, or does its speed change? Justify your guess.' Learners then share predictions with a shoulder partner (Think-Pair-Share). Selected predictions are recorded on a class Predict Wall (sticky notes or whiteboard section) to be revisited after the observation phase. This activates prior knowledge from Lessons 1–4 about Solar System structure and creates cognitive dissonance that the simulation will resolve.	Display the Lake Victoria time-lapse image on the projector. Say: 'Before we open any simulation or data table, I want you to commit to a prediction. Use your journal — three questions, three sketches or written answers. You have 4 minutes. This is individual thinking time.' [Allow 4 minutes of silent individual writing.] After 4 minutes: 'Now turn to your shoulder partner. Share your sketches. Do you agree? Where do you differ? 2 minutes.' [Allow 2 minutes.] Cold-call 3 pairs — choose one who sketched a circle, one who sketched an ellipse, and one unsure. Ask: 'Tell us what you drew and why.' [Wait 10 seconds after each response before commenting.] Record key prediction phrases on the board verbatim, e.g., 'Wanjiku says: all orbits are perfect circles because the planets look evenly spaced.' Say: 'We are going to keep these predictions here. By the end of today, you will either confirm or revise every single one of them with evidence.'	Think-Pair-Share with a Predict Wall. This strategy externalises prior conceptions, making them visible and revisable. Recording predictions verbatim gives learners ownership and creates a concrete record of conceptual change — a key feature of NGSS Science and Engineering Practice 6 (Constructing Explanations).	Teacher circulates during individual writing time and scans journals. Observable indicators of readiness: (1) learner has sketched an orbit shape (any shape — correctness not assessed here); (2) learner has attempted a justification for their speed prediction. Flag learners with blank journals for targeted support. Note the proportion of the class predicting circular vs elliptical orbits — this informs the depth of scaffolding needed in Phase 2.
Observe Phase 📹 **VIDEO:** 3D divergence theorem intuition Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners work in pairs at devices running the PhET 'Gravity and Orbits' simulation (gravity-and-orbits, HTML5 version). They follow a structured investigation guide with three tasks: TASK A — Orbit Shape: Set the simulation to 'Sun and Planet' mode. Drag the planet to different distances from the Sun and press Play. Sketch the resulting orbit shape. Increase	Before opening devices: 'You have a three-task investigation guide. Read Task A first — do not skip ahead. I will come around to check your orbit sketches before you move to Task B.' [Circulate actively. After ~8 minutes, pause the class.] Say: 'Pause — everyone look at your sketch from Task A. Hands up if you drew an ellipse, not a circle.' [Count hands.]	Structured Inquiry with a Simulation (NGSS Science and Engineering Practice 3 — Planning and Carrying Out Investigations). The three-task structure scaffolds observation from qualitative (orbit shape) to semi-quantitative (speed variation) to pattern-recognition (period vs distance). Using the PhET simulation provides controlled, repeatable evidence	Teacher checks orbit sketches after Task A — observable indicator: sketch shows a clearly non-circular, elongated ellipse with the Sun offset from centre. If majority show circular orbits, pause whole class and demonstrate using the simulation with an exaggerated eccentricity. After Task B, listen for the phrase 'faster near the Sun' in pair

Kepler's Laws of Planetary Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the initial velocity — what changes? Decrease it. Record observations. TASK B — Speed Variation (Kepler's Second Law): Enable the 'Path' and 'Velocity' vector options. Observe how the length of the velocity arrow changes as the planet moves around its orbit. Record: 'Where is the planet moving fastest — nearest to or farthest from the Sun?' Sketch two positions and their velocity vectors. TASK C — Period vs Distance (Kepler's Third Law — Qualitative): Switch to 'Solar System' mode. Observe Mercury, Earth, and Jupiter orbiting simultaneously. Use the on-screen timer to estimate the relative orbital periods. Record: 'Which planet completes an orbit fastest? Which is slowest? What pattern do you notice between distance from the Sun and orbital period?' Each pair records all observations in a structured data table in their journals. After the simulation, the teacher distributes a printed/projected data table with real values of T (years) and r (AU) for six Solar System planets.	'Compare that to what you predicted on our Predict Wall. Interesting, yes? Keep going — Task B now.' [After Task B, ~6 more minutes, pause again.] Ask the class (cold-call 3 students): 'Amina — where was your planet moving fastest?' [Wait 12 seconds.] 'Otieno — why do you think it speeds up near the Sun?' [Wait 10 seconds.] 'Chebet — what force do you think is responsible?' [Wait 12 seconds.] After Task C: 'Before I give you the real data table, tell your partner the pattern you observed. Use the sentence starter: The farther a planet is from the Sun, the _____ its orbital period because ___.' [Allow 2 minutes.] Distribute the real data table and say: 'Now you have real numbers. In Phase 3 we will use maths to test whether this pattern is exact.'	that mirrors how real scientists use computational models when direct observation is impossible — directly connecting to the anchoring phenomenon's theme of indirect evidence.	discussions — this is the observable indicator of Kepler's Second Law understanding. After Task C, cold-call responses should include a statement linking greater distance to longer period. Record names of students who articulate causal reasoning (mentioning gravity or force) for stretch questioning in Phase 3.
Explain Phase 📹 **VIDEO:** 3D divergence theorem intuition Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Kepler's Laws of Planetary Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar	Learners are formally introduced to Kepler's Three Laws by name, and immediately connect each law to what they just observed in the simulation. They then conduct a quantitative verification of Kepler's Third Law using the real planetary data table. Working in pairs, learners: (1) complete a calculation table — for each of six planets, calculate T^2 (years²), r^3 (AU³), and T^2/r^3; (2) observe that $T^2/r^3 \approx 1.00$ for all planets (in units of years and AU); (3) plot T^2 (y-axis) vs r^3 (x-axis) on graph paper or a provided grid, observe the	Begin by saying: 'You have now gathered evidence. Let me give these patterns the names scientists use.' Write the three laws on the board with diagrams. Say: 'Notice — you discovered these patterns before I told you the names. That is exactly how Kepler worked.' Introduce the calculation task: 'Calculate T^2, r^3, and T^2/r^3 for each planet in your table. Use your calculator. You have 8 minutes. Work in pairs but each person writes their own table.' [Circulate, checking arithmetic. Prompt struggling pairs: 'What is 1.88	Claim-Evidence-Reasoning (CER) Framework combined with graphical analysis. CER makes the logical structure of scientific argument explicit and visible (NGSS Science and Engineering Practice 6 — Constructing Explanations and Designing Solutions). The graph of T^2 vs r^3 provides a visual, pattern-based verification that complements the numerical T^2/r^3 table, reinforcing that Kepler's Third Law is a proportional relationship — not just a numerical coincidence. Connecting to the Kenya Space	Collect or photograph each pair's calculation table. Observable indicators of mastery: (1) T^2/r^3 values for all six planets fall between 0.95 and 1.05 (arithmetic errors caught and corrected); (2) graph shows a straight line passing through or near the origin; (3) CER paragraph contains a specific data citation (e.g., 'Jupiter's T^2/r^3 = 0.999') and a reasoning sentence that explains why a constant ratio supports the law. Flag learners whose CER paragraphs contain only a claim and evidence but no reasoning —

readings	straight line through the origin, and interpret this as evidence of a proportional relationship; (4) write a claim-evidence-reasoning (CER) paragraph: 'Claim: Kepler's Third Law holds for all planets in our Solar System. Evidence: [their T^2/r^3 table]. Reasoning: [explanation of what a constant ratio and straight-line graph indicate].' Learners then revise their Solar System diagrams from earlier lessons — replacing any circular orbits with ellipses and annotating them with each planet's orbital period. The teacher formally names and defines all three laws, connecting them to Johannes Kepler's work in the early 1600s and noting that Kenyan space scientists at the Kenya Space Agency (KSA) use these same laws today when calculating satellite orbits around Earth.	squared? Try again.'] After 8 minutes, ask the class: 'Raise your hand if your T^2/r^3 values are all close to 1.0.' [Count.] Cold-call one pair: 'Kamau — read me your value for Jupiter.' [Wait 10 seconds.] 'Does everyone agree? What does it mean that all six planets give us nearly the same number?' [Wait 12 seconds before taking responses. Accept 2–3 answers.] Say: 'This constant ratio is not a coincidence — it is a law of nature. Now write your CER paragraph. You have 6 minutes.' [After paragraphs are written, cold-call 2 students to read aloud.] Say: 'Now open your Solar System diagram from Lesson 3. Every circular orbit you drew — revise it. Planets travel in ellipses. Annotate each planet with its orbital period.'	Agency grounds abstract orbital mechanics in a contemporary Kenyan institutional context.	target these for the DQB phase discussion.
Driving Question Board (DQB) Creation 📹 **VIDEO:** 3D divergence theorem intuition Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Kepler's Laws of Planetary Motion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. The board shows the anchoring phenomenon description and the two key inquiry questions. Each pair receives two sticky notes: one green ('Evidence we now have') and one orange ('New questions this raises'). Learners write: GREEN: One specific piece of evidence from today's lesson that helps answer either key inquiry question. They must reference today's lesson explicitly (e.g., 'Kepler's Third Law — T^2/r^3 is constant — shows that all planets follow the same gravitational rule, so the Solar System is a structured, predictable system, not a random arrangement.'). ORANGE: One new question that today's lesson raised (e.g., 'What force actually	Say: 'Take two sticky notes — green and orange. You have 3 minutes to write. Green: one piece of evidence from today that helps us answer our driving question. Orange: one new question today raised for you.' [Allow 3 minutes.] 'Come post your notes on the DQB.' [Allow 2 minutes for posting.] Read 3–4 green notes aloud. Ask: 'How does this evidence connect to what we saw in our very first lesson — the Big Bang?' [Wait 12 seconds. Cold-call 2 students.] Then read 2–3 orange notes. Say: 'These orange questions — look at them. Several of you are asking what force causes these orbits. What do you think we will investigate in Lesson 6?' [Wait 10 seconds.] 'You are thinking like scientists. Our DQB is growing. The more evidence we	Driving Question Board with colour-coded sticky notes (NGSS Science and Engineering Practice 1 — Asking Questions and Defining Problems). The DQB makes the cumulative, iterative nature of scientific inquiry visible. By physically adding to a shared, growing board, learners experience the DQB as a living document — not a static display. The distinction between green (evidence) and orange (new questions) reinforces that good evidence in science simultaneously answers questions and generates new ones.	Read all green sticky notes before the next lesson. Observable indicator of quality: the note cites a specific piece of evidence (data, observation, or law name) rather than a vague statement ('we learned about planets'). Observable indicator of concern: a note that restates the driving question rather than answering it partially. Count the proportion of orange notes asking causal questions (about gravity/force) vs procedural questions — a majority of causal questions indicates the class is ready for Lesson 6 (Newton's Law of Universal Gravitation) without additional bridging.

	causes planets to follow Kepler's Laws?' or 'Do moons around Jupiter also follow $T^2 \propto r^3$?'). Pairs post their sticky notes on the DQB. The teacher facilitates a 5-minute whole-class review, reading selected notes aloud, grouping related questions, and explicitly connecting today's evidence to the broader storyline: the night sky above Nairobi that looks static and eternal is in fact a highly ordered, mathematically governed system — one that has been running by these laws since the Solar System formed 4.6 billion years ago.	gather, the more we understand — and the more precisely we can ask new questions.'		
Model Building Phase 📹 **VIDEO:** [3D divergence theorem intuition] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Kepler's Laws of Planetary Motion (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative Solar System model — a diagram they have been building and revising across Lessons 3, 4, and 5. Using today's evidence, they make three specific revisions: REVISION 1 — Orbit Shape: Redraw each planetary orbit as an ellipse (even slightly elliptical for near-circular orbits like Venus and Earth; more elongated for Mercury and Mars). Annotate: 'Kepler's First Law — elliptical orbit, Sun at one focus.' REVISION 2 — Speed Annotation: Add two annotations per orbit showing where the planet moves fastest (perihelion — closest to Sun) and slowest (aphelion — farthest from Sun). Use arrow length to indicate relative speed. Annotate: 'Kepler's Second Law — equal areas in equal times.' REVISION 3 — Period Labels: Add each planet's orbital period (in Earth years) next to its orbit. Write the ratio $T^2/r^3 \approx 1$ as a model caption. Learners then write a one-paragraph 'Model Explanation' beneath their diagram: 'My model shows… The evidence that supports this is…	Say: 'Open your Solar System model from your portfolio. You built this in Lesson 3. Look at the orbits you drew then. Based on what you discovered today — what needs to change?' [Wait 10 seconds. Take 2 hands.] 'Exactly. Today we have three revisions to make. I will write them on the board — Revision 1, 2, and 3.' [Write the three revision prompts clearly.] 'You have 10 minutes to revise your model and write your Model Explanation paragraph. Make sure your final sentence tells us what your model still cannot explain.' [Circulate. Prompt learners who are drawing very circular ellipses: 'Mercury's orbit has a real eccentricity of 0.21 — make it noticeably oval.'] After 10 minutes, cold-call 2 students to read their 'One thing my model still cannot explain is…' sentence. Say: 'Store these carefully in your portfolio. We will open them again in Lesson 6 and add the force that drives all of this.'	Iterative Model Revision with annotated diagrams and a written Model Explanation (NGSS Science and Engineering Practice 2 — Developing and Using Models). The cumulative, portfolio-based model makes conceptual growth visible over time. The 'one thing my model still cannot explain' prompt is a deliberate epistemic move — it trains learners to treat scientific models as provisional and incomplete, and it creates a forward-looking bridge to Lesson 6. Connecting the revised model to the anchoring phenomenon reinforces that the ordered, elliptical orbits visible in the night sky above Nairobi are not accidental but mathematically lawful.	Circulate and scan revised models before portfolios are closed. Observable indicators of mastery: (1) at least one orbit is visibly non-circular (ellipse); (2) perihelion/aphelion speed arrows are present and correctly labelled (faster at perihelion); (3) orbital period values are annotated and $T^2/r^3 \approx 1$ is noted as a model caption. Observable indicator of concern: orbit shapes remain circles with no revision — flag these learners for a targeted 2-minute check-in at the start of Lesson 6. Read the 'Model Explanation' paragraphs for quality of the 'still cannot explain' sentence — it should reference force, gravity, or 'what keeps planets in orbit' to confirm readiness for Lesson 6.

	One thing my model still cannot explain is…' The final sentence is key — it should reference the question of what force governs these orbits, setting up Lesson 6. Models are stored in learners' science portfolios for revision in subsequent lessons.			

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 6: (1) What proportion of learners accurately drew elliptical orbits in their model revision? If fewer than 70% revised their orbit shapes, consider opening Lesson 6 with a 5-minute simulation revisit using exaggerated eccentricity. (2) Did the CER paragraphs contain genuine reasoning, or did most learners stop at evidence? If reasoning was weak, introduce the CER scaffold more explicitly at the start of Lesson 6's explain phase. (3) Were the orange DQB sticky notes asking causal questions about force and gravity? If so, the class is intellectually primed for Newton's Law of Universal Gravitation. If most orange notes asked factual or definitional questions, plan a brief bridging discussion at the start of Lesson 6 to re-establish the causal thread. (4) Which pairs struggled most with the T^2/r^3 calculation — was the difficulty arithmetic, unit conversion, or conceptual? Prepare a worked example for the first 3 minutes of Lesson 6 if arithmetic errors were widespread. (5) Did the Kenya Space Agency reference land with learners? Consider inviting a follow-up discussion about UHURU — Kenya's first satellite, launched in 1970 — as a local application of orbital mechanics that connects Kepler's Laws to Kenyan scientific heritage.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using the PhET 'Gravity and Orbits' simulation, learners observed that planetary orbits are ellipses (not perfect circles), that planets move faster when closer to the Sun and slower when farther away, and that planets farther from the Sun take longer to complete one orbit. They also observed real orbital data for six Solar System planets (Mercury through Saturn).
What did I learn?	Learners learned Kepler's Three Laws of Planetary Motion: (1) First Law — planets orbit the Sun in ellipses with the Sun at one focus; (2) Second Law — a planet sweeps out equal areas in equal times, meaning it speeds up near the Sun; (3) Third Law — $T^2 \propto r^3$, meaning the square of a planet's orbital period is proportional to the cube of its semi-major axis. They mathematically verified the Third Law by calculating $T^2/r^3 \approx 1$ for Earth, Mars, Jupiter, and three other planets.
How does this explain the phenomenon?	Using Kepler's Three Laws, learners can explain why Venus and Jupiter appear to move at different rates across the night sky above Nairobi — they are at different distances from the Sun and therefore have different orbital periods governed by $T^2 \propto r^3$. The laws explain that the Solar System is a mathematically ordered system, not a random arrangement, and that its structure has been stable and predictable since its formation 4.6 billion years ago. The constant T^2/r^3 ratio is evidence that a single underlying force — yet to be fully explained — governs all planetary orbits.

LESSON 6 (40 minutes): Gravitational Forces in Space	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will deepen their understanding of gravity as the universal force that governs all orbital motion and investigate how the Moon's gravitational pull produces measurable tidal effects on Lake Victoria and the Kenyan coast at Mombasa.
Knowledge	Learners will be able to: (1) state Newton's Law of Universal Gravitation and explain that gravitational force increases with mass and decreases with the square of distance; (2) explain how gravity keeps the Moon in orbit around Earth and Earth in orbit around the Sun; (3) describe how the Moon's gravity causes tidal bulges and explain why tides occur twice daily; (4) connect tidal forces on Lake Victoria and Mombasa to the same gravitational law that governs planetary orbits.
Skills	Learners will: analyse gravitational force data for Earth-Moon and Earth-Sun systems; calculate relative gravitational forces using the inverse-square relationship; interpret a tidal data table for Mombasa port; construct and annotate a diagram showing Earth, the Moon, and tidal bulges; and update their cumulative universe model to include gravitational force arrows.
Attitudes	Learners will appreciate that a single universal law — gravity — links the Moon visible above Nairobi to the orbital mechanics of the entire Solar System, and will develop curiosity about how invisible forces shape everyday Kenyan environments including Lake Victoria fishing and Mombasa shipping.
Key Inquiry Question	How do the motions of celestial bodies in the Solar System affect life on Earth?
Purpose in Storyline	Lessons 4 and 5 established that planets orbit the Sun in ellipses and obey Kepler's Laws. Lesson 6 now explains WHY — revealing gravity as the underlying force. By connecting the Moon's pull to Lake Victoria tides and Mombasa coastal patterns, this lesson grounds abstract gravitational theory in observable Kenyan evidence, directly advancing the driving question by showing that the night sky does not just look beautiful but actively shapes life on Earth. The lesson also sets up Lesson 9 (Space Phenomena and Effects on Earth) by introducing the idea that space forces reach all the way down to Earth's surface.
Safety Notes	No physical hazards. If students are asked to move desks for a role-play simulation, remind them to move carefully. Ensure no student is singled out if they have never seen the ocean; frame Mombasa and Lake Victoria as shared Kenyan cultural knowledge.

B. LESSON OVERVIEW

In 40 minutes learners move through a Predict-Observe-Explain cycle centred on the question: Why does Lake Victoria have detectable tides and why does Mombasa harbour show a predictable tidal pattern? They begin by predicting what force keeps the Moon orbiting Earth rather than flying away. They then observe two data sources — a printed tidal height table for Mombasa port (Kenya Meteorological Department data) and a diagram of Lake Victoria water-level micro-tides — and connect these observations to Newton's Law of Universal Gravitation. In the Explain phase they apply the inverse-square law qualitatively and quantitatively (ratio problems only, no calculus), annotate a tidal-bulge diagram, and articulate how the same force that holds the Solar System together reaches down to affect Kenyan fishing communities on Lake Victoria. The lesson closes with model updates and a Driving Question Board revision adding new gravitational evidence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Line integrals and vector fields](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravitational Force and Inclined Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects or draws a simple image: the Moon orbiting Earth above a stylised outline of Kenya — Nairobi at the centre, Lake Victoria to the west, Mombasa to the east. The teacher poses two questions and asks learners to write individual predictions in their notebooks before any discussion: (1) What stops the Moon from flying off into space in a straight line? (2) Does the Moon have any effect on water bodies here in Kenya — like Lake Victoria or the Indian Ocean at Mombasa? Learners write for 2 minutes, then share predictions with a shoulder partner for 1 minute. Selected pairs share with the class. The teacher records key prediction words on the board without judging them: common expected responses include gravity, magnetic pull, the Earth holds it, tides. The teacher then asks the class to raise hands if they have ever seen or heard of tides at the Mombasa coast or noticed fishing communities on Lake Victoria discussing water levels. This activates prior cultural knowledge and sets up the observe phase.	Opening prompt (display on board or say aloud): 'Last lesson we proved that planets follow Kepler's Laws — they orbit in ellipses and T-squared equals r-cubed. But Kepler himself admitted he did not know WHY. Today we find out why, and we will discover that the answer affects Lake Victoria and the port of Mombasa every single day.' WAIT TIME: After posing prediction question 1, wait 60 seconds in silence before asking anyone to speak. After posing prediction question 2, wait 45 seconds. Teacher move — cold call two students who have not yet spoken this week: 'Akinyi, what did you write for question two?' Follow up: 'Interesting — what evidence from everyday life made you think that?' If a student says gravity, probe deeper: 'Good — but what do you mean by gravity? Is it the same gravity that makes a mango fall from a tree?' Record the word GRAVITY on the board with a question mark next to it: 'We will decide together by the end of the lesson whether gravity is truly the answer and whether it is the same force everywhere.'	Think-Pair-Share followed by class prediction wall. Teacher writes the two prediction questions on a corner of the board that remains visible for the entire lesson so learners can return to check their predictions after observing evidence. This prediction anchoring technique makes the cognitive shift in the Explain phase more visible to learners.	Teacher circulates during the 2-minute individual writing time and reads 4 to 5 notebooks to check for prior knowledge and misconceptions. Key misconception to watch for: learners believing the Moon is held in orbit because gravity pulls it toward Earth but that centrifugal force pushes it away equally — this is partly correct but needs precise language later. Note which students mention tides versus those who say the Moon has no effect on Earth water — this gap informs Explain phase targeting.
Observe Phase 📹 **VIDEO:** Line integrals and vector fields Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravitational Force and Inclined Plane	Learners receive a one-page data sheet (teacher-prepared, printed or hand-drawn on the board if no printing is available) containing two evidence sources. Source A is a simplified tidal height table for Mombasa Old Port showing four readings per day across a 3-day period, with times and heights in metres (for example: Day 1 — 00:12 h 3.1 m, 06:24 h 0.4 m, 12:48 h 3.0 m, 18:36 h 0.5 m). Source B is a labelled diagram showing the Earth-Moon system	Distribute data sheet and say: 'You have 6 minutes. Do not look for answers yet — only record what you notice in the data and the diagram. Scientists call this disciplined observation.' WAIT TIME: Do not intervene in groups for the first 3 minutes. After 3 minutes, circulate and ask probing questions only — no answers: 'How many hours separate the two high tides on Day 1? What does that remind you of?' and 'Look at the diagram — if the Moon is	Data analysis with a structured observation table. Learners use the sentence stem: 'I notice that... which suggests that...' This separates observation from inference and builds scientific reasoning habits. The diagram with Africa silhouette makes the phenomenon geographically personal — Lake Victoria and Mombasa are not generic ocean locations but places Kenyan students recognise.	Teacher listens to group discussions during the 6 minutes and targets one key diagnostic question to each group: 'Why are there TWO tidal bulges and not just one?' Correct reasoning should mention that the Moon pulls the near-side water toward itself AND that Earth itself is pulled slightly toward the Moon, leaving the far-side water behind. Groups that cannot explain the second bulge will need extra scaffolding in the Explain phase. Teacher makes

[(Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	with an exaggerated tidal bulge on the side of Earth facing the Moon and a second smaller bulge on the opposite side, with Lake Victoria marked on the Africa silhouette and Mombasa marked on the east coast. Learners work in groups of three to complete three observation tasks: (1) Identify the pattern in the Mombasa tidal data — how many high tides per day and roughly how many hours between them; (2) On the diagram, identify which part of Earth is experiencing high tide and which is experiencing low tide at any given moment; (3) Discuss and write one sentence answering: what must be causing the water to bulge on two opposite sides of Earth simultaneously? Groups record observations in their notebooks. After 6 minutes of group work, one spokesperson per group shares observation 3 with the class.	pulling water toward itself on the near side, what is causing the bulge on the FAR side?' After group sharing, ask: 'Did anyone notice that Mombasa gets two high tides roughly 12 hours apart? What does that tell us about how often Earth rotates relative to the Moon?' Teacher then transitions: 'You have now observed the effect. In the Explain phase we will use Newton's Law of Universal Gravitation to explain the cause. But first — has your prediction changed? Look back at what you wrote.' Give learners 30 seconds to re-read their predictions silently.		a mental note of which groups need this targeted support.
Explain Phase 📹 **VIDEO:** Line integrals and vector fields Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravitational Force and Inclined Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces Newton's Law of Universal Gravitation using precise but accessible language: every object with mass attracts every other object with mass, and the force is proportional to the product of their masses and inversely proportional to the square of the distance between them. The teacher writes the relationship on the board: F is proportional to (M times m) divided by d-squared. Learners copy this and label each variable. The teacher then leads the class through three connected explanations. Explanation 1 (Moon orbit): The Moon is pulled toward Earth by gravity — this is the centripetal force keeping it in a curved path. Without gravity it would travel in a straight line into	Write the proportionality on the board slowly and say: 'This equation was published by Newton in 1687. It explains the Moon above Nairobi, the tides at Mombasa, and the orbits we calculated with Kepler last lesson — all with one law. That is the power of a universal law.' WAIT TIME: After writing the law, pause for 45 seconds and ask: 'Before I say anything else, what does the word universal mean in this context? Write one word or phrase.' Cold call three students. For Problem A, give groups 2 minutes then ask: 'Who got a factor of 4? Show me your working.' Praise correct inverse-square reasoning explicitly: 'Exactly — doubling the distance does not halve the force, it	Worked ratio problems with scaffolded sentence frames connect abstract mathematics to observable Kenyan phenomena. The use of the Kisumu fisherman narrative grounds gravitational theory in a culturally familiar Kenyan livelihood, making the physics feel relevant rather than foreign. The final explanation sentence template ensures every learner produces a complete causal explanation linking force, mechanism, and observable effect.	Collect the two ratio problem solutions by asking learners to hold up notebooks — teacher does a quick visual scan for correct answers (factor of 4 for Problem A). For learners with wrong answers, teacher pairs them with a peer who has the correct working. The final explanation sentence is the primary formative artefact: teacher circulates and reads 6 to 8 sentences, checking that learners correctly identify the Moon as the cause, the inverse-square law as the mechanism, and twice-daily tides as the observable effect. Common error to address: learners who write that the Sun has no tidal effect — teacher redirects using the Problem B result showing the Sun produces approximately 46 percent of the

	space as Newton's First Law requires. Learners annotate a Moon-Earth diagram with force arrows. Explanation 2 (Inverse-square ratio activity): Learners solve two ratio problems in their notebooks. Problem A — if the Moon were moved to twice its current distance from Earth, by what factor would the gravitational force decrease? (Answer: 4 times weaker — inverse-square law.) Problem B — the Sun is approximately 390 times farther from Earth than the Moon, yet the Sun has 27 million times more mass than the Moon. Which produces a larger tidal force on Earth? (Using the proportionality, the Sun actually produces a tidal force approximately 46 percent of the Moon's — teacher provides scaffolded working.) Explanation 3 (Kenya connection): The teacher explicitly connects the calculation back to Mombasa and Lake Victoria: 'The tidal data you read from Mombasa port is direct evidence of Newton's Law operating in real time above Kenya. Fishermen on Lake Victoria notice micro-tides of a few centimetres — too small to be dangerous but detectable with sensitive instruments — caused by the same Moon you see in the Nairobi night sky.' Learners write a final explanation sentence: 'Tidal forces at [Mombasa / Lake Victoria] occur because [Newton's Law of Universal Gravitation / the Moon's gravity] causes [water bulges on two sides of Earth], which means [as Earth rotates, any coastal point passes through high and low tide approximately twice per day].'	quarters it. That is the inverse-square relationship.' For the Kenya connection, say: 'I want you to imagine you are a fisherman leaving Kisumu harbour at 4 am. You need to know if the water level will allow your boat to clear the shallow channel. The Moon's gravity — the same force keeping the Solar System organised — is what you are depending on.' WAIT TIME: After this statement, allow 30 seconds of quiet reflection before asking: 'How does knowing the physics behind tides change how you think about the night sky above Nairobi?'		Moon's tidal force, which is why solar and lunar tides combine during full and new moons (spring tides).

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Line integrals and vector fields] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravitational Force and Inclined Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board — the shared display that has accumulated questions since Lesson 1. The teacher reads aloud the driving question: 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' Each learner writes on a sticky note (or a small slip of paper) one new question that this lesson has raised for them and one question from the board that they now feel they can partially answer using gravitational theory. Learners post new questions and mark answered questions with a tick or star. The teacher then leads a 2-minute whole-class review of the board, noting which questions have migrated from unanswered to answered and which new ones connect to upcoming lessons (for example, questions about solar flares affecting Earth connect to Lesson 9, questions about telescopes detecting gravity waves connect to Lesson 7).	Say: 'The Driving Question Board is our collective brain. Last lesson you added questions about WHY planets orbit the way they do. Today we answered some of those — gravity is the WHY. Now — what new questions does gravity raise for you? What do you still not understand?' WAIT TIME: Give learners 90 seconds of silent individual thinking before they write on their sticky notes. After posting, the teacher reads 3 new questions aloud and says: 'These are exactly the questions real physicists ask. Question 3 — can gravity from a distant galaxy affect Earth? — is something researchers at the Square Kilometre Array in South Africa are actually investigating. We will return to that in Lesson 8.' This connects the DQB to the storyline and to African science infrastructure.	The DQB update serves a metacognitive function — learners see their own knowledge growing across lessons. By physically moving questions from unanswered to answered, they experience intellectual progress, which sustains motivation across a 12-lesson sequence. The teacher's reference to the Square Kilometre Array localises frontier science to Africa.	The sticky notes are a formative artefact. Teacher photographs the board or quickly reads new questions to identify persistent misconceptions (for example, learners who think gravity only exists between Earth and Moon, not universally) and lingering gaps (for example, confusion about the difference between gravitational force and the tidal force differential). These inform Lesson 7 and Lesson 9 planning.
Model Building Phase 📹 **VIDEO:** [Line integrals and vector fields] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravitational Force and Inclined Plane (Flexbook)]	Learners open their cumulative universe model — the visual diagram they have been building since Lesson 2. Today they add a gravitational layer to the Solar System section of their model. Specifically, they: (1) draw curved force arrows between the Sun and each planet, labelling the arrows with F proportional to 1 over d-squared to indicate the inverse-square relationship; (2) draw the Earth-Moon system as a sub-diagram showing the Moon in orbit with a force arrow pointing toward Earth and a second arrow showing	Say: 'Your cumulative model is growing. In Lesson 4 you added structure and scale. In Lesson 5 you changed circular orbits to ellipses. Today you add the invisible threads — gravitational force arrows — that explain why any of those orbits exist at all.' Circulate during the 5 minutes of model building and ask targeted questions: 'Why is your force arrow between the Sun and Jupiter drawn longer or shorter than the one between the Sun and Earth? Should it be?' (Expected answer: the Sun-Jupiter force arrow should	Cumulative model building makes learning visible across the unit. Each lesson deposits a new layer onto the model, and by Lesson 12 the full diagram will tell the complete story from the Big Bang to tonight's sky above Nairobi. The peer feedback protocol (one clear thing, one more precise thing) teaches scientific communication without creating anxiety. The closing phenomenon callback — connecting the Moon to Mombasa tides in real time — ensures the anchoring phenomenon is advanced meaningfully in this	Teacher examines 4 to 5 models during circulation and checks three specific features: (1) force arrows are present between Sun and planets AND between Earth and Moon; (2) the inverse-square label is included; (3) the Kenya inset panel links the Moon to a named Kenyan water body. Models missing any of these three features receive a verbal prompt from the teacher to add them before the lesson ends. The partner feedback also surfaces misunderstandings that the teacher notes for future correction.

Source: CK-12 🔍 Search ARES for similar readings	Earth's motion would be a straight line without gravity; (3) add a small inset panel showing the African continent with Lake Victoria and Mombasa labelled, connected by a dotted line to the Moon with the label 'tidal force reaches Earth surface'; (4) write one update sentence at the bottom of their model page: 'Gravity is the universal force that organises the Solar System seen above Nairobi and produces measurable effects on Kenyan water bodies.' Learners share their updated model diagrams with a partner for 1 minute of peer feedback using the prompt: 'One thing that is clear in your model and one thing that could be more precise.'	be shorter because Jupiter is farther away and the force is weaker at that distance per the inverse-square law.) After peer feedback, close the lesson by returning to the anchoring phenomenon: 'Tonight, if you look up at the Moon above Nairobi, you are seeing the same object that is right now pulling the water at Mombasa port into a high tide. The universe is not just above us — it is acting on us, here, in Kenya, at this moment.'	lesson.	

D. TEACHER REFLECTION

After the lesson, reflect on the following questions. (1) Did learners successfully connect the abstract inverse-square law to the Mombasa tidal data, or did the ratio problems expose significant gaps in proportional reasoning that need revisiting? (2) Were learners able to explain the second tidal bulge — the one on the far side of Earth from the Moon — or did most only understand the near-side bulge? This distinction is important for Lesson 9 when spring and neap tides are revisited in the context of space phenomena affecting Earth. (3) Did the Kisumu fisherman narrative make the physics feel relevant to Kenyan learners, or did some students indicate that Lake Victoria tides are too small to matter? If the latter, prepare a brief extension showing scientific instruments used to measure micro-tides on large lakes, referencing the Kenya Meteorological Department lake monitoring programme. (4) How many new Driving Question Board entries connected to upcoming lessons, and which questions will require the most careful addressing in Lessons 7 through 9? (5) Were girls equally represented in cold-call responses and group spokesperson roles? If not, adjust questioning strategies in Lesson 7 to ensure gender equity in participation as required by the PCI framework.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Tidal height data from Mombasa Old Port showing two high tides and two low tides per day, and a diagram of Earth-Moon tidal bulges with Lake Victoria and Mombasa marked on the African continent.
What did I learn?	Newton's Law of Universal Gravitation states that gravitational force is proportional to the product of two masses and inversely proportional to the square of the distance between them. This universal force keeps the Moon orbiting Earth, Earth orbiting the Sun, and causes the Moon to pull Earth's water into tidal bulges that produce measurable twice-daily tides at Mombasa and detectable micro-tides on Lake Victoria.
How does this explain the phenomenon?	The night sky above Nairobi is not static or separate from life on Earth — the same gravitational force that organises the Solar System reaches down to pull water at the Kenyan coast and on Lake Victoria, meaning that the universe visible above Nairobi is

	actively shaping everyday Kenyan environments right now.

LESSON 7 (40 minutes): Telescopes: Windows to the Universe

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain how telescopes enable scientists to gather evidence about the universe without physically travelling to distant objects, directly addressing the driving question of how we know what we know about the night sky above Nairobi.
Knowledge	Learners will be able to: (1) distinguish between refracting telescopes (using lenses), reflecting telescopes (using mirrors), and radio telescopes (detecting radio waves); (2) explain why telescopes are essential when direct travel to distant celestial objects is impossible; (3) describe how the Hubble and James Webb Space Telescopes have provided evidence about the early universe, stellar formation, and galaxy structure; (4) connect telescope observations to the Big Bang theory and cosmic evolution.
Skills	Learners will practise: interpreting telescope images as evidence; comparing the capabilities of different telescope types; evaluating the quality of evidence gathered remotely; annotating diagrams of optical paths in refracting and reflecting telescopes; and updating their cumulative model with new observational evidence.
Attitudes	Learners will appreciate: that human curiosity and technological innovation can extend our senses far beyond direct experience; that indirect evidence gathered through instruments can be just as reliable as direct observation; and that African scientists and institutions participate in cutting-edge telescope science through projects such as the Square Kilometre Array.
Key Inquiry Question	How did the universe begin and how has it evolved over time? — addressed by showing that telescopes provide the observational evidence (redshift, galaxy images, CMB mapping) that supports the Big Bang model and evolutionary timeline built in Lessons 1 and 2.
Purpose in Storyline	Lesson 7 answers a critical methodological question raised implicitly since Lesson 1: if no human has visited distant stars or witnessed the Big Bang, how do scientists actually gather their evidence? By mastering how telescopes work and what they reveal, learners gain the epistemic foundation needed to trust the evidence underpinning every prior lesson. This lesson also transitions the unit toward Lesson 8 (space exploration history) by showing that telescope technology is itself a form of exploration.
Safety Notes	Strictly warn learners never to point any optical instrument — including binoculars, refracting telescopes, or even naked eyes — directly at the Sun. Solar observation requires certified solar filters. During simulation and video work, ensure screen brightness is comfortable and learners who experience photosensitivity are seated appropriately.

B. LESSON OVERVIEW

Learners begin by predicting what limits our ability to learn about distant objects in space, anchoring the problem in the night sky above Nairobi that has driven the whole unit. They then observe: (a) a brief teacher-led ray-diagram demonstration of how a convex lens refracts parallel light to a focal point; (b) a video profile comparing the Hubble Space Telescope and James Webb Space Telescope missions; and (c) still images from the Virtual Telescope Project and JWST showing galaxies, nebulae, and exoplanet atmospheres. In the Explain phase, learners construct a three-column comparison table (refracting, reflecting, radio) and write a structured evidence paragraph. The Driving Question Board is updated with new answers and new questions about what telescopes still cannot reveal. Learners close by adding a telescope icon and annotation to their cumulative universe model, showing how humans observe from Earth what they cannot touch.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Video: Bending Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Refraction: Light Entering Water (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher projects a single photograph of the night sky taken from the Rift Valley near Nakuru — the same anchoring phenomenon image used in Lesson 1 — alongside this prompt written on the board: 'You want to study the star Proxima Centauri, which is 4.24 light-years from Nairobi. You cannot travel there. List three problems this creates for a scientist and suggest one way to overcome each problem.' Learners write individual responses in their science notebooks for 3 minutes, then share with a shoulder partner for 1 minute. Pairs are cold-called to share one problem-solution pair each. The teacher records all suggested solutions on the board under the heading 'How might we gather evidence from a distance?' without evaluating them yet. Learners are likely to suggest: using cameras, sending probes, using binoculars, or simply looking carefully. The teacher accepts all responses and highlights that even the most powerful rocket would take approximately 18,000 years to reach Proxima Centauri, making a telescope humanity's only practical window.	Teacher opens with: 'Cast your minds back to Lesson 1 — you stood under the night sky near Nakuru and asked how we could possibly know anything about a universe 13.8 billion years old. Six lessons later, you have learned about the Big Bang, stellar evolution, planetary motion, and gravity. But I want to ask you a deeper question today: how did scientists actually gather the evidence for all of those claims?' After posing the board prompt, teacher sets a 3-minute individual think time — no talking — then says: 'Now turn to your partner and compare your lists.' After 1 minute of pair talk, teacher cold-calls three pairs using name cards. Teacher records responses in two columns: Problems | Proposed Solutions. Teacher then reveals: 'Proxima Centauri is 4.24 light-years away. At the speed of the fastest spacecraft ever built — NASA Parker Solar Probe at about 700,000 km/h — the journey would take roughly 6,300 years. We are not going there. So our evidence must come from light and other radiation that travels to us. Today we learn the instrument that captures that evidence.' WAIT TIME instruction: after posing the travel-time fact, wait 10 seconds in silence to let the scale sink in before continuing.	Predict-then-share activates prior knowledge from all six previous lessons and frames today's learning as answering a fundamental epistemological question: how do we know what we know? The Nakuru sky image re-anchors learners in the driving phenomenon and creates cognitive dissonance — we have made many claims, but today we interrogate the basis of those claims.	Teacher scans notebooks during pair-share to check whether learners are distinguishing between observation tools (telescopes, cameras) and travel-based exploration. Misconception to watch: learners may believe all space knowledge comes from astronauts visiting places. Teacher notes any learner who already mentions electromagnetic radiation or light, as this can be drawn on in the Observe phase.
Observe Phase 📹 **VIDEO:** [Video: Bending Light] Source: PhET Interactive	The observation phase has three sequential components, each building on the previous. Component 1 (8 minutes): Teacher draws a ray diagram on the board showing a convex lens refracting parallel rays of starlight to a focal	During Component 1 ray diagram, teacher narrates while drawing: 'Watch where the parallel rays — representing light coming from a star millions of kilometres away — are bent by the lens. They converge here at the focal point.	The three-part observe sequence moves from mechanism (how telescopes work) to evidence (what they have revealed) to access (how learners themselves can participate). This scaffolds from abstract optics to concrete,	Teacher circulates during diagram copying and checks that learners correctly label focal point — a common error is placing the label on the lens itself. During video note-taking, teacher scans tables to check that learners are writing

Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	point, then a second lens (eyepiece) magnifying the image. Learners copy the annotated diagram into their notebooks, labelling: objective lens, focal point, focal length, eyepiece lens, and eye. Teacher then draws a second diagram showing a concave mirror collecting and reflecting light to a focal point (reflecting telescope), labelling: primary mirror, secondary mirror, and eyepiece. Learners annotate: 'Refracting telescopes bend light using lenses. Reflecting telescopes bounce light using mirrors.' Teacher briefly shows a photograph of the Lovell Radio Telescope in the UK and one of the MeerKAT radio telescope in South Africa — part of the Square Kilometre Array project — explaining that radio telescopes collect invisible radio waves emitted by galaxies and hydrogen clouds, using large dish antennas rather than lenses or mirrors. Component 2 (7 minutes): Teacher plays a 5-minute curated video compilation: first 2 minutes from NASA showing the Hubble Space Telescope launch, its flawed mirror, and the correction mission, with key images of the Hubble Deep Field (thousands of galaxies in a patch of sky the size of a grain of sand held at arms length); final 3 minutes showing JWST images: the Pillars of Creation in infrared, the Carina Nebula, and the first confirmed detection of carbon dioxide in an exoplanet atmosphere. Learners record in a two-column table: 'What the telescope showed | Why this matters as evidence.' Component 3 (3 minutes): Teacher displays three still images from the Virtual	This is where the image forms. The eyepiece then acts like a magnifying glass to make that image visible to your eye.' After drawing, teacher asks: 'What do you notice is different about the reflecting telescope design?' WAIT TIME: 8 seconds. Teacher expects: mirrors replace lenses. Teacher confirms: 'Correct — and this matters because very large lenses are heavy and sag under their own weight. Mirrors can be much larger and still hold their shape precisely. That is why all the world biggest telescopes today are reflectors.' Before playing the video in Component 2, teacher instructs: 'You have a two-column table ready. As you watch, record two things: what specific object or phenomenon the telescope revealed, and why that observation matters as evidence for our driving question. Pause your pencil for emotional moments — you are allowed to just watch — but resume recording immediately after.' After the video, teacher asks: 'Which single image surprised you most? Turn and tell your partner — 30 seconds.' Teacher then cold-calls two learners to share. During Component 3, teacher emphasises: 'The Virtual Telescope Project means that a student sitting in Mathare or Kisumu can, tonight, watch a professional astronomer in Italy track Jupiter in real time. The universe is no longer accessible only to wealthy nations. Africa is part of this story through the SKA in South Africa.'	emotionally resonant images, and closes by localising the technology in an African context. The two-column evidence table trains learners to distinguish observation from inference — a core scientific literacy skill.	specific observations (e.g., 'Hubble Deep Field showed thousands of galaxies in a tiny patch of sky') rather than vague statements (e.g., 'it showed space'). Teacher provides immediate written feedback on two or three notebooks by writing a single guiding question such as: 'What does this image tell us about how old the universe is?'

	Telescope Project livestream archive (accessible free at virtualtelescope.eu) — a real-time tracked image of Jupiter, a deep-sky galaxy (M101), and a comet — explaining that professional-grade telescopes can be operated remotely from Nairobi via the internet, and that Kenyan students can access live sky data without owning any equipment.			
Explain Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work individually for 5 minutes to complete a three-column comparison table in their notebooks with headings: Telescope Type | How It Gathers Evidence | What It Can Detect. They fill in rows for: Refracting Telescope, Reflecting Telescope, and Radio Telescope. They may use their ray diagrams, video notes, and any class notes from the Observe phase. After completing the table, learners write a structured evidence paragraph (minimum 5 sentences) responding to this sentence starter provided by the teacher: 'Scientists studying the night sky above Nairobi cannot travel to distant stars, but they can gather evidence by...' The paragraph must include: (1) at least two telescope types; (2) one specific observation made by the Hubble or James Webb Space Telescope; (3) a connection to either the Big Bang theory from Lesson 1 or the evolutionary timeline from Lesson 2; and (4) a mention of Africa or Kenya in the telescope story. Learners who finish early are challenged with an extension question on the board: 'Why do astronomers launch telescopes into space rather than just using ground-based telescopes? List at	Teacher writes the sentence starter on the board and reads it aloud: 'Scientists studying the night sky above Nairobi cannot travel to distant stars, but they can gather evidence by...' Teacher says: 'This is not a summary of what we did today. This is your scientific explanation — you are answering part of our driving question. Use specific evidence. Use correct vocabulary. Make the connection to lessons we have already done.' Teacher circulates and offers targeted prompts: to a learner who is vague, teacher says: 'You wrote that Hubble showed galaxies. Which specific observation, and what does it tell us about the age or structure of the universe?' To a learner who has not mentioned Africa, teacher asks: 'Where in Africa is a major telescope facility being built right now, and what type of telescope is it?' Teacher does not give answers but redirects learners to their notes. After 5 minutes, teacher selects two paragraphs (one strong, one developing) to read aloud anonymously with permission, then facilitates a brief class discussion: 'What did the first paragraph do well? What could be added to the second?' WAIT TIME: after asking the class to critique	The comparison table forces systematic categorisation — a key scientific practice — while the paragraph requires learners to synthesise across the table, the video, and prior lessons. Requiring an Africa or Kenya connection in the paragraph ensures the local context is not superficial but integrated into the scientific explanation. The anonymous paragraph critique builds a classroom norm of evidence-based peer feedback without personal risk.	Teacher uses the paragraph writing as a real-time formative check. Key indicators of understanding: (1) learner distinguishes telescope types correctly; (2) learner names a specific JWST or Hubble observation; (3) learner connects telescope evidence to Big Bang theory or the evolutionary timeline; (4) learner mentions SKA or MeerKAT in the Africa connection. Teacher notes any learner who conflates refraction and reflection, or who believes radio telescopes detect sound waves — both common misconceptions to address in the whole-class critique.

	least two reasons.'	the developing paragraph, wait 12 seconds before accepting responses, as evaluative thinking requires processing time.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. The teacher reads aloud the original driving question: 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' Teacher highlights the second part of the question: 'how do scientists gather evidence about a universe they cannot directly touch or travel to?' and asks: 'Has today changed your answer to this part? Write one sentence on a sticky note beginning with the words: Scientists gather evidence by...' Learners post sticky notes on the DQB under a new section labelled 'Evidence-Gathering Methods.' Teacher then asks: 'What new questions did today raise that we have not yet answered?' Examples that typically emerge: Can telescopes detect life on other planets? Why is the universe dark if there are so many stars (Olbers Paradox)? How far can a telescope see? Could Kenya ever build its own space telescope? Learners write one new question each on a different coloured sticky note and add it to the DQB. Teacher flags two or three of these new questions as ones that will be partially addressed in Lessons 8 and 9, building anticipation.	Teacher stands at the DQB and gestures to the entire board: 'Look at this board. In Lesson 1, almost all of this was blank and full of questions. Today we are filling in the second half of our driving question — the how-do-we-know half. Every piece of evidence we have discussed across seven lessons — from the CMB to gravitational lensing to tidal patterns on Lake Victoria — came to scientists through instruments, not through personal visits to space.' Teacher reads two or three sticky notes aloud from earlier lessons to show the progression of understanding. After learners post new questions, teacher draws a circle around the question 'Could Kenya build its own space telescope?' and says: 'Flag this one. In Lesson 8, we will meet the Kenya Space Agency and ask exactly this kind of question.' WAIT TIME: after reading the new questions aloud, pause for 8 seconds and say: 'Is there a question here that connects to something you personally have wondered about the night sky at home — whether in Nairobi, Eldoret, or your upcountry home?' Allow 2 voluntary responses before moving on.	The DQB update serves two functions: it provides a formative metacognitive check (learners articulate what they now know) and it maintains narrative momentum (new questions build anticipation for Lessons 8 and 9). Splitting sticky note colours between answers and new questions visually shows the class that learning generates further inquiry — a key epistemic value of science.	Teacher reads all posted 'Scientists gather evidence by...' sticky notes during the transition to the Model Building phase and mentally categorises them into: full understanding (mentions specific methods and instruments), partial understanding (mentions telescopes generically), and incomplete (vague statements). Teacher uses this to decide whether to open Lesson 8 with a brief revisit of this concept.
Model Building Phase	In the final 5 minutes, learners open their cumulative universe model — the visual representation	Teacher gives clear, timed instructions: 'You have 5 minutes. Your model must answer the	Adding telescopes as observation instruments to the cumulative model makes explicit that all prior	Teacher checks that the dotted timeline arrows are pointing to the correct eras of cosmic history — a

📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	they have been building and updating since Lesson 2. They add a new section to their model representing Earth as an observation platform. Inside this section, learners draw and label: (1) a refracting telescope on Earth's surface pointing at the night sky; (2) the Hubble Space Telescope in low Earth orbit with an arrow labelled 'observes visible and ultraviolet light'; (3) the James Webb Space Telescope at L2 orbit with an arrow labelled 'observes infrared light from the early universe'; and (4) a radio telescope dish (labelled MeerKAT or SKA, South Africa) with an arrow labelled 'detects radio waves from galaxies and hydrogen clouds.' Learners draw a dotted line from each telescope back along the timeline in their model — the JWST arrow reaching back to the 'first stars and galaxies' era, the Hubble arrow reaching to 'galaxy formation,' and the radio telescope arrow reaching to 'hydrogen epoch after recombination.' This visually represents that telescopes do not just observe the present sky but look back in time. Learners write a two-sentence model annotation at the bottom of the new section: 'This shows that... because...'	visual question: how does humanity gather evidence about a universe it cannot physically explore? Add the four telescope elements I have written on the board, with their labels, and draw the dotted timelines back into your cosmic history model.' Teacher writes the four elements on the board so learners can copy labels accurately. Teacher circulates and asks probing questions to individual learners: 'If the JWST can observe infrared light from 13.5 billion years ago, what does that tell us about light and distance?' and 'Where on your Big Bang timeline should this telescope arrow point back to?' Teacher ends the phase by selecting one learner to hold up their model and narrate the new telescope section to the class in 30 seconds, modelling the kind of explanation expected in the final Lesson 12 presentation. Teacher closes the lesson by returning to the anchoring phenomenon: 'Tonight, if you stand on a rooftop in Nairobi and look up, you are doing what every astronomer does — gathering light that has travelled across the universe to reach your eye. The difference between you and Hubble is precision and aperture. The curiosity is exactly the same.'	evidence — galaxy images, CMB maps, redshift data — was gathered through these instruments. The dotted timeline arrows make the concept of lookback time concrete and visual. The closing quote ties the technological lesson back to the human, Kenya-grounded experience of the anchoring phenomenon, reinforcing that science is a human activity accessible to every learner in the room.	frequent error is drawing all arrows back to the Big Bang rather than to the appropriate epoch. Teacher also checks the two-sentence annotation for the structure 'This shows that... because...' to ensure learners are not just labelling but explaining. Any model that lacks the MeerKAT or SKA element is flagged for the learner to complete as homework, ensuring the African telescope connection is not omitted.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners demonstrate understanding that telescopes gather different types of electromagnetic radiation — not just visible light — and that this expands the range of phenomena scientists can study? (2) Were learners able to connect specific telescope observations (Hubble Deep Field, JWST Pillars of Creation) to the Big Bang timeline and evolutionary model from Lessons 1 and 2, or did they treat telescope images as isolated facts? (3) Did the Africa and Kenya telescope connection (MeerKAT, SKA, Virtual Telescope Project access) feel integrated and substantive, or did it feel like an add-on? If the latter, plan to open Lesson 8 by asking learners to find the KSA and SKA on a map before beginning the space exploration timeline. (4) Review the DQB new questions: do any signal misconceptions that need addressing before Lesson 8? Common unresolved misconceptions to watch: learners may believe radio telescopes detect radio broadcasts

from alien civilisations rather than natural astrophysical emissions; learners may believe that looking through a telescope in real time means seeing what is happening now in distant stars, not realising they are seeing light that left the source thousands or millions of years ago.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the night sky above Nairobi, thousands of stars, the Milky Way band, the Moon, and visible planets appear static and permanent. Yet every piece of scientific evidence we have discussed across seven lessons — the Big Bang, stellar evolution, galaxy formation, planetary orbits, gravitational effects — was gathered by scientists using telescopes and other remote-sensing instruments, not by travelling to these objects. Telescopes reveal that the apparently still night sky is in fact a window into billions of years of cosmic history, with each point of light representing a different moment in time depending on how far away it is.
What did I learn?	Telescopes gather electromagnetic radiation — visible light, infrared, ultraviolet, or radio waves — that travels from distant objects to Earth, allowing scientists to study the universe without physically travelling to it. Refracting telescopes use convex lenses to bend and focus light; reflecting telescopes use concave mirrors to collect and focus light and can be built much larger. Radio telescopes use large dish antennas to detect radio waves emitted by galaxies, hydrogen clouds, and other astrophysical sources. Because light takes time to travel, telescopes looking at distant objects are also looking back in time — the JWST can observe light from galaxies that formed just 300 million years after the Big Bang. Africa participates in world-leading telescope science through the MeerKAT and SKA project in South Africa.
How does this explain the phenomenon?	The night sky above Nairobi looks complete to the naked eye, but telescopes reveal it is only a sliver of what is actually there — limited by the small aperture of the human pupil and by our eyes' narrow sensitivity to visible light alone. Telescopes overcome both limits: larger apertures gather far more light from faint distant objects, and instruments tuned to infrared, ultraviolet, or radio wavelengths reveal sources our eyes cannot detect at all. Because light takes time to travel, every telescope is also a time machine — the JWST sees galaxies as they were just 300 million years after the Big Bang, evidence we could never reach by travelling. This is how humanity has built a detailed scientific account of a universe we cannot physically visit, with Africa contributing through MeerKAT and the SKA project in South Africa.

LESSON 8 (80 minutes): History of Space Exploration: From Sputnik to the James Webb Telescope — and Kenya's Place in the Cosmos

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To trace the historical timeline of human space exploration and evaluate how each milestone mission has contributed evidence toward answering how we know what we know about the universe — connecting Kenya and Africa's emerging role in global space science.
Knowledge	Learners will be able to: (a) identify and sequence key milestones in the history of space exploration from Sputnik (1957) to the James Webb Space Telescope (JWST, 2021); (b) describe what scientific evidence each landmark mission contributed to our understanding of the universe; (c) explain the role of the Kenya Space Agency (KSA) and Africa's contribution to the Square Kilometre Array (SKA) in advancing space science.
Skills	Learners will be able to: (a) construct and annotate a timeline of space exploration milestones; (b) analyse mission data summaries to extract evidence relevant to the driving question; (c) evaluate the significance of indirect evidence-gathering methods used in space science (telescopes, rovers, radio arrays).
Attitudes	Learners will: (a) appreciate that scientific knowledge about the universe is built incrementally through the work of many nations and generations; (b) value Kenya's and Africa's growing participation in global space science; (c) demonstrate curiosity and open-mindedness about humanity's capacity to explore what cannot be directly touched.
Key Inquiry Question	How do the missions and instruments humans have sent into space provide evidence about a universe we cannot directly travel to — and what does Kenya's involvement in space science tell us about who gets to ask these questions?
Purpose in Storyline	In Lessons 1–7, learners built understanding of the Big Bang, cosmic expansion, the structure of the Solar System, stellar life cycles, and gravity. They now ask: how did we actually gather all that evidence? Lesson 8 shifts the lens from WHAT we know to HOW we came to know it — tracing the human story of space exploration and showing that the night sky above Nairobi is not merely something to observe passively, but something Kenyan and African scientists are now actively investigating. This lesson positions learners to evaluate evidence quality and source in Lessons 9–12.
Safety Notes	No physical hazards. If learners use devices to access Stellarium or online infographics, ensure screen-time guidelines are followed. Remind learners to respect copyright when reproducing mission images in their timeline artefacts.

B. LESSON OVERVIEW

Learners begin by revisiting the Driving Question Board and asking: 'We have claimed to know so much about the universe — but how did we actually find it out?' The lesson is anchored in a short 'mystery object' provocation — a photograph of a piece of hardware — before learners are revealed it is part of Sputnik 1. From there, learners engage a structured timeline-building activity using mission data cards (Sputnik, Vostok 1, Apollo 11, Voyager 1 & 2, Hubble Space Telescope, Mars rovers, JWST) in which they must identify, for each mission, WHAT was discovered and HOW that discovery advanced the driving question. A focused reading and discussion on the Kenya Space Agency (KSA) and the Square Kilometre Array (SKA) in South Africa grounds the global story locally. Learners revise their cumulative explanatory models to incorporate the idea that human instruments — not just natural observation — are the primary tools of modern space science. The lesson uses the NGSS practice of Obtaining, Evaluating, and Communicating Information as its central sensemaking engine.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Intro to theoretical probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a close-up photograph of a metallic sphere covered in antenna rods — no labels, no context. They are asked to write individually (2 minutes, silent): 'What is this object? Where is it? What is it doing?' They then share predictions with a partner (Think-Pair-Share). After pairs share out, the teacher reveals: this is Sputnik 1, launched by the Soviet Union on 4 October 1957 — the first human-made object ever to orbit Earth. Learners are then asked: 'Before Sputnik, how did humans gather evidence about space? After Sputnik, what changed?' They record a prediction in their science notebooks under the heading: 'Before I learn the history — I predict that space exploration changed our knowledge of the universe by…'	Display the Sputnik image on screen without any label. Say: 'Look carefully at this object. In your notebook, write: what is it, where do you think it is, and what do you think it is doing? You have 2 minutes — silent writing.' [WAIT 2 minutes — circulate, do not hint.] 'Now turn to your partner. Share your prediction. You have 90 seconds.' [WAIT 90 seconds.] Cold-call 3 pairs — choose one confident pair, one uncertain pair, one who wrote something unexpected. Say exactly: 'Thank you. Hold your prediction — we are about to find out.' Reveal the image label: 'This is Sputnik 1. Launched from Kazakhstan in 1957. It was the size of a basketball. And it changed everything.' Pause 10 seconds for effect. Then say: 'In your notebook, respond to this question: Before Sputnik, how did humans learn about space? Write for 90 seconds.' Cold-call 2 learners. Then frame the lesson: 'Today we ask not just WHAT scientists discovered — but HOW they discovered it, and what that means for us here in Nairobi.'	Think-Pair-Share with a Mystery Object Provocation. This strategy activates prior knowledge about observation and evidence-gathering before instruction begins, surfaces misconceptions (e.g., 'scientists always knew this'), and creates cognitive dissonance — the leap from ground-based stargazing to orbital spacecraft is jarring and memorable. Connecting the mystery to the anchoring phenomenon reminds learners that the night sky above Nairobi is the same sky Sputnik first orbited.	Teacher circulates during silent writing and reads 4–5 notebooks. Observable indicator: learners' predictions reveal whether they distinguish between ground-based astronomy (telescopes, naked eye) and space-based exploration. Flag any learner who writes 'scientists used computers' or 'they guessed' — these learners need scaffolding in the Observe phase. Cold-call responses reveal baseline understanding of the pre-space-age era.
Observe Phase 📹 **VIDEO:** Intro to theoretical probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners work in groups of 4. Each group receives a set of 7 Mission Data Cards (printed or displayed digitally), one per landmark mission: (1) Sputnik 1 — 1957, first satellite, proved orbit was possible; (2) Vostok 1 — 1961, Yuri Gagarin, first human in space; (3) Apollo 11 — 1969, first Moon landing, returned lunar rock samples; (4) Voyager 1 & 2 —	Distribute Mission Data Cards (or display on screen for digital classrooms). Say: 'Each card tells the story of one mission. Your job as a group is to be detectives — not just reading what was found, but asking: HOW did this mission gather evidence, and does this evidence help us answer our driving question about the night sky above Nairobi?' [Allow 18–20	Structured Data Card Analysis with an Evidence-Evaluation Table. This is an instantiation of the NGSS Science and Engineering Practice of Obtaining, Evaluating, and Communicating Information. By requiring learners to explicitly connect each mission to the driving question, the strategy prevents passive information consumption and builds the habit	Observable indicator 1: Groups' observation tables show specific, mission-linked entries (e.g., 'Hubble's deep field images showed galaxies 13 billion light-years away, which is evidence the universe has been expanding for a very long time') rather than vague summaries ('Hubble saw far away things'). Observable indicator 2: During cold-call justifications,

📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	1977, first detailed images of outer planets, now in interstellar space; (5) Hubble Space Telescope — 1990–present, deep field images, evidence for accelerating expansion; (6) Mars rovers (Curiosity & Perseverance) — 2012–present, evidence for ancient water on Mars; (7) James Webb Space Telescope — 2021–present, infrared imaging of the earliest galaxies. Each card contains: mission name, date, key instrument used, one key discovery, and one image or data visualisation. Groups complete a structured observation table with three columns: MISSION | WHAT WAS DISCOVERED | HOW THIS HELPS ANSWER OUR DRIVING QUESTION. Groups also read a 1-page briefing on the Kenya Space Agency (KSA) — founded 2017, headquartered in Nairobi, first Kenyan satellite (1KUNS-PF) launched from the ISS in 2018 — and on Africa's role in the Square Kilometre Array (SKA) based in South Africa, the world's largest radio telescope array. Learners add KSA and SKA as two additional rows in their table.	minutes for group work.] Circulate to each group. At each table, ask one probing question — rotate through: 'What instrument did Hubble use, and why does that matter?'; 'Why do you think it took until 1957 for humans to send anything into space?'; 'What does the word infrared mean on the JWST card, and why would that help us see ancient galaxies?'; 'How is the SKA different from a traditional telescope — what does a radio telescope detect?' [WAIT 10 seconds after each probe before accepting an answer.] At the 15-minute mark, say: 'In 3 minutes, each group must agree on ONE mission they believe gave the most important evidence for our driving question. Be ready to justify your choice.' Cold-call 3 groups to share their choice and justification. Write the group choices on the board without comment — revisit in Phase 3.	of treating exploration history as an evidence trail rather than a list of facts. The inclusion of KSA and SKA ensures that learners see this evidence-gathering enterprise as something Kenya and Africa are actively part of — not merely spectators to a Western narrative.	learners reference the driving question explicitly. Red flag: if a group's 'most important mission' choice has no justification connected to evidence about the universe's origin or the methods of space science — pull that group aside in the Explain phase for targeted questioning.
Explain Phase 📹 **VIDEO:** [Intro to theoretical probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12	The class comes together for a whole-group sensemaking discussion. The teacher facilitates a 'Mission Evidence Gallery Walk' in which each group's completed table is displayed (on desks or walls). Learners do a 5-minute silent gallery walk, placing a sticky dot next to the mission entry they personally find most compelling as evidence for the driving question. After the gallery walk, the class reconvenes. The teacher facilitates a structured discussion around three anchor questions: (1) 'What	Facilitate the gallery walk by saying: 'Walk silently. Read each group's table. Place a sticky dot next to the single mission entry you find most convincing as evidence. You have 5 minutes.' [WAIT — do not guide dot placement.] After gallery walk, display the dot-tallied tables and say: 'Look at where the dots clustered. Why do you think that is?' [WAIT 10 seconds.] Cold-call 3 learners who placed dots on different missions — ask each: 'Tell me specifically what evidence	Dot Democracy Gallery Walk followed by Socratic Whole-Class Discussion. The dot-voting externalises individual reasoning and creates a class-wide data set that can itself be analysed ('why did most dots go to Hubble or JWST?'). The Socratic questioning then pushes learners from description to explanation — they must articulate WHY evidence is compelling, not just WHAT it shows. The 3-sentence Evidence Summary is a writing-to-learn strategy that consolidates	Observable indicator: 3-sentence Evidence Summaries are collected (or photographed if digital). Proficient response explicitly names a specific instrument or data type (e.g., 'infrared imaging,' 'lunar rock samples,' 'radio wave detection'), links it to the driving question (origin/expansion of universe, or methods of evidence gathering), and articulates a non-trivial reason for Kenya's importance (e.g., 'the SKA's location in the Southern Hemisphere gives it a view of the

🔍 Search ARES for similar readings	pattern do you notice in HOW each mission gathered evidence — what tools or instruments keep appearing?'; (2) 'The JWST can see galaxies that existed only 300 million years after the Big Bang — what does this tell us about light and distance?'; (3) 'Kenya's 1KUNS-PF satellite was launched in 2018 — what questions could a Kenyan satellite help answer about the universe above Nairobi?' Learners then write a 3-sentence 'Evidence Summary' in their notebooks: (a) The most powerful piece of evidence gathered by human space exploration is…because…; (b) This evidence helps answer our driving question by…; (c) Kenya's/Africa's role in space science matters because…	that mission produced, and how it connects to what we see in the Nairobi night sky.' After responses, pose anchor question 2: 'The JWST is looking at light that left a galaxy 13.5 billion years ago. What does that tell us about the relationship between distance and time in the universe?' [WAIT 15 seconds — this is a hard question.] Cold-call 2 learners, then rephrase if needed: 'If light takes time to travel, and we can see light from 13.5 billion years ago — what does that mean about what the universe looked like then versus now?' Synthesise by saying: 'Every mission we studied gathered evidence indirectly — no human has touched a star, visited another galaxy, or seen the Big Bang directly. But our instruments have. That is the central story of space science.' Prompt the 3-sentence Evidence Summary: 'Write three sentences. Sentence one: your most compelling piece of evidence and why. Sentence two: how it answers our driving question. Sentence three: why Kenya's role matters. You have 4 minutes.' [WAIT 4 minutes.] Cold-call 2 learners to read their summaries aloud.	understanding before model revision, and the Kenya sentence ensures local identity is woven into the scientific narrative, not appended as an afterthought.	Milky Way's core that Northern Hemisphere telescopes cannot access'). Developing response: lists facts without causal connection. Emerging response: describes missions without connecting to evidence or the driving question — these learners receive targeted verbal feedback during DQB phase.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to theoretical probability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners return to the class Driving Question Board. They first review sticky notes added in Lessons 1–7 — questions about the Big Bang, expansion, stellar life cycles, the Solar System, and gravity. Each learner then does the following: (1) Places a GREEN dot next to any DQB question they now feel Lesson 8's evidence (space exploration history) has helped answer or partially answer; (2) Adds at least ONE new sticky	Direct learners to the DQB. Say: 'Look at the questions we have been collecting since Lesson 1. Today, we focused on the second half of our driving question — HOW do scientists gather evidence. Which of these questions did today's lesson help answer?' [WAIT 15 seconds.] 'Place a green dot on those questions. Then write at least one NEW question — something today sparked that you still don't fully	Driving Question Board Curation with Confidence Metacognition. The DQB is a running public record of the class's growing understanding — curating it with green dots makes visible which questions have been resolved and which remain open. The confidence meter is a metacognitive sensemaking tool: learners must not only rate their understanding but justify it, which surfaces the difference between	Observable indicator: New DQB sticky notes move beyond recall ('what is JWST?') toward methods and epistemology ('how does infrared light help us see older galaxies?', 'what can radio telescopes detect that optical ones cannot?'). Confidence meter justifications reveal whether learners can connect specific missions to specific evidence types. Red flag: a learner who gives themselves a 5 but cannot

📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	note with a question sparked by today's lesson — ideally about methods, instruments, or Kenya/Africa's role (e.g., 'Can the SKA detect signals from the Big Bang?', 'Why didn't African countries have space agencies earlier?', 'What will the JWST find that we haven't predicted yet?'); (3) Reviews the overarching driving question — 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' — and updates a personal 'confidence meter' (a scale of 1–5 drawn in their notebook) for how well they can now answer the second half of that question: 'how do scientists gather evidence about a universe they cannot directly touch or travel to?'	understand.' [Allow 4 minutes.] After the sticky note activity, say: 'In your notebook, draw a simple 1-to-5 scale. One means: I cannot yet answer how scientists gather evidence about the universe. Five means: I can explain this fully and confidently. Mark where you are right now. This is not a grade — it is a thinking tool.' [WAIT 2 minutes.] Cold-call 3 learners to share their confidence rating AND the reason for it — not just the number. Say: 'Your number means nothing without your reasoning. Tell me WHY you gave yourself that rating.' Briefly note which DQB questions received the most green dots — these become the focus of teacher-led consolidation in future lessons.	feeling like you understand and being able to explain. This is particularly important in Lesson 8, where the content is rich and narrative-heavy — learners may feel they 'learned a lot' without being able to articulate specific evidence-method connections.	name a specific instrument or mission — probe with 'Tell me one specific way scientists gathered evidence they could not gather from the ground in Nairobi.'
Model Building Phase 📹 **VIDEO:** [Intro to theoretical probability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative explanatory model — the annotated diagram they have been building since Lesson 1, showing the universe from the Big Bang to the present-day night sky above Nairobi. In previous lessons, learners added: the Big Bang event, cosmic expansion arrows, formation of stars and galaxies, the Solar System, stellar life cycles, and gravitational interactions. In this lesson, learners add a new layer to their model: a 'HUMAN EVIDENCE LAYER' — a timeline bar drawn along the bottom or side of their model, showing at least 5 space exploration milestones and annotating each with: (a) the instrument/mission used; (b) the type of evidence gathered; (c) an arrow connecting that evidence to	Say: 'Get out your cumulative model. Today we are adding the human story — the evidence layer. Without human missions and instruments, everything we have drawn in this model would be conjecture. The missions are the PROOF.' Display a sample model on the board or projector showing where the timeline bar could go. Say: 'Draw a horizontal timeline bar along the bottom of your model. Mark at least 5 missions. For each one, draw an arrow to the part of your universe model that mission helped us understand. Label the arrow with the type of evidence — was it light? Radio waves? Rock samples? Photographs?' [Allow 12 minutes for model revision.] Circulate and give specific feedback: 'I see you have Hubble pointing to distant	Cumulative Model Revision with an Evidence-Source Layer. Adding a 'Human Evidence Layer' to the model is a powerful sensemaking move because it makes explicit the epistemological chain: from cosmic event → to physical signal → to instrument → to human interpretation. Learners who draw these arrows are not just recalling history — they are building a causal-evidential framework that connects the phenomenon (the Nairobi night sky) to the methods that produced our knowledge of it. The Kenya/Africa inset personalises the model and prevents the history of space exploration from feeling like a story that happens elsewhere, to other people.	Observable indicator: Models contain correctly directed arrows linking named missions to specific parts of the universe model (e.g., Voyager → outer planets; JWST → earliest galaxies; Apollo → Moon/Solar System age). Captions for the Kenya/Africa inset make a non-trivial connection to the driving question (e.g., 'The SKA can detect radio waves from the early universe, helping us answer how the universe began'). Developing response: missions are listed but not connected by arrows to the model's content areas. Emerging response: timeline bar is present but annotations are mission names only, with no evidence type or connection. Collect (or photograph) 5–6 models as an exit artefact — these inform Lesson 9 planning.

	the part of the universe model it helped reveal (e.g., an arrow from 'JWST — infrared imaging' pointing to 'earliest galaxies' in the model). Learners also add a small inset box labelled 'KENYA & AFRICA IN SPACE SCIENCE' containing: the KSA logo area, the year 1KUNS-PF was launched, and the SKA location (South Africa). A written caption of 2–3 sentences accompanies the inset: 'This matters to our driving question because…'	galaxies — can you label WHAT Hubble detected and WHY that connects to expansion?'; 'Your Apollo 11 arrow — where does it point in your model? What did lunar rocks tell us about the age of the Solar System?' At the 10-minute mark, say: 'Add your Kenya & Africa inset box. Write 2–3 sentences explaining why this matters to our driving question. You have 3 minutes.' Close by asking 2 learners to share their models under a document camera or hold them up. Ask the class: 'What is one thing this learner's model shows that yours doesn't yet — and what could you add?' [WAIT 10 seconds before taking responses.]		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) EVIDENCE QUALITY: Did learners' 3-sentence Evidence Summaries demonstrate the ability to name specific instruments and evidence types, or did they stay at the level of general mission narrative? If most summaries were vague, plan a brief 10-minute retrieval warm-up at the start of Lesson 9 using mission flashcards. (2) KENYA CONNECTION: How did learners respond to the KSA and SKA content? Did it generate genuine curiosity or feel 'bolted on'? If engagement was low, consider inviting a guest speaker (virtually) from the Kenya Space Agency for a future lesson — the KSA has an active public outreach programme. (3) MODEL ARROWS: Were learners able to draw meaningful arrows from missions to model components? If arrows were missing or random, the epistemological concept (instrument → evidence → knowledge) needs explicit re-teaching. Consider using a whole-class model-building session at the start of Lesson 9 before learners work individually. (4) DQB QUALITY: Did new sticky notes reflect methods-level thinking ('how does infrared imaging work?', 'what is a radio wave in this context?') or remain at recall level ('who built JWST?')? Methods-level questions signal readiness for Lessons 9–10, which focus on electromagnetic spectrum and telescope types. (5) PACING: The Observe phase (Mission Data Cards) is content-rich. If groups did not finish all 7 cards plus the KSA/SKA reading in 20 minutes, consider pre-assigning 2 cards per group and using a jigsaw structure in the next iteration of this lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined 7 Mission Data Cards (Sputnik 1, Vostok 1, Apollo 11, Voyager 1 & 2, Hubble Space Telescope, Mars rovers, James Webb Space Telescope) and a briefing on the Kenya Space Agency (KSA) and the Square Kilometre Array (SKA) in South Africa. They recorded what each mission discovered and how that discovery contributes evidence toward the unit's driving question about the night sky above Nairobi.
What did I learn?	Students learned that human knowledge of the universe is not based on direct travel or touch, but on an incremental chain of evidence gathered by instruments — from Sputnik's proof of orbit in 1957, to the Hubble deep field images confirming cosmic expansion, to the JWST's infrared imaging of galaxies from only 300 million years after the Big Bang. They also learned that Kenya

	launched its first satellite (1KUNS-PF) from the ISS in 2018 and that Africa hosts the SKA — the world's largest radio telescope — positioning the continent as an active participant in answering the deepest questions about the universe.
How does this explain the phenomenon?	Students explained, in their 3-sentence Evidence Summaries and updated cumulative models, that each space mission provided a specific type of evidence (light waves, radio waves, rock samples, infrared imaging) that revealed a specific aspect of the universe's history or structure. By adding a 'Human Evidence Layer' to their models — with arrows connecting missions to the parts of the universe each revealed — students demonstrated that our knowledge of what we see in the Nairobi night sky is the product of over 65 years of instrument-based, indirect evidence-gathering by scientists from many nations, including Kenya and Africa.

LESSON 9 (40 minutes): Space Phenomena and Effects on Earth: Solar Flares, Geomagnetic Storms, Eclipses, and Tidal Forces	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate how dynamic space phenomena — solar flares, geomagnetic storms, eclipses, and tidal forces — directly affect life on Earth, using the 2024 geomagnetic storm that disrupted Kenyan radio communications as the supporting phenomenon. This lesson answers the driving question by showing that the universe is not a static backdrop but an active system whose events reach down and touch daily Kenyan life.
Knowledge	Learners will know: (1) The Sun produces solar flares and coronal mass ejections (CMEs) that release charged particles and electromagnetic radiation into space. (2) Earth's magnetosphere deflects most solar wind but geomagnetic storms occur when CMEs interact strongly with the magnetosphere, disrupting radio signals, power grids, and satellite communications. (3) Solar and lunar eclipses result from the geometric alignment of the Sun, Earth, and Moon, and are predictable using orbital mechanics from Lesson 5. (4) Tidal forces from the Moon (and to a lesser extent the Sun) cause the twice-daily tidal cycle observed at Mombasa and influence lake-level fluctuations on Lake Victoria affecting fishing and agriculture.
Skills	Learners will be able to: (1) Analyse a timeline of the May 2024 geomagnetic storm and link each event (CME emission, solar wind arrival, magnetosphere compression, radio blackout) to a physical cause. (2) Construct a cause-and-effect diagram connecting solar activity to observable impacts on Kenyan infrastructure, farming calendars, and communications. (3) Predict how future geomagnetic storms might affect Kenya using a provided solar activity forecast graphic. (4) Use eclipse geometry diagrams to explain why a total solar eclipse is only visible from a narrow path on Earth's surface.
Attitudes	Learners will appreciate that Earth is not isolated from the cosmos — the universe actively shapes daily Kenyan life — and will develop a sense of responsibility toward understanding space weather as a matter of national infrastructure resilience and scientific literacy.
Key Inquiry Question	How do the motions and events of celestial bodies in the Solar System affect life on Earth?
Purpose in Storyline	Lessons 1 through 8 established what the universe is, how it formed, how we explore it, and how Kenya participates. Lesson 9 closes the loop by showing that space is not merely something to observe — it reaches back and affects the observer. The 2024 geomagnetic storm disrupting Kenyan radio communications is direct, lived evidence that the Sun (a star visible in our night sky) is an active force in Kenyan daily life. This advances the driving question by providing the most visceral, personal evidence yet that the universe is dynamic and interconnected with human existence on Earth.
Safety Notes	No physical hazards. Remind learners never to look directly at the Sun during any discussion of eclipses or solar observations. If showing the Sun through any optical device in a future practical, certified solar filters are mandatory. All activities in this lesson are paper-based and digital.

B. LESSON OVERVIEW

In this 40-minute lesson, learners begin by predicting which space events they think could affect their lives in Nairobi or Kisumu. They then observe a structured case study of the May 2024 geomagnetic storm — the strongest storm in 20 years — and its documented effects on HF radio communications used by Kenya Meteorological Department stations and amateur radio operators across East Africa. Using a cause-and-effect card-sorting activity, they construct explanations linking solar physics to tangible Kenyan impacts. The lesson also covers eclipse geometry and tidal forces on Lake Victoria and the Mombasa coast. Learners update their Driving Question Boards and revise their cumulative universe models to show the Sun as an active, dynamic force — not merely a light source — whose behaviour travels across 150 million kilometres of space to affect a student sitting in a Nairobi classroom.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** New localities lead to new biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a half-page scenario card (teacher-printed or read aloud): 'It is May 11, 2024. A radio operator at Kenya Meteorological Department in Dagoretti, Nairobi, notices that all high-frequency radio communications have gone silent. Farmers in the Rift Valley who rely on weather forecasts broadcast on these frequencies receive no warning of an approaching storm. In Mombasa, a fishing community notices unusually high tides. On social media, people in South Africa and Rwanda are posting photos of coloured lights in the sky — lights that have no business appearing over the equator. What is happening? Where is the cause?' Working in pairs for 3 minutes, learners record two predictions in their notebooks: (1) What do you think is causing these events? (2) Could something happening on the Sun reach all the way to a radio station in Nairobi? Explain your reasoning using evidence from previous lessons.	Distribute or read the scenario card. Say: 'You have studied the Big Bang, planetary orbits, gravity, telescopes, and space exploration. Now I want you to use that knowledge to explain something that actually happened in Kenya last year. With your partner, write two predictions. You have 3 minutes. Use what you already know — do not worry about being wrong.' After 3 minutes, use cold calling to sample 4 pairs. Record their predictions verbatim on the board in two columns labelled SPACE CAUSE and EARTH EFFECT. Do not correct errors yet. Say: 'These are your predictions. By the end of this lesson, you will know whether you were right — and why.' WAIT TIME: After asking each pair to share, wait 5 full seconds before prompting or moving to the next pair. This silence is productive — it signals that thinking is valued.	Prediction journaling activates prior knowledge from Lessons 6 and 7 (gravitational forces, electromagnetic radiation from stars). The two-column board display externalises the class's collective prior knowledge and creates cognitive dissonance between varied predictions, motivating the Observe Phase.	Teacher scans notebooks during the 3-minute pair work. Listen for whether learners spontaneously connect solar radiation or the Sun's energy output (Lessons 1-2) to Earth effects. Flag pairs who mention electromagnetic radiation for targeted questioning in the Explain Phase. Note pairs who have no connection to offer — these will need additional scaffolding during the case study.
Observe Phase 📹 **VIDEO:** New localities lead to new biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners receive a one-page structured case study handout: 'The May 2024 Geomagnetic Storm: What Happened and What Kenya Felt.' The handout contains four labelled panels: (A) Solar Event — A diagram of the Sun showing a coronal mass ejection (CME) on May 10, 2024, with a brief description: 'The Sun released a massive burst of charged particles and magnetic	Distribute the handout. Say: 'Read panel A silently for 90 seconds. Then I will explain what you are looking at.' After reading, explain Panel A in plain language: 'The Sun is not a quiet ball of fire. It has an atmosphere that erupts. When it does, it fires a cloud of charged particles across the Solar System at hundreds of kilometres per second. That cloud reached Earth.' Pause for questions before moving	The structured handout uses a spatial layout that mirrors the causal chain: Sun to space to Earth to Kenya. This visual sequence scaffolds the cause-and-effect reasoning learners will formalise in the Explain Phase. Annotating the handout (underline a cause, circle an effect) is a low-stakes sensemaking strategy that prepares learners for the card-sort activity. Connecting to the Kenya	Collect annotated handouts at the end of the phase for a quick scan. Look for whether learners correctly identify the CME (not the light flash) as the cause of radio blackouts. Common misconception to intercept: learners may think the Sun's heat caused the blackout rather than the magnetic disturbance from charged particles. Use this to prepare a targeted correction in the Explain

[Solar Power (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	field energy travelling at 800 km per second toward Earth.' (B) Journey Through Space — A diagram showing the CME crossing 150 million kilometres of space in approximately 17 hours, with labels for solar wind, electromagnetic radiation (arriving in 8 minutes at the speed of light), and the CME particle cloud (arriving later). (C) Earth Impact — A diagram of Earth's magnetosphere being compressed on the sunward side, with labels for aurora borealis and aurora australis extending toward the equator, radio blackout zones over Africa, and satellite drag increasing. (D) Kenya Connections — A factual summary box listing: HF radio blackouts affecting Kenya Meteorological Department and East African aviation weather communication; amateur radio operators in Nairobi reporting complete signal loss for 6 hours; an unusual geomagnetic reading recorded at the Mbita Point Research and Training Institute on Lake Victoria; tidal gauge data from Mombasa showing a 12 cm anomaly in predicted versus actual sea level during the storm. Learners also watch a 3-minute NASA video clip (Space Weather: Storms from the Sun — publicly available on NASA YouTube) showing real CME footage from the Solar Dynamics Observatory and a world map of the May 2024 blackout zones with East Africa highlighted. Learners annotate their handout, underlining one cause and one Kenya effect in each panel.	to Panel B. For Panels C and D, use a think-pair-share: 'Look at Panel D. With your partner, identify one effect on Kenya that surprised you. You have 90 seconds.' Cold-call two pairs. Then play the NASA video. After the video, say: 'Look back at the predictions you wrote 10 minutes ago. Underline anything you got right. Put a star next to anything the video or handout changed in your thinking.' WAIT TIME: After the video ends, wait 8 full seconds before speaking. Allow the information to settle. Do not rush to explain — let learners sit with the evidence first.	Meteorological Department and Mbita Point Institute grounds the global phenomenon in recognisable Kenyan institutions.	Phase.
Explain Phase 📹 **VIDEO:**	Learners complete a cause-and-effect card-sort activity in groups of	Distribute card sets face-down. Say: 'Do not flip the cards until I	The card-sort externalises the causal chain and forces learners to	During card sorting, circulate and listen for whether groups place

New localities lead to new biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	three. Each group receives 10 pre-cut cards (teacher-prepared) with the following events printed one per card: (1) Sun develops active sunspot region. (2) Coronal mass ejection launches charged particles into space. (3) Electromagnetic radiation (radio waves and X-rays) leaves the Sun at the speed of light. (4) EM radiation reaches Earth in 8 minutes — causes immediate radio blackout on sunlit side of Earth. (5) CME particle cloud travels for 17 hours and compresses Earth's magnetosphere. (6) Magnetosphere deflects most particles but allows some to spiral toward poles. (7) Charged particles collide with atmospheric gas — aurora forms (seen in South Africa and Rwanda). (8) Geomagnetic field fluctuations disrupt HF radio used by Kenya Met Department. (9) Satellite orbits experience increased drag — GPS accuracy reduces over East Africa. (10) Fishing communities at Lake Victoria and Mombasa observe tidal anomalies linked to combined solar-lunar tidal forcing. Groups arrange the cards into a logical causal chain on their desk, then draw connecting arrows and write a one-sentence explanation for each arrow ('This causes this because...'). After 7 minutes, each group shares one arrow-and-explanation with the class. Teacher builds the master chain on the board. The teacher then introduces two additional phenomena using diagrams drawn on the board: (a) Eclipse Geometry — Teacher draws Sun, Moon, and Earth in umbra-penumbra alignment and asks:	say go. When I say go, your group has 7 minutes to arrange these 10 cards into the correct causal sequence — from the Sun to Kenya. Every arrow you draw must have a reason written next to it. If you disagree in your group, try both arrangements and argue for yours using evidence from your handout.' After 7 minutes: 'Each group — tell me one arrow. Just one. Which card causes which other card, and why?' Build the master chain on the board from group contributions, asking the class to confirm or challenge each link. Say: 'Notice that this chain has a beginning in space and an ending in a Kenyan fishing village. The universe is not separate from us.' For eclipse geometry, draw slowly and ask: 'If the Moon were twice as far away from Earth as it actually is, would we still see total solar eclipses? Think about it for 30 seconds before answering.' WAIT TIME: 30 full seconds of silence after the eclipse question. Then cold-call. For tidal forces, connect explicitly to Lesson 6: 'We already know the Moon causes tides. What is new today is understanding that this is not just a coastal fact — it is a space physics fact that affects inland lakes and farming calendars in Kenya.' WAIT TIME: After asking learners to write their tidal sentence, give 2 full minutes of silent individual writing.	argue for sequencing using evidence — a high-order sensemaking move. The master chain built collectively on the board becomes a shared reference artefact. The eclipse geometry question applies Lesson 5 knowledge (orbital distances, geometry) to a new context, promoting transfer. The explicit tidal-food-security connection models scientific thinking as socially relevant, not abstract.	Card 3 (EM radiation) and Card 4 (immediate radio blackout) in the correct position relative to Card 2 (CME launch) and Card 5 (CME arrival 17 hours later). The most common error is conflating the two separate emissions (light-speed EM and slower particle cloud) into one event. Gently ask groups with this error: 'If light takes 8 minutes to reach Earth, but the particle cloud takes 17 hours, can they arrive at the same time? What does that mean for the sequence?' For the eclipse question, assess whether learners apply the concept of angular size from orbital distance (Lesson 5 content). Accept partial reasoning and build on it.

	'Why is a total solar eclipse only visible from a narrow strip of Earth? Use your knowledge of orbital distances from Lesson 5.' Learners respond in writing for 2 minutes. (b) Tidal Forces — Teacher revisits Lesson 6 content: 'The Moon's gravity pulls on Lake Victoria. The Sun also pulls, but less strongly. When the Sun and Moon align — as during a new or full moon — their combined tidal force is strongest. Fishermen on Lake Victoria time their nets to these tidal rhythms. This is space physics affecting Kenyan food security.' Learners add a sentence to their notebooks connecting tidal forces to a specific Kenyan livelihood.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [New localities lead to new biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Solar Power (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one sticky note (or paper slip) for the Driving Question Board. They choose from three prompts: (A) 'Today I found new evidence that answers the driving question: ...' (B) 'A new question this lesson raised for me is: ...' (C) 'Something about space phenomena that I still do not understand: ...' Learners post their notes in the corresponding column of the Driving Question Board. The teacher reads 3-4 notes aloud (anonymously) and categorises them: evidence gathered, questions answered, and open questions remaining. The class briefly discusses: 'How does knowing that solar flares can cut off Kenya Met Department radio change your view of the night sky above Nairobi? Is the sky still just a beautiful backdrop?'	Say: 'The Driving Question Board tracks our growing understanding. We have been building evidence since Lesson 1. Today you found that the universe is not just something we look at — it acts on us. Write one thing: a new piece of evidence, a new question, or something still unclear. One minute to write.' After posting: read 3-4 aloud. For each, say either: 'This is evidence that moves us closer to answering the driving question — because...' or 'This is a great open question. We may address it in Lesson 11 or it may be something scientists are still working on.' Close by saying: 'The night sky above Nairobi looked static and permanent in Lesson 1. Does it still look that way to you?' WAIT TIME: After this closing question, wait 6 full seconds. Let the silence do its work before taking 2-3 verbal responses.	The DQB update externalises the class's growing evidence base and metacognitive awareness. The three-prompt structure prevents the common student response of 'I have no questions' by offering an evidence-sharing option for confident learners. Reading notes aloud without attribution builds psychological safety for sharing uncertainty.	Scan sticky notes for three things: (1) Are learners attributing effects to causes correctly (CME causing radio blackouts rather than, for example, the Moon)? (2) Are questions becoming more sophisticated compared to earlier lessons — moving from 'what is the Big Bang' to 'how does space weather affect satellites'? (3) Are any learners still fundamentally confused about the Earth-Sun relationship? Flag these for one-to-one follow-up before Lesson 10.
Model	Learners open their cumulative	Say: 'Your model has been	Cumulative model building is the	Assess models for three key

Building Phase 📹 **VIDEO:** New localities lead to new biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	universe model — the visual representation they have been building since Lesson 2. In today's update, they add a new section labelled 'Dynamic Sun: Space Weather and Earth Effects.' In this section they must include: (1) An arrow from the Sun toward Earth labelled with two separate labels: 'EM radiation — 8 minutes' and 'CME particle cloud — 17 hours'. (2) A small diagram of Earth's magnetosphere with labels: 'deflects most charged particles' and 'compressed during geomagnetic storm'. (3) One Kenya-specific impact written in a speech bubble or annotation box — learners choose from radio blackout (Nairobi), tidal anomaly (Mombasa), aurora sighting (South Africa/Rwanda), or GPS disruption (East Africa). (4) A small eclipse geometry sketch showing umbra and penumbra with the label 'total eclipse only visible from narrow path.' Learners who finish early add a second annotation connecting today's tidal force content to Lesson 6, noting how the Sun reinforces or reduces the Moon's tidal effect depending on alignment.	growing since Lesson 2. Every lesson we have added something. Today the Sun changes from a static warm light source in your model to an active, dynamic force. Add the four elements I have listed on the board. You have 5 minutes. Make sure your model tells a story someone else could read and understand.' Circulate and look at models. For each learner, give one specific positive observation and one targeted prompt: 'I can see you have drawn the CME arrow — good. Now, what label shows me that it takes longer to arrive than the light? Where is that on your model?' WAIT TIME: Do not rush the model-building phase. Resist the urge to fill silence with instruction. If a learner is stuck, ask: 'What did the card closest to Kenya say? Can you draw that near your Earth diagram?' Close with: 'In Lesson 12 you will present this full model. Today is one of the most important updates — because today the universe stopped being something out there and became something happening here, in Kenya, right now.'	primary epistemic tool of this unit — it externalises the learner's evolving scientific understanding as a physical artefact. Today's update is particularly powerful because it requires learners to draw a directional causal arrow from the Sun to a specific Kenyan location, making the abstract connection between space physics and daily life concrete and spatially explicit. Encouraging early finishers to connect to Lesson 6 content rewards synthesis and prevents dead time.	elements: (1) Two separate arrows or labels distinguishing EM radiation speed from CME particle speed — this tests whether learners have grasped the two-event structure of a solar storm. (2) A magnetosphere label that shows it as protective rather than passive — learners who omit this may still hold a misconception that Earth is defenceless against solar wind. (3) A Kenya-specific annotation — this tests whether learners can move from global to local scale, a key science practice. Photograph two or three models (with permission) to use as exemplars in Lesson 11 consolidation.

D. TEACHER REFLECTION

After the lesson, the teacher should consider: (1) Did learners successfully distinguish between the two separate emissions from a solar flare (immediate EM radiation and the slower CME particle cloud)? If not, begin Lesson 10 with a brief 2-minute clarification using a racing analogy: light is like a sprinter reaching Earth in 8 minutes; the CME particle cloud is like a lorry reaching Earth 17 hours later. (2) Did the Kenyan contexts (Dagoretti radio station, Mbita Point Institute, Mombasa tidal gauge, Rift Valley farmers) land as real and relevant — or did learners still treat the events as abstract global phenomena? If the latter, consider beginning future lessons with a Kenyan news headline as the entry point rather than a global diagram. (3) Were there learners who made unusually sophisticated connections — linking today's solar activity content to Lesson 7 (how Hubble and JWST observe in non-visible wavelengths including X-ray and UV, exactly the wavelengths that cause radio blackouts)? If so, celebrate these publicly in Lesson 10 as examples of scientific thinking. (4) Did the card-sort activity reveal persistent misconceptions about the Sun being a stable, unchanging body? If so, this is an important conceptual challenge worth flagging for the Lesson 11 consolidation discussion. (5) Consider whether gender dynamics in group work were equitable — did all learners handle the cards and contribute to the causal chain, or did dominant voices control the activity? Assign specific roles (card-holder, arrow-drawer, scribe, spokesperson) in future group tasks to ensure participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed the Sun emitting a coronal mass ejection on May 10, 2024, in footage from NASA Solar Dynamics Observatory, and we observed documented evidence that this storm disrupted radio communications across East Africa, was recorded by the Mbita Point Research and Training Institute, and caused a measurable tidal anomaly in Mombasa — all traceable to a single event 150 million kilometres away on the Sun.
What did I learn?	We learned that the Sun produces two types of emissions during a solar storm: electromagnetic radiation (arriving at Earth in 8 minutes at the speed of light) and a coronal mass ejection particle cloud (arriving 17 hours later). Earth is partially protected by its magnetosphere, but strong CMEs compress it and allow charged particles to disrupt radio communications, satellite signals, and power infrastructure. Eclipses result from precise geometric alignments of Sun, Moon, and Earth, and tidal forces from both the Moon and Sun affect coastal and lake communities in Kenya.
How does this explain the phenomenon?	We explained that space is not a passive backdrop to life on Earth but an active system whose events — solar flares, geomagnetic storms, eclipses, and tidal forces — directly affect Kenyan infrastructure, communications, agriculture, and fishing. The universe visible in the night sky above Nairobi is not static or separate from us: it reaches down and shapes our daily lives, which is the most direct evidence yet that the driving question — how did everything in the night sky come to exist and how does it affect us — has a deeply personal answer for every Kenyan.

LESSON 10 (80 minutes): Careers in Space Physics: African and Kenyan Voices in the Cosmos	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To connect the evidence gathered throughout the unit to real-world career pathways in space physics, with emphasis on African and Kenyan professionals who study the very phenomena students have been investigating.
Knowledge	Learners identify specific career pathways in space physics (astrophysics, satellite engineering, meteorology, telecommunications, space law, data science) and name the institutions in Kenya and Africa that employ space professionals, including the Kenya Space Agency (KSA), Kenya Meteorological Department, universities such as University of Nairobi and Strathmore University, and regional bodies such as the African Space Agency.
Skills	Learners research, synthesise, and present structured career profiles using print and digital media; they evaluate sources for credibility; they design and display a career infographic or poster that highlights the connection between sub-strand content and a chosen career pathway.
Attitudes	Learners appreciate that space science is an achievable career for any Kenyan student regardless of gender, ethnicity, or county of origin, and they demonstrate open-mindedness by engaging respectfully with diverse career stories, including those of female scientists and engineers.
Key Inquiry Question	How do the motions of celestial bodies in the Solar System affect life on Earth? / How did the universe begin and how has it evolved over time?
Purpose in Storyline	This lesson is Lesson 10 of 12 in the evidence-gathering sequence. Students have spent nine lessons accumulating scientific evidence about the origin, structure, and dynamics of the universe visible above Nairobi. Now they pause to ask: who does this work professionally, and could I be one of them? By attaching real human faces — especially African and Kenyan faces — to the science, learners deepen their sense that the driving question is not merely academic but is the daily work of living scientists. Gender equity is foregrounded as a PCI to dismantle the myth that space science is not 'for' girls or students from rural counties.
Safety Notes	No physical hazards. If internet research is conducted, remind learners to use school-approved platforms only and to evaluate sources critically. Ensure any guest speaker (virtual or in-person) has been vetted by the school administration in advance.

B. LESSON OVERVIEW

Students work in small research groups to build and present a structured 'Career Profile' of a real space scientist or engineer with African or Kenyan connections. Each group is assigned or selects a different career cluster (astrophysics, satellite engineering, meteorology, space law/policy, data science for space, telecommunications). Groups use provided infographics, curated web links, and — where connectivity allows — the Kenya Space Agency website and university course pages to gather evidence. Profiles are presented to the class as short 'Ted-Talk style' pitches, followed by a whole-class gallery walk. The lesson closes with students individually writing one sentence connecting their chosen career to the anchoring phenomenon: the night sky above Nairobi. Gender Disparity is addressed explicitly through PCI discussion mid-lesson, and the Driving Question Board is updated with new entries: 'Who is working to answer our driving question right now?'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students sit in their existing groups. Each student receives a blank 'Career Prediction Card' — a half-page template with three prompts: (1) 'Name one job you think exists in space science.' (2) 'Do you think there are Kenyan professionals doing this job? Why or why not?' (3) 'Could you see yourself in a space science career? What would excite or worry you about it?' Students complete cards individually and silently for 4 minutes, then share within their group for 3 minutes. Groups note any disagreements or surprises on a sticky note and post it on the Driving Question Board under a new column headed 'Who studies this?'	Distribute Career Prediction Cards as students settle. Say exactly: 'Before we look at any sources today, I want to know what you already believe about who does space science and whether it is something Kenyans — people like you — actually do for a living. Write honestly. There are no wrong answers here.' Allow 4 minutes of silent individual writing — enforce no talking. After 4 minutes say: 'Now share your cards inside your group. You have 3 minutes. Listen carefully — I will cold-call three people to share something surprising a group member said.' After group sharing, cold-call exactly 3 students (ensure at least one female student is called). Ask: 'What surprised you most about what a group member predicted?' Allow 10 seconds of wait time after each question before accepting responses. As sticky notes go up on the DQB, read two aloud and say: 'We are going to see by the end of today whether your predictions were right.'	Elicitation of Prior Beliefs (Prediction Cards): This strategy surfaces existing assumptions and misconceptions about who does science before evidence is introduced. By making predictions public on sticky notes, students create a personal stake in being 'right' or 'corrected' — motivating engagement with the research phase. Connecting to the phenomenon: the anchoring image of a student gazing at the night sky above Nairobi is invoked to ask — has any Kenyan professionally studied that same sky?	Observable indicator: Every student has completed all three prompts on their Career Prediction Card before group sharing begins. Teacher circulates and checks cards — look specifically for students who write 'no' to the question about Kenyan professionals, as these students hold the key misconception this lesson targets. Note names for targeted cold-calling during the Explain phase.
Observe Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Gravity and Space Problems (Flexbook)] Source: CK-12 🔍 Search ARES	Each group is assigned one of six career cluster folders (physical or digital). Each folder contains: (a) a 1-page curated career profile of a real African or Kenyan space professional; (b) a list of 3–4 Kenyan/African institutions connected to that career; (c) a short list of university courses or qualifications required; (d) one or two 'connection questions' linking the career to sub-strand content studied in Lessons 1–9. Career clusters and featured professionals are: Cluster 1 —	Distribute or open career cluster folders. Say: 'You have 18 minutes to read your folder carefully and fill in your Career Profile Poster. Every member of the group must be able to explain every part of the poster — I will assign who presents randomly at the end.' Circulate continuously. At the 5-minute mark, pause the class for 30 seconds and say: 'I am seeing some groups skip the connection question at the bottom of your template. That question is the most important one — it links your	Jigsaw Evidence Gathering: Each group becomes the class 'expert' on one career cluster. This mirrors how actual scientists specialise and then share findings at conferences — connecting the strategy to real scientific practice (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information). The curated profiles serve as primary evidence documents, modelling how scientists use secondary sources while evaluating credibility. Connection to phenomenon: each	Observable indicator: The 'Connection to our unit content' field on every group's Career Profile Poster must include at least one specific concept from Lessons 1–9 (e.g., Big Bang, cosmic microwave background, planetary motion, stellar classification, Doppler effect). Teacher checks this field during circulation — if it is blank or vague ('it is related to space'), prompt the group with: 'Which specific lesson or idea from our unit does this career use every day? Think about the evidence we

for similar readings	Astrophysics/Cosmology (featuring Dr. Hannington Alatenge Otieno, Kenyan physicist); Cluster 2 — Satellite Engineering (featuring the Kenya Space Agency's BIRD-5 CubeSat team); Cluster 3 — Space Meteorology (featuring Kenya Meteorological Department professionals who use satellite data from EUMETSAT); Cluster 4 — Telecommunications & Satellite Technology (featuring Safaricom engineers who rely on geostationary satellites); Cluster 5 — Space Policy & Law (featuring the African Union Space Policy and Kenyan legal professionals in the sector); Cluster 6 — Data Science for Space (featuring researchers at Strathmore University's @iLabAfrica using satellite data for agriculture in the Rift Valley). Groups read, discuss, and collaboratively complete a Career Profile Poster template: Name of career / Key daily tasks / Kenyan institution employing this professional / Education pathway from Form 1 to job / Connection to our unit content / One quote or fact that surprised us. Groups have 18 minutes for this evidence-gathering phase.	career to everything we have studied about the universe since Lesson 1. Do not skip it.' At the 12-minute mark, visit any group that has not yet discussed gender representation in their cluster and prompt: 'Look at your career profile — is the professional male or female? Does that match what you predicted on your card?' Ensure the PCI prompt is raised organically in at least four groups before the full-class discussion. With 2 minutes remaining, say: 'Begin deciding who will present which part of the poster. I will choose the presenter randomly, so everyone must be ready.'	career profile explicitly includes a 'connection to our unit' field, ensuring students trace every career back to the question of how we know what we know about the night sky above Nairobi.	have gathered.'
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES	Groups take turns delivering 3-minute 'Ted-Talk style' pitches presenting their Career Profile Poster to the class. After all six presentations, the teacher facilitates a 10-minute structured whole-class PCI discussion on gender equity in space careers, using data from the posters and from a short infographic displayed on screen: 'Women in African Space Science — 2024 Statistics.' Students then complete an individual 'Exit Ticket Reflection' (3	Before pitches begin, say: 'You have 3 minutes per group. I will stop you at 3 minutes exactly. The rest of us are listening for two things: first, a connection to our driving question about the night sky above Nairobi; second, whether the career pathway is realistic for a student sitting in this room today.' Use a visible timer. After each presentation, ask the audience exactly one question — rotate the question type: 'What surprised you?' / 'How does this	Ted-Talk Peer Teaching + Structured PCI Discussion: Peer teaching (NGSS SEP 8) consolidates group knowledge and builds communication competency. The structured PCI discussion uses real data (gender statistics) rather than opinion alone — modelling the scientific value of evidence-based argument. The Exit Ticket Reflection closes the cognitive loop by requiring each individual student to personally articulate connections —	Observable indicator: Exit Ticket Reflection sentence 1 must name a specific career AND give a specific reason linked to unit content (e.g., 'The astrophysicist career is most connected because they study the cosmic microwave background radiation that is evidence for the Big Bang — the origin of the stars we see above Nairobi'). Collect all Exit Tickets before students leave. Flag any student whose sentence 1 is non-specific for a brief check-in

for similar readings	sentences): (1) The career I found most connected to our driving question is ___ because ___. (2) One thing I was wrong about in my Prediction Card was ___. (3) If I were to pursue a space career, I would need to study ___ at ___ [institution].	career use evidence we gathered in Lesson [X]?' / 'Would this career be open to a student from Homa Bay or Garissa — what would they need?' Allow 10 seconds of wait time before cold-calling. After all presentations, display the gender infographic and say: 'Look at this data. In Kenya's space sector, approximately 28% of professionals are women. Does that reflect ability, or does it reflect something else? I want honest answers — take 10 seconds to think before anyone speaks.' Cold-call at least 2 female students and 2 male students. Validate all responses and guide towards the conclusion: 'The myth that physics is not for girls is exactly that — a myth. The Kenya Space Agency actively recruits women. Your ability determines your path, not your gender.' Close by distributing Exit Ticket Reflection slips.	preventing passive observation during group presentations. Connection to phenomenon: every presentation must reference the anchoring night sky, reinforcing that each career is a professional attempt to answer the driving question.	conversation at the start of Lesson 11.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the Driving Question Board as a class. The teacher adds a new column header: 'Who is working to answer our driving question RIGHT NOW?' Students are given 3 sticky notes each — one green (something we now know that we did not know before this lesson), one orange (a new question this lesson raised), one blue (a connection between today's career and a previous lesson's evidence). Students post all three notes in the appropriate DQB column. The class reads 3–4 notes aloud together and the teacher maps visible connections between today's career profiles and concepts from earlier lessons (e.g., 'The meteorology career connects to Lesson 6 on planetary	Hand out three sticky notes per student. Say: 'Green note — one new thing you know. Orange note — one new question today raised. Blue note — one link between a career we heard about today and evidence we gathered in a previous lesson. You have 4 minutes. Write one idea per note — be specific, not vague.' After posting, read 2 green, 1 orange, and 1 blue note aloud. For the blue note, explicitly trace the connection on the DQB by drawing a line or arrow between the new career card and the relevant earlier lesson evidence card. Say: 'This is what scientists call convergent evidence — multiple different types of work all pointing towards the same questions about the universe. Every career we	Three-Colour DQB Protocol: The tri-colour sticky note system differentiates between new knowledge (green), new uncertainty (orange), and cross-lesson synthesis (blue) — making three distinct cognitive moves visible and trackable. This mirrors how scientific knowledge is structured: not just accumulation of facts but recognition of remaining gaps and integration across domains. The teacher's explicit arrow-drawing models the scientific practice of identifying patterns and connections (NGSS SEP 6: Constructing Explanations).	Observable indicator: Every student posts all three sticky notes before time is called. The blue 'connection' note is the highest-order task — teacher scans these specifically. A strong blue note names both a career AND a specific concept from a prior lesson (e.g., 'The KSA satellite engineer connects to Lesson 3 where we saw how telescopes gather electromagnetic radiation'). Weak blue notes ('it connects to space') are flagged for follow-up.

	motion and how Earth's tilt, studied through satellite data, causes our seasons in Nairobi').	heard about today is another way of investigating the same night sky we started this unit with.'		
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Gravity and Space Problems (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative 'How We Know' model — the annotated diagram of the universe that has been built across Lessons 1–9. Today they add a new layer: a 'Human Network' overlay. Using a different coloured pen, students annotate their model diagram with small icons or labels showing WHERE in the universe's structure each career operates. For example: cosmologists work at the scale of the entire observable universe (Big Bang, CMB); astrophysicists work at the scale of galaxies and stars; satellite engineers work at the scale of Earth orbit; meteorologists work at the scale of Earth's atmosphere; telecommunications engineers work at the scale of geostationary orbit. Students also add one new sentence to their model's caption that begins: 'The evidence about the universe above Nairobi is gathered by real scientists who…' and complete it based on today's lesson.	Say: 'Open your cumulative model. In a new colour — I suggest red — add at least three career labels to the correct scale on your diagram. A cosmologist does not work at Earth orbit level — they work at the largest scale our model shows. Be precise.' Circulate and check that career labels are placed at scientifically appropriate scales. Prompt misconceptions: if a student places 'meteorologist' at the galaxy scale, ask: 'What data does a meteorologist at the Kenya Meteorological Department actually use? At what distance from Earth is that data collected?' Allow 10 minutes for model annotation. With 3 minutes remaining, ask students to write their caption sentence. Cold-call 3 students to read their sentence aloud — ensure one sentence is from a student who earlier expressed doubt about Kenyan professionals in space science. Close by saying: 'Your model now shows not just what the universe is, but who studies it — including people who grew up in Kenya, studied at Kenyan universities, and now answer the same question you asked when you first looked up at the night sky: how did all of this come to exist?'	Layered Model Annotation (Human Network Layer): Adding a 'Human Network' layer to a scientific diagram is a high-order synthesis task that requires students to integrate two knowledge domains: the structure of the universe (content) and the sociology of science (careers). This models authentic scientific practice — real universe maps and atlases include the institutions and teams that produced each data layer. The caption sentence forces individual written synthesis, producing a formative artefact that can be assessed. Connection to phenomenon: the model's final caption explicitly names the night sky above Nairobi as the object of study, closing the loop on the anchoring phenomenon.	Observable indicator: Each student's model has at least three career labels added in a new colour, each placed at a scientifically defensible scale on the universe diagram (not randomly distributed). The caption sentence must include: (1) a reference to real scientists or a real institution, AND (2) a reference to evidence or observation. Collect models at the end of the lesson for written feedback — return before Lesson 11 with one specific comment per model.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 11:

1. GENDER PCI: Did the structured PCI discussion feel genuine or performative? Did female students contribute freely, or did male students dominate? If the latter, consider a think-pair-share structure before open discussion in future lessons.

2. CAREER CONNECTIONS: Which career cluster produced the richest connections to the driving question? Which produced the weakest? The weakest cluster likely

needs a stronger 'connection question' in its folder — revise before next cohort.

3. MODEL QUALITY: Review the collected cumulative models. Are career labels placed at appropriate scales, or are students placing all careers at the 'Earth' level regardless of scope? If so, design a brief scale-review activity at the start of Lesson 11.

4. MISCONCEPTION CHECK: Review Exit Tickets. How many students still wrote non-specific connections in sentence 1? These students may need a structured scaffold (sentence starters) during Lesson 11's final explanation preparation.

5. ENGAGEMENT: Which groups were most engaged during the Ted-Talk pitches — the presenters or the audience? If the audience was passive, introduce a formal 'question card' system in Lesson 11 where audience members must write one question per presentation before the presenter opens for discussion.

6. DRIVING QUESTION PROGRESS: After 10 of 12 evidence-gathering lessons, can most students now articulate a partial answer to both KICD Key Inquiry Questions in their own words, connected to real careers? If not, Lessons 11 and 12 must build in more structured synthesis time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed curated career profiles of real African and Kenyan space professionals across six career clusters (astrophysics, satellite engineering, meteorology, space policy, data science, telecommunications). They observed gender representation data showing approximately 28% of Kenya's space sector professionals are women, and they observed real institutional names — Kenya Space Agency, Kenya Meteorological Department, University of Nairobi, Strathmore University — connected to the careers discussed.
What did I learn?	Students learned that the scientific questions anchoring this unit — how the universe began, how celestial bodies move, how we gather evidence about the cosmos — are actively investigated by real Kenyan and African professionals today. They learned specific career pathways, educational requirements, and institutional homes for space careers in Kenya. They learned that gender is not a scientific barrier to a space career, and that the Kenya Space Agency actively recruits women. They added a 'Human Network' layer to their cumulative universe model, placing each career at its correct cosmic scale.
How does this explain the phenomenon?	Students explained, through Ted-Talk pitches and individual Exit Ticket Reflections, how each career cluster connects to specific evidence gathered in Lessons 1–9 of this unit. They explained, using the structured PCI discussion, why gender disparities in the space sector reflect social myth rather than scientific ability. They updated the Driving Question Board with new knowledge, new questions, and explicit cross-lesson connections — advancing collective progress towards answering: 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?'

LESSON 11 (80 minutes): Consolidation and Evidence Review: Building a Coherent Narrative of the Universe	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To synthesise evidence gathered across all 11 lessons into a coherent, evidence-based narrative that answers the driving question: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?
Knowledge	Learners consolidate knowledge of the Big Bang, cosmic expansion, the formation of stars and planets, the Solar System, celestial body classification, planetary motion, and the methods scientists use to gather indirect evidence about the universe.
Skills	Learners synthesise multi-lesson evidence, construct structured written and visual arguments, evaluate the quality of scientific evidence, revise collaborative models, and communicate scientific narratives clearly and coherently.
Attitudes	Learners appreciate that scientific understanding is built incrementally through evidence gathered over time, and that unanswered questions are an invitation to further inquiry rather than a failure of science.
Key Inquiry Question	How did the universe begin and how has it evolved over time? How do the motions of celestial bodies in the Solar System affect life on Earth?
Purpose in Storyline	This lesson is the capstone of the 12-lesson evidence-gathering sequence. Learners pause before the final explanation lesson to review all evidence collected, complete the three-column synthesis template (Lesson Title | What Was Learned | How It Explains the Phenomenon), and revisit the Driving Question Board to address lingering questions. By the end, learners possess a full, internally consistent narrative — grounded in evidence — that explains the night sky above Nairobi from the Big Bang to the present moment.
Safety Notes	No physical hazards. Ensure respectful, inclusive discussion norms are maintained. Learners should be reminded that expressing uncertainty or unresolved questions is scientifically valid and valued.

B. LESSON OVERVIEW

In this 80-minute consolidation lesson, learners work collaboratively to synthesise all 10 previous lessons into a single, coherent evidence-based narrative answering the unit's driving question. Using a teacher-provided three-column synthesis template (Lesson Title | What Was Learned | How It Explains the Phenomenon), small groups reconstruct the logical chain of evidence from the Big Bang through to the motions of celestial bodies visible from Nairobi tonight. The Driving Question Board (DQB) is comprehensively reviewed: questions that have been answered are marked and evidenced, while unresolved questions are flagged for future study or the final explanation lesson. Learners also revisit and refine the cumulative model they have built across the unit, producing a final annotated diagram that maps the universe's history and structure. Digital tools including Stellarium, NASA visualisations, and Khan Academy resources are available for reference and verification. The lesson culminates in group gallery presentations of the synthesis template, with peers providing structured feedback using a WWW/EBI (What Went Well / Even Better If) protocol.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Without referring to any notes,	Distribute the prompt on a half-	Brain Dump / Prior Knowledge	Observable indicator: Can learners

📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [physics-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	learners individually write for 5 minutes in response to the prompt: 'Imagine you are standing on the shores of Lake Victoria on a clear night. A younger student — perhaps your sibling — points at the sky and asks: How did all of this get here? Using only what you now know, write your best answer.' Learners then share their responses with a shoulder partner, noting where their answers overlap or differ. Selected responses are shared with the whole class and displayed on a visible section of the board labelled 'Our Best Answer So Far.'	sheet of paper or display it on the board. Say verbatim: 'Do not open any books or look at any notes. This is not a test — it is a snapshot of where your thinking is right now. You have exactly 5 minutes. Go.' Start a visible timer. After 5 minutes, say: 'Turn to your shoulder partner. You have 2 minutes. Tell each other your answer and listen for anything your partner said that you missed or said differently.' After pair share, cold-call 4 students (aim for at least 2 female, 2 male voices) to share highlights. Write key phrases from their responses on the board. Do NOT correct or affirm yet — say: 'We are going to test how good this answer is today.' WAIT TIME: Allow 10–12 seconds of silence after the cold-call question before accepting responses.	Activation — by forcing retrieval without notes, learners surface their current mental model and identify gaps before the synthesis activity. Displaying responses publicly creates accountability and a reference point for measuring growth by the end of the lesson.	produce a multi-sentence, causally connected narrative (e.g., Big Bang → expansion → star formation → Solar System → Earth → life) without prompting? Partial or fragmented responses indicate which conceptual links need reinforcement during the synthesis activity. Note which lessons or concepts are absent from student responses — these become priority focus areas for Phase 2.
Observe Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [physics-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners are placed into groups of 3–4 and receive the three-column synthesis template: Lesson Title | What Was Learned | How It Explains the Phenomenon (the Nairobi night sky). Groups are given 25 minutes to collaboratively complete one row per lesson (Lessons 1–10), referencing their exercise books, any printed handouts, and available digital resources (Stellarium, Khan Academy Big Bang articles, NASA visualisation screenshots, PhET Gravity and Orbits simulation, and the 'Evolution of the Universe in 5 Minutes' YouTube summary). Each group is assigned two 'anchor lessons' that they must complete first and in greatest depth before moving to others. Anchor lesson pairs are: Group A — Lessons 1 & 2 (Big Bang,	Distribute the three-column template (pre-printed or digital). Say verbatim: 'This template is the backbone of your final explanation. Every row is one piece of evidence. When you are done, someone reading only this table should be able to understand how the universe went from nothing to the sky we see above Nairobi tonight. Your anchor lessons must be completed in full before you move on — those are your group's expert rows.' Circulate continuously. At each group, spend no more than 90 seconds before moving on. Ask targeted probing questions: 'Which row are you least confident about? What evidence from your exercise book supports what you have written in column 3? If I removed column 2, could a reader still understand	Jigsaw Synthesis with Gallery Walk — assigning anchor lessons distributes expertise across groups, ensuring that every lesson is treated with depth by at least one group. The gallery walk introduces peer review as a metacognitive tool, prompting learners to evaluate the logical coherence and evidence quality of each row — not just recall facts. This mirrors how scientists review and critique each other's evidence claims.	Observable indicator: Does each completed row in column 3 explicitly reference the anchoring phenomenon (the Nairobi night sky)? Are causal connectives used (e.g., 'this means that,' 'this is evidence that,' 'this explains why')? Groups whose column 3 entries are merely restatements of column 2 (without connecting to the phenomenon) need teacher redirection. Target check: At least 8 of 10 rows should have a distinct, phenomenon-linked explanation by end of gallery walk.

	evidence); Group B — Lessons 3 & 4 (expansion, cosmic background radiation); Group C — Lessons 5 & 6 (star formation, stellar classification); Group D — Lessons 7 & 8 (Solar System formation, planetary motion); Group E — Lessons 9 & 10 (celestial body classification, space exploration and indirect evidence methods). After 25 minutes, groups do a 5-minute 'gallery walk' to read other groups' templates and add sticky-note comments or additions to rows they can improve.	column 3?' During the gallery walk, prompt learners: 'Find one row that is missing a connection to the phenomenon and add a sticky note suggesting how to improve it.' WAIT TIME: When a group is stuck, say the question and wait 12 seconds before offering a hint. Hint format: 'Think about what you observed in that lesson — what did the data show?'		
Explain Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **PDF:** [physics-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 [Search ARES for similar readings]	The class reconvenes. The physical or digital Driving Question Board (DQB) is displayed in full. The teacher reads each question aloud (or a student volunteer does so). For each question, the class votes using a simple system: Green card (we can now answer this with evidence), Yellow card (we have partial evidence but gaps remain), Red card (this is still unanswered or has become a new question). Each green-carded question is 'closed' by a student volunteer who states the answer and names the lesson and evidence that supports it. Yellow-carded questions are discussed briefly — what evidence do we have, and what is missing? Red-carded questions are transcribed onto a new section of the board: 'Questions for Future Study.' Following the DQB review, each group uses their completed synthesis template to construct a 5-sentence 'Universe Narrative' paragraph that tells the full story from the Big Bang to tonight's Nairobi sky, written as if explaining to a Form 1 student who has never	Display the DQB prominently. Say verbatim: 'We have spent 10 lessons gathering evidence. Now it is time to see what that evidence actually answers. I am going to read each question. Hold up your card — green, yellow, or red. Be honest. Yellow is not a failure; it means science is still working.' After each green card, cold-call one student: 'Tell us the answer and the evidence. Which lesson? What did we observe or read?' After yellow cards, say: 'What would we need to find out to turn this yellow into a green? Where might that evidence come from?' After red cards, say: 'Write this on a sticky note and put it on the Future Questions section. Scientists have lists like this too.' For the narrative paragraph, say: 'Five sentences. Every sentence must move time forward. Start with: Approximately 13.8 billion years ago… and end with: …and that is why, on a clear night above Nairobi, we can see…' Circulate and check that narratives are causally linked, not just a list. Cold-call 3 groups to read their	DQB Traffic Light Review + Guided Narrative Construction — the traffic light protocol externalises metacognitive monitoring, making visible to both teacher and learner which aspects of the driving question have been resolved by evidence and which remain open. The constrained narrative paragraph (5 sentences, causal progression, specific start and end) forces learners to prioritise and sequence evidence rather than list disconnected facts — a key scientific reasoning skill.	Observable indicator: For DQB — are students accurately categorising questions (not over-confidently greening questions for which they cannot name evidence)? For narrative paragraphs — do all 5 sentences use causal language and advance chronologically? Does the final sentence explicitly name something observable in the Nairobi night sky (e.g., the Milky Way band, the Moon, Venus, Jupiter, star colours)? Flag any narrative that ends with a generic conclusion not grounded in the phenomenon.

	studied the topic. Groups write the paragraph on a large sheet of manila paper and display it.	paragraph aloud. WAIT TIME: After each group reads, wait 10 seconds before asking the class: 'What is one thing this group said particularly well? What is one thing they could add?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [physics-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Using the traffic light outcomes from Phase 3, the class collaboratively creates a 'Final DQB Status Summary' — a two-column chart: Questions We Can Now Answer With Evidence | Questions That Remain Open. For answered questions, learners write a one-sentence evidence summary beside each. For open questions, learners write a brief note on what kind of evidence or study would be needed to answer them (e.g., 'We would need data from a space telescope,' 'This may be answered in a future unit on modern physics'). Individually, each learner then writes a personal 'Knowledge Gain Statement' in their exercise book: 'At the start of this unit, I thought the night sky above Nairobi was… Now I understand it is actually… The most surprising piece of evidence I encountered was… One question I still have is…'	Provide the two-column chart template verbally or on the board. Say verbatim: 'Column one is our victories — questions science has answered, and we have the evidence to prove it. Column two is our horizon — questions that are still open, either because science has not answered them yet, or because we have not studied the evidence yet. Both columns matter equally.' Guide the class through each question from the DQB vote, scribing the summary onto the board or a shared digital doc. For open questions, ask: 'What kind of scientist would investigate this? What tool would they need? We have talked about telescopes, spectroscopy, the cosmic microwave background — which of these points toward an answer?' For the personal Knowledge Gain Statement, say: 'This is private — just for you and me. Be completely honest. If you are still confused about something, write it down. That is the most useful thing you can write.' Cold-call 2–3 volunteers to share their 'most surprising piece of evidence' sentence only. WAIT TIME: 12 seconds after the prompt before accepting volunteers. Affirm with: 'That is exactly the kind of thinking scientists do — surprise is a signal that your mental model has just been updated.'	Two-Column DQB Synthesis + Reflective Writing — separating answered from open questions teaches learners that science is a cumulative, ongoing process, not a closed body of facts. The personal Knowledge Gain Statement is a structured metacognitive reflection tool that surfaces individual misconceptions and growth, giving the teacher actionable data before the final explanation lesson.	Observable indicator: Is the student's 'At the start I thought… Now I understand…' statement demonstrating genuine conceptual shift (e.g., moving from 'the universe has always existed' to 'the universe began 13.8 billion years ago from a hot dense state and has been expanding ever since')? Are students able to name a specific piece of evidence (not just a lesson topic) as their most surprising? Generic or unchanged statements signal a learner who needs individual follow-up before Lesson 12.
Model Building	Each learner retrieves their cumulative model — the annotated	Say verbatim: 'Your model has grown with every lesson. Today	Cumulative Model Revision with Peer WWW/EBI Protocol — model	Observable indicator: Does the final model include all major

Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [physics-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	diagram that has been revised across all previous lessons. Using their completed synthesis template and DQB summary, learners spend 10 minutes making final revisions: adding any missing labels, correcting any inaccurate annotations, adding arrows that show causal or temporal relationships (e.g., Big Bang → expansion → cooling → first atoms → stars → Solar System → Earth → life → human observer in Nairobi), and writing a 2–3 sentence 'Model Caption' at the bottom that summarises what the model shows and how it answers the driving question. Learners then do a paired model exchange — swapping with a partner, reading the caption, and giving one specific WWW (What Went Well) and one EBI (Even Better If) comment in pencil on a sticky note attached to the model. Final models are collected by the teacher for review before Lesson 12.	you are completing it. It should now tell the full story of the universe — from the Big Bang to you, standing on the shores of Lake Victoria, looking up. Every arrow matters. Every label is evidence. The caption is your voice — it should sound like a scientist explaining their diagram.' Circulate during the 10-minute revision. Look for: missing temporal arrows, incorrect sequencing, vague labels ('stuff happened' instead of 'nuclear fusion in first stars'), and captions that do not reference the phenomenon. Prompt where needed: 'Where does the formation of our Solar System sit on this timeline? Can I see that in your model? How does this part of your model connect to what we can actually observe from Nairobi tonight?' For the paired exchange, model the WWW/EBI protocol with an example: 'WWW: Your model clearly shows the arrow from the Big Bang to cosmic expansion. EBI: If you labelled the Milky Way formation separately from the Solar System formation, it would be even clearer.' Collect all models at the end. Say: 'I will review these before our final lesson. If there is something important to address, I will write a note on your model.' WAIT TIME: During revision, enforce a quiet working norm for the full 10 minutes — no group talk, just individual focus.	revision is a core scientific practice (NGSS SEP 2: Developing and Using Models). The final revision integrates all prior sensemaking into a single visual artifact that externalises the learner's complete mental model of the universe. The WWW/EBI peer review introduces constructive scientific critique as a social learning norm, and the collection of models gives the teacher a formative data set to inform the final explanation lesson (Lesson 12).	milestones (Big Bang, expansion, CMB, first stars, galaxy formation, Solar System formation, Earth, Moon, planetary motion)? Are causal arrows present and correctly directed? Does the model caption explicitly answer the driving question in 2–3 sentences? Models that are missing more than 3 major milestones or lack causal arrows indicate incomplete conceptual integration and require targeted feedback before Lesson 12.

D. TEACHER REFLECTION
After collecting the synthesis templates and cumulative models, review them before Lesson 12 with three specific questions in mind: (1) Which lesson rows in the synthesis template have the weakest 'How It Explains the Phenomenon' entries — these are the conceptual links that need reinforcement in the final explanation? (2) Which DQB questions remained red-carded by the majority of learners — these should be explicitly addressed at the opening of Lesson 12? (3) Which learners'

personal Knowledge Gain Statements show little or no conceptual shift from the beginning of the unit — these students need targeted individual support or a differentiated task during Lesson 12. Note also: Were learners able to construct causally linked narrative paragraphs, or did they tend to list facts without connecting them? If the latter, the final explanation lesson should explicitly model the difference between a list of facts and a causal scientific argument. Celebrate aloud any moments in this lesson where learners expressed genuine intellectual surprise or revised a prior misconception — these are the most important learning events in the entire unit and should be named and affirmed in Lesson 12 as examples of how science (and scientists) actually work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What evidence from across all 10 previous lessons did learners draw on today — including Big Bang evidence, cosmic expansion data, stellar formation, Solar System structure, planetary motion, and the methods scientists use to gather indirect evidence about the universe?
What did I learn?	How does each of the 10 lessons contribute a distinct, necessary piece of evidence that together builds a complete explanation of how the universe — and everything visible in the night sky above Nairobi — came to exist and continues to evolve?
How does this explain the phenomenon?	How does the full body of evidence gathered across this unit answer the driving question: the night sky above Nairobi looks static and eternal, yet science shows it erupted from a hot dense point 13.8 billion years ago and has been expanding, evolving, and structuring itself ever since — and scientists know this not by travelling to the universe but by reading the light, radiation, and motion it sends to us here on Earth?

LESSON 12 (80 minutes): Model Building and Final Presentation: From the Big Bang to the Night Sky Above Nairobi	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all unit learning by completing, presenting, and critiquing cumulative visual models of the universe — from the Big Bang to the present night sky above Nairobi — and to articulate a full, evidence-based Final Explanation that answers the unit Driving Question.
Knowledge	Learners demonstrate integrated knowledge of: the Big Bang and cosmic timeline (~13.8 billion years); stellar evolution (nebula → main sequence → red giant → supernova/white dwarf/neutron star/black hole); Solar System structure and classification of celestial bodies; orbital mechanics and gravitational forces (Newton's Law of Universal Gravitation); types of telescopic and electromagnetic evidence (visible light, radio waves, CMB); and key milestones in space exploration history including contributions by African and Kenyan scientists.
Skills	Learners apply scientific argumentation (NGSS SEP 7) to construct and defend a Final Explanation; use modelling (NGSS SEP 2) to revise and present a cumulative visual model; provide structured peer feedback using evidence-based criteria; compare their Lesson 1 pre-instructional model with their final model to identify conceptual growth; and synthesise multiple sources of evidence into a coherent scientific narrative.
Attitudes	Learners appreciate the collaborative and cumulative nature of scientific knowledge-building; demonstrate intellectual humility by acknowledging how their initial ideas were incomplete or incorrect; show respect for peer contributions during structured feedback; and develop a sense of wonder and Kenyan identity in relation to the universe observed above their own skies.
Key Inquiry Question	1. How did the universe begin and how has it evolved over time? 2. How do the motions of celestial bodies in the Solar System affect life on Earth?
Purpose in Storyline	This is the capstone lesson of the unit. All previous lessons have built evidence, concepts, and partial models in service of this moment. In Lesson 1, learners sketched a naïve model of what they thought the night sky above Nairobi contained and why. Over eleven lessons they gathered evidence — from Big Bang theory and CMB radiation, to stellar evolution, gravitational forces, orbital mechanics, telescopic data, and space exploration. In Lesson 12, learners complete their cumulative visual model, present it to peers with a structured Final Explanation, receive and act on feedback, and formally close out the Driving Question Board. This lesson answers the anchoring phenomenon: the night sky above Nairobi appears static, but everything in it has a dynamic, evidence-supported origin story that science can reconstruct.
Safety Notes	No laboratory hazards in this lesson. If learners are using digital devices, ensure responsible and respectful online conduct. Remind learners that structured peer feedback must remain constructive and respectful; no personal criticism — only evidence-based commentary on the model content.

B. LESSON OVERVIEW

In this 80-minute capstone lesson, learners finalise their cumulative visual models of the universe and deliver structured Final Explanation presentations to their peers. The lesson opens with a brief Predict Phase revisiting learners' Lesson 1 naïve models and the original Driving Question Board entries. In the Observe Phase, learners conduct a rapid evidence audit — reviewing key resources including the NASA CMB visualisation, Stellarium sky data for Nairobi, the PhET Gravity and Orbits simulation summary, and their own unit notes — to ensure their models are evidence-complete. The Explain Phase centres on gallery-walk presentations: each group presents their cumulative visual model and delivers a 4-minute Final Explanation structured around the unit Driving Question, while peers provide written feedback using a structured CER (Claim–Evidence–Reasoning) rubric. The DQB Creation Phase formally closes the Driving Question Board: the class revisits every question posted across the unit, categorises each as 'answered with evidence', 'partially answered', or 'still open', and discusses what further investigation would be needed. The Model

Building Phase involves a final individual model revision incorporating peer feedback, followed by a whole-class synthesis discussion that constructs a shared, authoritative Final Explanation for the anchoring phenomenon. The lesson closes with individual written reflections comparing Lesson 1 and Lesson 12 understanding.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Big bang introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners receive back their original Lesson 1 sketches of the night sky above Nairobi and their initial written responses to the Driving Question: 'How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to?' Working individually for 3 minutes, they silently read their Lesson 1 model and annotate it in a different colour pen, marking: (a) one idea they held in Lesson 1 that was incomplete or incorrect, (b) one idea they held in Lesson 1 that was broadly correct, and (c) one concept they now understand that they had no language for in Lesson 1. They then share annotations briefly with a shoulder partner (Think-Pair-Share, 2 minutes). This activates metacognitive awareness of their learning journey and frames the lesson as a culminating meaning-making event connected directly to the anchoring phenomenon.	Distribute Lesson 1 model sketches and Driving Question cards (prepared in advance). Say verbatim: 'Before we build our final models today, I want us to travel back in time — not 13.8 billion years, just to Lesson 1. Look at what you drew and wrote on that very first day. Read it carefully. Use your red pen to annotate it silently — I want you to find three things: one idea that was incomplete, one idea that was already on the right track, and one concept you now have language for that you simply didn't know existed back then. You have 3 minutes of silent thinking time — starting now.' [WAIT — full 3-minute silence, circulate, do not prompt.] After 3 minutes: 'Now turn to your shoulder partner. Each of you has 90 seconds to share your annotations. Partner A goes first — go.' [1 min 30 sec.] 'Partner B — your turn now.' [1 min 30 sec.] Cold-call 3 pairs: 'Tell me one concept you now have that you didn't have in Lesson 1.' Accept responses, write key concepts on the board (Big Bang, stellar evolution, orbital mechanics, gravitational force, electromagnetic spectrum evidence, CMB). Say: 'These are the ideas your final model must incorporate. Let's make sure the evidence is solid before we present.'	Metacognitive Model Annotation — Learners use a colour-coded annotation protocol to make their own thinking visible and to recognise conceptual change. This strategy builds understanding by externalising the difference between prior and current knowledge, making the learning journey concrete and motivating learners to ensure their final model reflects genuine conceptual growth rather than surface changes.	Observable indicator: Every learner has at least three colour-coded annotations on their Lesson 1 model within the 3-minute window. Teacher scans models during circulation and listens to pair-shares. Red flag: any learner whose annotations are superficial or who cannot identify a new concept — teacher makes a note to check this learner's Final Explanation for depth. Whole-class indicator: the 3 cold-called pairs produce at least 4 distinct scientific concepts on the board, confirming broad unit coverage.

Observe Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scientific Notation: Light Years to the Centaurus Constellation (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Working in their established groups of 4, learners conduct a structured 12-minute 'Evidence Audit' of their cumulative visual model using a printed Evidence Checklist. The checklist contains 10 evidence anchors drawn from across the unit: (1) Big Bang timeline with approximate age of universe (~13.8 billion years); (2) CMB radiation as evidence for the Big Bang; (3) Hubble's redshift observations and expanding universe; (4) at least three stages of stellar evolution with named examples; (5) the role of gravity in forming stars, planets, and galaxies; (6) Solar System structure showing all 8 planets, asteroid belt, and the Moon; (7) Newton's Law of Universal Gravitation formula and its application to orbital mechanics; (8) at least two types of telescopes or electromagnetic spectrum bands used in astronomy; (9) at least one African or Kenyan connection to space science (e.g., Kenya's equatorial launch advantage, the Square Kilometre Array radio telescope project in South Africa, or the work of Ethiopian astronomer Tessema); (10) a clear visual representation of the night sky above Nairobi as observed from the Rift Valley or Lake Victoria shores, incorporating the Milky Way and at least two named celestial bodies. Groups tick completed items, flag gaps, and spend the remaining time filling in missing elements using their unit notes, the NASA CMB infographic (printed), the Stellarium screenshot of Nairobi's sky, and the 'Evolution of the Universe in 5 Minutes' YouTube summary notes. Each group's	Distribute the printed Evidence Checklist to each group. Say verbatim: 'You have 12 minutes to audit your model against this checklist. Every item on this list is evidence or a concept we have built up across the past eleven lessons. Your model should show all ten. If something is missing, use your notes, the NASA printout, or the Stellarium screenshot on your desk to fill it in now. This is your last chance to strengthen your model before you present it to the class. Go.' [WAIT — circulate actively. Spend no more than 2 minutes per group. Prompt with: 'Where is your CMB evidence?' / 'Can I see Newton's formula on your model?' / 'Where is Nairobi in your model — is it clear that this is the sky above Kenya?' / 'Where is your African connection to space science?'] At 10 minutes: 'Two minutes remaining — finalize your labels and make sure every team member can explain every part of the model, because anyone in your group may be asked to present any section.' At 12 minutes: 'Evidence audit complete. Models face-down on your desks. We move to presentations now.'	Structured Evidence Audit (Checklist Protocol) — Learners use a criterion-referenced checklist to systematically evaluate the completeness of their model against unit-wide evidence standards. This strategy builds understanding by transforming a creative artifact (the model) into a scientific argument: every visual element must be traceable to a specific piece of evidence gathered during the unit, mirroring how professional scientists validate theoretical models against observational data.	Observable indicator: Each group's checklist has all 10 items ticked or annotated with a plan before presentations begin. Teacher spot-checks 3 groups during circulation, asking one member to verbally explain one checklist item in their model. Red flag: groups missing more than 3 items at the 8-minute mark — teacher directs them to the specific resource that addresses the gap. Overall indicator: all groups' models are physically complete (fully labelled, legible, structured from Big Bang to present night sky) when the gallery walk begins.

	model must be physically finalised (drawn, annotated, labelled) before the gallery walk begins.			
Explain Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scientific Notation: Light Years to the Centaurus Constellation (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Groups post their cumulative visual models on the classroom walls, forming a gallery. Each group selects one spokesperson to remain at their model for the first rotation; all other members become 'visitors'. In a structured gallery walk (two rotations of 6 minutes each), visitors move to two other groups' models and listen to a 3-minute Final Explanation delivered by the spokesperson, then spend 3 minutes providing written feedback on a structured CER (Claim–Evidence–Reasoning) Peer Feedback Card. Feedback cards ask: (1) 'What is the clearest and most evidence-supported claim in this model?' (2) 'Name one piece of evidence in this model that you found compelling and explain why.' (3) 'Identify one part of the model or explanation that would benefit from stronger evidence or clearer reasoning — be specific.' After two rotations (12 minutes), all learners return to their own model. Spokespersons from each visited group share the most useful piece of feedback received (whole-class, 4 minutes). Groups then have 4 minutes to make one final evidence-based revision to their model based on peer feedback before the whole-class Final Explanation synthesis.	Set up the gallery by directing groups to post models. Say verbatim: 'This is your Final Explanation moment. When you are the spokesperson, you have 3 minutes to walk visitors through your model and answer the Driving Question out loud: How did everything we see in the night sky above Nairobi come to exist, and how do scientists gather evidence about a universe they cannot directly touch or travel to? Use your model as your evidence board. Visitors: you are scientists peer-reviewing a colleague's model. Fill in your CER Feedback Card honestly and specifically — no vague comments, only evidence-based observations.' [Signal Rotation 1 start.] Circulate during gallery walk, listening to Final Explanations. After each 3-minute explanation, prompt visitors: 'Ask one clarifying question before you write your feedback.' [Signal Rotation 2.] After rotations: 'Return to your own model. Read the feedback cards you received. Spokespersons — I will cold-call four groups: tell the class the single most useful piece of feedback you received and how you will act on it.' [Cold-call 4 groups.] 'You now have 4 minutes to make one evidence-based revision. Make it count.' [4-minute revision window.] Circulate, verify revisions are evidence-linked, not cosmetic.	CER Gallery Walk (Claim–Evidence–Reasoning Peer Review) — Learners function as both presenters and peer reviewers, using a structured CER framework to evaluate scientific models. This mirrors authentic scientific peer review and builds understanding by requiring learners to articulate not just what a model shows but whether the evidence is sufficient and the reasoning is sound — directly connecting to NGSS SEP 7 (Engaging in Argument from Evidence). Presenting to peers also consolidates the presenter's own understanding through the act of teaching.	Observable indicator: Every spokesperson can deliver a coherent 3-minute Final Explanation that answers the Driving Question using at least 3 named pieces of evidence from their model. Every visitor completes all three prompts on the CER Feedback Card with specific, model-referenced comments (not generic praise). Teacher evaluates 4 cold-called feedback summaries for specificity and scientific accuracy. Red flag: any Final Explanation that cannot connect the Big Bang to the present night sky above Nairobi in a causal chain — teacher notes this group for the synthesis discussion to address the gap publicly.
Driving Question Board (DQB)	The class reconvenes facing the Driving Question Board (DQB) that has been maintained across all 12	Direct learners' attention to the full DQB. Say verbatim: 'This board has been growing since Lesson 1.	Driving Question Board Synthesis (Traffic-Light Categorisation) — Learners use a colour-coded	Observable indicator: Every DQB question receives a colour-coded dot with a one-sentence

Creation 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scientific Notation: Light Years to the Centaurus Constellation (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	lessons. The DQB contains questions generated by learners in every lesson — from 'What caused the Big Bang?' to 'Why does the Moon affect ocean tides in Mombasa?' to 'Can a Kenyan ever go to space?' and 'How do we see light from stars that died billions of years ago?' Working as a whole class (teacher-facilitated, 8 minutes), learners systematically categorise every question on the DQB using three sticky-dot colours: GREEN = 'We can now answer this with evidence from our unit'; YELLOW = 'We have partial evidence — this question needs more investigation'; RED = 'This question is still open — it may be at the frontier of current science.' Learners take turns approaching the board, placing a coloured dot on each question, and giving a one-sentence justification. After all questions are categorised, the class identifies the two most important RED (still open) questions and discusses what kind of evidence or investigation would be needed to begin answering them — connecting to how real scientists at observatories in South Africa, NASA, or the James Webb Space Telescope team work to answer open cosmological questions.	Every question on it was asked by someone in this room — and every question mattered. Now it's time to find out how far we've come. We're going to categorise every question using three colours: green means we can answer it with evidence right now; yellow means we have some evidence but the question isn't fully closed; red means this is still open — maybe even for the world's best scientists. I need volunteers to come up, place a dot, and give one sentence of justification. Who wants to start?' [Accept volunteers; if no volunteers after 5 seconds, cold-call by name.] Move through all DQB questions in 8 minutes (keep pace). After categorisation: 'Look at our red questions — the ones that are still open. Which two are most important to you?' [Show of hands to vote — tally.] 'For our top two: what kind of evidence would a scientist need to begin answering this? What instrument, what observation, what data?' [Cold-call 2 learners per question.] Close by saying: 'Some of these questions may be answered in your lifetime. The James Webb Space Telescope, the Square Kilometre Array in South Africa — these are real instruments gathering real evidence right now. Your curiosity is pointing in the same direction as the world's best science.'	categorisation protocol to systematically assess the explanatory power of the unit's accumulated evidence against the full landscape of questions they raised. This strategy builds understanding by making epistemic progress visible: learners can physically see that green questions have multiplied since Lesson 1, demonstrating that science produces genuine answers — while red questions honour the authentic open-endedness of frontier science and position learners as future contributors, not passive recipients.	justification from a learner (not the teacher). The class correctly categorises at least 80% of green questions with an accurate evidence reference. Teacher listens for misconceptions during justifications — if a learner incorrectly categorises a question (e.g., marks as green a question for which no evidence was gathered), teacher prompts: 'What specific evidence from our unit supports that answer?' Red flag: fewer than 60% of questions are marked green — this indicates insufficient evidence integration and teacher should note for post-lesson review and possible supplementary session.
Model Building Phase 📹 **VIDEO:** [Big bang introduction] Source: Khan Academy (English - US curriculum)	In the final phase (15 minutes), learners complete two tasks. TASK 1 — Individual Written Reflection (8 minutes): Each learner independently writes a personal Final Explanation in their science notebook, responding to the unit Driving Question in full paragraphs. The explanation must:	Transition to individual writing. Say verbatim: 'Now it is your turn — individually, in silence, in your science notebook. Write your personal Final Explanation for the Driving Question. You have 8 minutes. Your explanation must travel the full journey: from the Big Bang, through the formation of	Sequential Collaborative Authorship (Sentence-by-Sentence Synthesis) — Learners construct the definitive Final Explanation one sentence at a time in a cold-call sequence, each sentence required to be scientifically accurate and logically connected to the previous one.	Observable indicator: Individual written Final Explanations include all 7 required components (a–g) and are written in learner's own words in full paragraphs (not bullet points). Teacher reads at least 6 notebooks during the writing window and flags any missing components. Whole-class

🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	(a) begin with the Big Bang (~13.8 billion years ago) as the origin of space, time, matter, and energy; (b) trace the formation of the first stars and galaxies through gravitational collapse of hydrogen clouds; (c) explain stellar evolution including how heavier elements (carbon, iron) were forged in stellar cores and distributed by supernovae — the atoms now in every Kenyan's body were made in stars; (d) describe the formation of the Solar System from a collapsing nebula ~4.6 billion years ago; (e) explain how Newton's Law of Universal Gravitation governs the orbital motion of planets, the Moon, and how that motion affects life in Kenya (tides in Mombasa, seasons, the 24-hour day); (f) identify at least two forms of electromagnetic evidence scientists use to study a universe they cannot travel to (CMB, redshift, radio telescopes); and (g) conclude by describing what a person standing in the Rift Valley on a clear night in Nairobi is actually seeing — ancient light from objects whose histories span billions of years — and why that view is not static but is a snapshot of a dynamic, evolving universe. TASK 2 — Whole-Class Synthesis (7 minutes): Teacher facilitates a rapid, structured whole-class construction of the definitive class Final Explanation on the whiteboard, cold-calling learners to contribute one sentence each in sequence, building the explanation from Big Bang to present night sky above Nairobi. The final written explanation remains on the board as the class's collective scientific achievement for the unit.	stars and planets, through the forces that govern orbital motion, through the evidence scientists use to know what they know — all the way to you, standing in the Rift Valley tonight, looking up. The atoms in your hand were forged in stars. The light you see left those stars thousands of years ago. Write that story, with evidence.' [WAIT — 8-minute silent writing. Circulate, read over shoulders, do not interrupt. Note 2-3 strong sentences you observe to quote back to class.] After 8 minutes: 'Pens down. Now we build the class Final Explanation together on the board. I will cold-call learners one by one. Each person contributes one sentence that continues the explanation — it must be scientifically accurate and it must connect to the next idea. Who begins us at the Big Bang?' [Cold-call Learner 1 — write their sentence verbatim on the board.] Continue cold-calling, building the explanation sentence by sentence: Big Bang → first atoms → first stars → stellar evolution → supernovae and element formation → Solar System formation → gravitational orbital mechanics → telescopic evidence → the night sky above Nairobi. After the final sentence, step back and say: 'Read what we have written together. This is 12 lessons of physics, 13.8 billion years of cosmic history, and every one of you contributed a sentence. That board belongs to this class.'	This strategy builds understanding by transforming individual knowledge into a co-authored scientific narrative, making the logical causal chain from Big Bang to present night sky explicit and visible. It also reflects how scientific knowledge is cumulative and community-built — no single scientist, and no single sentence, tells the whole story.	synthesis indicator: the board-built Final Explanation forms a coherent causal chain from Big Bang to Nairobi night sky with no logical gaps — if a gap appears (e.g., no sentence connecting stellar evolution to Solar System formation), teacher prompts the next cold-called learner: 'The previous sentence left us with supernovae. What happens to all that expelled material — and what does it have to do with our Sun?' Red flag: any learner unable to contribute a scientifically accurate sentence — teacher scaffolds by asking a simpler sub-question and returns to the learner for a second attempt before moving on.

D. TEACHER REFLECTION
After delivering this capstone lesson, reflect on the following questions before the next academic cycle: 1. MODEL GROWTH: When you compared learners' Lesson 1 models with their Lesson 12 final models side-by-side, what was the most striking conceptual change you observed? Were there any concepts that remained shallow or underdeveloped across the unit (e.g., stellar evolution stages, the role of CMB as evidence)? If so, which lesson earlier in the unit should be restructured to build that concept more robustly? 2. FINAL EXPLANATION QUALITY: During the gallery walk, did learners' Final Explanations successfully answer the Driving Question with a complete causal chain from Big Bang to the night sky above Nairobi? Or did explanations tend to address some sections (e.g., Solar System structure) more confidently than others (e.g., electromagnetic evidence, gravitational mechanics)? Note which concepts need additional lesson time or earlier formative checkpoints in the next iteration. 3. DQB CATEGORISATION: What proportion of DQB questions were marked GREEN at the end of Lesson 12? If fewer than 70% were green, this suggests that evidence-building across the unit was insufficient or not explicitly connected back to the DQB during earlier lessons. Consider making DQB review a 3-minute standing routine at the start of every lesson in the next cycle. 4. KENYA CONTEXT INTEGRATION: Did learners make spontaneous connections to Kenyan contexts (Rift Valley observation, Mombasa tides, equatorial launch advantages, Square Kilometre Array in Africa, Kenyan seasons) in their Final Explanations and model presentations — or did these connections require teacher prompting? If prompting was needed, plan to embed Kenya-specific evidence more deliberately in lessons 3–8 so that by Lesson 12 these connections are learner-driven. 5. PEER FEEDBACK QUALITY: Were CER Feedback Cards specific and evidence-based, or were they generic? If generic, introduce the CER framework explicitly in Lesson 3 and practise it in at least two intermediate lessons so learners are fluent in the protocol by Lesson 12. 6. EQUITY CHECK: Were all learners able to contribute a sentence to the whole-class Final Explanation synthesis? If some learners were silent or unable to contribute, review your cold-call tracking records and ensure that in the next unit, quieter learners receive more scaffolded opportunities to articulate scientific ideas aloud in earlier lessons — so that this final public moment is accessible to every learner, not only the most confident.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners re-examined their Lesson 1 naïve models of the night sky above Nairobi and annotated them to identify conceptual change. They audited their cumulative visual models against a 10-item Evidence Checklist covering the Big Bang, CMB radiation, stellar evolution, Solar System structure, gravitational forces, orbital mechanics, telescopic evidence, and African connections to space science. They participated in a structured CER Gallery Walk, delivering Final Explanation presentations and providing written peer feedback. They categorised all Driving Question Board questions using a traffic-light protocol. They wrote an individual Final Explanation and contributed to a whole-class synthesis explanation built sentence-by-sentence on the whiteboard.
What did I learn?	Learners synthesised the full unit narrative: the universe began approximately 13.8 billion years ago in the Big Bang, producing space, time, matter, and energy; the first stars formed through gravitational collapse of hydrogen clouds; stellar evolution produced heavier elements distributed by supernovae; the Solar System formed from a collapsing nebula ~4.6 billion years ago; Newton's Law of Universal Gravitation governs orbital mechanics, affecting life in Kenya through tides, seasons, and the lunar cycle;

	scientists gather evidence about a universe they cannot travel to using CMB radiation, redshift observations, radio telescopes, and space probes; the night sky above Nairobi — seen from the Rift Valley or the shores of Lake Victoria — is not static but is a snapshot of a dynamic, evolving universe whose light carries billions of years of history.
How does this explain the phenomenon?	Learners constructed and defended a complete, evidence-based Final Explanation for the anchoring phenomenon and the unit Driving Question: the apparently permanent and unchanging night sky above Nairobi is in fact a window into 13.8 billion years of cosmic history. Every star is a sun undergoing nuclear fusion; the Milky Way is one of hundreds of billions of galaxies formed from primordial hydrogen; the planets orbit the Sun because of gravitational forces first mathematically described by Newton; the atoms in every Kenyan's body were forged in the cores of long-dead stars. Scientists know this not by travelling to the universe but by decoding the light, radiation, and gravitational signals it sends — using telescopes, satellites, and space probes that extend human senses across cosmic distances. The universe is not eternal and unchanging: it had a beginning, it has evolved, and it is still expanding today.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.